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Derivation of Effective Equations for a Microscopic Model of Chemotaxis with Creation of Chemotaxin Particles

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Abstract

The aim of this master's thesis is to derive the effective equations for a microscopic system (a system of stochastic differential equations) for *chemotaxis* of N amoebas attracted to the molecule of cAMP (cyclic adenosine monophosphate) that they released in random pulses. The attraction is given by a Gaussian-like potential, and the creation of cAMP particles follows a Poisson process. This microscopic model is inspired from [11], which is also the main source of the techniques that we used in the proof, together with [5]. Furthermore, we discuss the similarities and differences of the macroscopic description found with the classic model for chemotaxis, namely the *Keller-Segel* equation.

Zusammenfassung

Das Ziel dieser Masterarbeit ist die Herleitung von effektiven Gleichungen für ein mikroskopisches Modell (ein System von stochastischen Differentialgleichungen) für *Chemotaxis* von N Amöben, die vom Molekül von cAMP (zyklisches Adenosinmonophosphat) mit einem gaussartigen Potential angezogen werden. Diese Partikeln von cAMP werden wellenartig mit einem Poissonprocess abgegeben. Das mikroskopische Modell ist von [11] inspiriert, das auch die Quelle der vielen Techniken ist (zusammen mit [5]), die hier benutzt wurden, um das Endresultat zu beweisen. Darüber hinaus erörtern wir hier die Ähnlichkeiten und die Unterschiede dieses Modells zu dem klassichen makroskopischen Modell für Chemotaxis, nämlich die Gleichung von *Keller-Segel*.

Sommario

Lo scopo di questa tesi magistrale è di derivare le equazioni effettive per un modello microscopico (un sistema di equazioni differenziali stocastiche) per la chemotassi di N amebe attratte dalla molecola di cAMP (adenosina monofosfato ciclico) che esse stesse secernono in istanti ditribuiti casualmente nel tempo tramite un processo di Poisson. L'interazione è espressa mediante un potenziale di tipo Gaussiano. Il modello microscopico è ispirato a [11], che, insieme a [5], è la fonte principale delle tecniche adottate nella dimostrazione del risultato principale. Inoltre discutiamo delle somiglianze e delle differenze che la descrizione macroscopica che abbiamo derivato ha con il modello macroscopico classico per la chemotassi, cioè l'equazione di *Keller-Segel*.

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Tübingen, den 22. Mai 2026 Carlo Lanzi Luciani

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1. Introduction

If nature were not beautiful it would not be worth knowing, and life would not be worth living.

H.P., [33]

This master's thesis' aim is to analyze mathematically the phenomenon of *chemotaxis*, i.e., the movement of an organism in response to a chemical stimulus. This is a mechanism used by a considerable amount of microorganisms (various bacteria and amoebas as well as some eukaryots) to search for nutrients and escape toxins, but also by cells of a multicellular living beings for more complex tasks (e.g., leukocytes rely on it to find infections in the body [31]). Naturally, the study of this process has wide applications in medicine, since the variation of the chemotactical behaviour of various human cells can help the onset of a number of illnesses [21], but also because many drugs act by altering the chemotactic behaviour of specific agents [34, Conclusions and perspectives]. There is considerable research on this theme on a cellular level (how does the microorganism perceive the chemical stimulus? Which receptors are responsible for it? [39]), but we are interested to inquire about the behaviour of a population of microorganisms: what happens when all these corpuscles start to follow a specific chemical stimulus? Can we predict what we will observe about the spatial distribution of the group, for example through the lenses of a microscope?

When examining mathematically the phenomenon on such a scale, there are two main points of view: one can see the population macroscopically, as a continuum, representing it therefore with a density a, or can see it microscopically, i.e., as a finite group of individuals $\{A^1, \dots, A^N\}$. Of course it would be optimal that the two descriptions somehow match, meaning that if we construct some empirical density from $\{A^1, \dots, A^N\}$, this is in some way close to a, meaning that one can mathematically derive the macroscopic system from the microscopic one. For this procedure, which is often seen as adding insights and validity to a macroscopic description, is common to speak of *propagation of chaos* and of *effective equations* of the microscopic model. This was the topic of Cañizares García in her PhD thesis [11], who derived the classical macroscopic model for chemotaxis, the *Keller-Segel* equation, from a microscopic model for the amoeba *Dictyostelium Discoideum*. The master's thesis focuses on studying what happens if one would construct a more "microscopic" system than the one of [11], where the molecules of the chemical stimuli are taken into account as a different population of individuals spawned at random times.

1.1. Chemotaxis...

In Ancient Greek "τάξις" means "order" or "arrangement", but in biology it is used to indicate the movement of an organism in response to an external stimulus. Depending on the type of such triggers the phenomenon is classified into different subcategories, e.g., *phototaxis* or *thermotaxis*. The τάξις can be *positive*, meaning directed towards the stimulus, or *negative*, away from it. *Chemotaxis* (from the Ancient Greek "χημεία", meaning "chemistry" - or more precisely "transmutation") is the specific case where the reason of the movement is some chemical substance. Many lifeforms rely on this mechanism for

their survival, the most simple examples being unicellular organisms searching for nutrients (e.g., the bacterium *Escherichia coli* follows sources of sugar, [2]) or fleeing toxins (e.g., *Escherichia coli* being repelled by acetate, [40]). More evolved paradigms can be observed in multicellular organisms: white blood cells are able to find infections by following formyl peptides, that is molecular byproducts of many bacteria [20]. The phenomenon is even more interesting when the *chemotaxin*, the substance triggering the reaction, seems to be released as a form of signaling. Many kinds of growth are controlled in this way: the axons of neurons develop following extra-cellular signaling substances [38], as it is the case for the sprouting of new capillaries *angiogenesis*. The latter example is particularly relevant in medicine because many tumors secrete the corresponding chemotaxin VEGF (Vascular Endothelial Growth Factor), stimulating the proliferation of blood vessels in their direction and effectively hacking the body machinery to steal nutrients [41]. Other important examples from medicine can be found in [21].

1.1.1. ...and Dictyostelium Discoideum

In order to study a phenomena occurring so frequently and in so many forms, it is necessary to find a case free of great complications and which can be observed in a controlled environment: the amoeba *Dictyostelium discoideum* is perfect for such a role. This organism proliferates till there are available nutrients. When these finish, the cells first disperse, searching for other food resources, but if after a while these are not found, some individuals start to secrete cAMP (3',5'-cyclic adenosine monophosphate) waves in quick pulses. Such substance act as a chemical signal, triggering the amoebas who come in contact to it to direct their movement towards its higher concentration and to release other cAMP themselves. This chemotactical mechanism aggregates the microorganisms into many *slugs*, each formed by several thousands members; these lumps then migrate to the appropriate spot thermotactically to undergo the last metamorphosis: they produce *fruiting bodies* which then release spores and die. The spores become then new individuals, and the life cycle is closed [1].

The *Dictyostelium discoideum* is fairly easy to cultivate in a lab [25, Chapter 4], in some stages of its life is haploid (i.e., has a single copy of its DNA) which allows for easy genetic manipulation, and its genome (which is approximately 34MB of data) is completely mapped and available to the public [35]. At the same time it is an eukaryote and presents complex behaviours, such as cell differentiation and apoptosis (programmed cell death) in the non-unicellular phases, thermotaxis in its slug phase [27]. And, more relevant to us, it is regarded as one of the basic example for eukaryotic chemotaxis [36].

1.2. Previous Models

1.2.1. The Classic Approach: Keller-Segel

The macroscopic *classic mathematical model for chemotaxis*, namely the Keller-Segel system of equations, was given in 1970 in [23] (a more beginner friendly and recent source to read about it is [17]), even though an heuristic microscopic derivation had already been given by Patlak in 1953 [32]. In the specific case in which the chemotaxin molecules diffuse much faster than the organism reacting to them (as it is the case for the aforementioned amoeba), one can derive from the general case this parabolic-elliptic system of equations [18]:

[1] Since sometimes an image is worth a thousand words, I advise you to take a look at the informal short documentary [1].

$$\begin{cases} \partial_t a = -\chi \nabla(a \nabla c) + \triangle a \\ -\triangle c = a. \end{cases} \tag{1.1}$$

Here a represents the density of the microorganisms, c of the chemotaxin and χ is a constant called *chemosensitivity*, which measures how much the cells are reacting to the chemical stimuli. Solving the system for c and setting $V_\lambda := \frac{x}{2\pi|x|^2}$ we obtain

$$\partial_t a = \chi \nabla((V_\lambda \star a)a) + \triangle a. \tag{1.2}$$

1.2.2. The PhD Thesis of Cañizares García

In [11] the author derived the set of equations 1.2.1 from a microscopic model previously seen in numerical approximation of solutions of Keller-Segel [16] as well as in other derivation of it [15]. The microscopic system describes N amoebas with their positions $\{A^1, \dots, A^N\} \in \mathbb{R}^2$ and encodes their attraction via cAMP with a Coulomb-like potential v. In equations we have

$$\forall p \in \{1, \dots, N\} \qquad A_t^p = A_0^p - \int_0^t \frac{\chi}{N} \sum_{q \neq p} V_\lambda(A_s^p - A_s^q) ds + \sqrt{2} B_t^p, \qquad A_0^p \sim a_0, \tag{1.3}$$

where $B^p : \mathbb{R}_+ \longrightarrow \mathbb{R}^2$ are i.i.d. (independent and identically distributed) B.m. (Brownian motion(s)), and at time $t = 0$ the microorganisms A_0^p are i.i.d. according to the given density a_0. From this system of SDEs (stochastic differential equations) she derived the Keller-Segel equations in their parabolic-elliptic form 1.2. This means that she proved, under the appropriate conditions, that the empirical measure

$$\mu_t^{A,(N)} := \frac{1}{N} \sum_{p=1}^{N} \delta_{A_t^p}$$

converges to a_t as a r.v. (random variable(s)) with values in $\mathcal{P}(\mathbb{R}^2)$ (the space of probability measures in $\mathbb{R}^2$), where a_t is the solution to 1.2 with initial condition $a_{t=0} = a_0$. Her technique involved the usage of a mesoscopic model given by the i.i.d. *mean-field particles*

$$\forall p \in \{1, \dots, N\} \qquad \overline{A}_t^p = \overline{A}_0^p - \int_0^t \frac{\chi}{N} (V_\lambda \star a_s)(\overline{A}_s^p) ds + \sqrt{2} B_t^p, \qquad \overline{A}_0^p \sim a_0, \tag{1.4}$$

coupled (i.e., with the same B.m.) with the original ones. She used the fact that the expression 1.2 is the *Kolmogorov forward equation* (Fokker-Planck) of 1.4 to obtain the convergence of the aforementioned $\mathcal{P}(\mathbb{R}^2)$-valued r.v. by showing that the probability of $(A^p)_p$ and $(\overline{A}^p)_p$ being far tends to 0 as N grows to infinity. More precisely, in [11, Theorem 2.1] she could prove a degree of convergence of this probability to 0 faster than $N^{-\gamma}$ for every $\gamma > 0$.

1.3. Chemotaxin in the Microscopic Description

The focus of this master's thesis is to modify the microscopic model used by Cañizares García to encode the cAMP as a different population of particles C^m being *created* at Poisson-distributed times T_m by the amoebas and wiggling of B.m., and see which consequences does this have on the macroscopic description. Which equation describes the

model on a large scale? Which interpretation can we give to the result? Can we obtain in this way the Keller-Segel equation 1.2?

One would expect this different microscopic framework to not yield exactly 1.2, because the cAMP molecules C^m, encoded as in our model, do not have a diffusion coefficient (and release frequency) which is inherently much higher than the one of the amoebas. And in fact, as we will see, this expectation will prove to be true. Still, one could guess that, in some limit case where we let the diffusion coefficient of C^m and the Poisson parameter λ (encoding the spawning frequency) tend to infinity, the law 1.2 appears, or something close to it. This hypothesis is not formally confirmed, but we expect that this is the case and we give a glimpse of how one might prove it.

1.3.1. Outline

Chapter 2 In this chapter we define the microscopic model with N amoebas A^p and creation of chemotaxin molecules C^m. The creation of particles happens via Poisson-distributed spawning process $(M_t^p)_{p=1}^N$, and at each spawning time T_m only one particle C^m is released. Such C^m moves purely of B.m. with diffusion σ_C and attracts the amoebas with a *Gaussian-like* potential v. Furthermore, we guarantee in this section the existence and uniqueness of a solution for this system of almost-SDEs.

Chapter 3 Here we explain what it means formally that a macroscopic model is derived by a microscopic one and we construct a valuable candidate for the first one. In order to do so we first calculate the mean-field of the microscopic description $X_t = ((A_t^p)_p, (C_t^m)_m)$ and we consider the set of i.i.d. particles $\overline{X}_t = ((\overline{A}_t^p)_p, (\overline{C}_t^m)_m)$ evolving under its influence (we call them *mean-field particles* or *mesoscopic system*). Then we use the Kolmogorov's forward equation of $\overline{X}_t$ as the candidate macroscopic equation, of which we sign the solution as $a(t,\cdot) = a_t$. Furthermore, this chapter shows that a sufficient condition to obtain that a_t is the effective description of the model is that X_t and $\overline{X}_t$ remain close with a high probability, namely proposition 3.4.1 (same argument as in [11]). In the final part we hint at possible ways to formally derive a macroscopic description also for the chemotaxin particles C^m.

Chapter 4 and 5 These chapters lay the ground for the final proof of the derivation (i.e., the proposition 6.0.1). Chapter 4 defines the Dini derivative $\overline{D}_t^+$ and yields a simple formula for the Dini derivation (with respect to time) of the distance between the microscopic and the mesoscopic description, i.e., for $\overline{D}_t^+ \left|X_t - \overline{X}_t\right|_\infty$. Chapter 5 will prove a law of large numbers for the mean-field particles via a "divide et impera" strategy. Both chapters develop tools which can be applied in more general settings than the one of this microscopic model.

Chapter 6 This section contains the proof of the final result 6.0.1:

Proposition. *Let $X_t^{(N)}$ be the solution to the microscopic model 2.6 and let $\overline{X}_t^{(N)}$ be the solution to the mean-field model 3.18, each with N amoebas. Let $T > 0$ and let $0 < \delta < \xi < \frac{1}{2}$. Then there exists an $\overline{N} > 0$ and an increasing continuous function $f^\uparrow : \mathbb{R}_+ \longrightarrow \mathbb{R}_+$ independent from $\overline{N}, N$ such that for every $N > \overline{N}$ we have that*

$$\mathbb{P}\left(\sup_{0\le t\le T} \left|X_t^{(N)} - \overline{X}_t^{(N)}\right|_\infty \ge N^{-\delta} \right) \le \; f^\uparrow(T)\; N^{-(\xi-\delta)}.$$

From this proposition it follows (by lemma 3.4.1) that 3.19 are the effective equations of the microscopic model considered.

Chapter 7 Here we expand the model treated in chapter 2: per each random time T_m we have L chemotaxin molecules released in radial directions with a constant drift $\frac{v_C}{N}$. Then we derive the effective equations 7.3 for this system, which are essentially the same as for the simple model, but with an addend of the equation scaled by a factor L:

$$\begin{cases} a(0,\cdot) = a_0 \\ \partial_t a = L\lambda\nabla\left[\left(\int_0^t V \star \Phi_{t-\tau}^{\frac{d\sigma_C^2}{2}} \star a(\tau,\cdot)d\tau\right) a\right] + \dfrac{d\sigma_A^2}{2} \triangle a. \end{cases} \tag{1.5}$$

Conclusion In this last chapter we discuss and interpret the effective equations of our microscopic model, and we compare them with the Keller-Segel model. Finally, we illustrate the many directions in which one might develop further the research on this topic.

2. The Basic model

This chapter will define a microscopic model for the Dictyostelium discoideum individuals (and its chemotaxin particles) and prove that it has a solution. A key difference with the real situation will be that the amoeba will release a single cAMP with no initial velocity for each spawning moment. The second important difference is that the release of cAMP will be regulated by a Poisson process, thus ignoring the fact that in reality the amoebas should respond to each trace of cAMP with the release of a chemical cry of more chemotaxin. Recall that the space is throughout the whole master thesis $\mathbb{R}^d$ with $d \in \{2, 3\}$. In general, it is advised to take a look to the section A.1 in the appendix before reading further: it will avoid the reader misunderstandings regarding the usage of specific notation for the particles, their superscripts, the norms, the generic positive constants and increasing positive continuous functions (and other amenities).

2.1. The Heuristics and the Definition

In words we can describe this design in this way:

1. There is a fixed number N of amoebas, each subjected to its own Brownian motion with constant diffusion coefficient σ_A
2. Each amoeba at time $t = 0$ is in the same random position $a_0 \in \mathcal{L}^1 \cup \mathcal{L}^\infty$.
3. Each amoeba A^p is attracted to the cAMP molecules with a potential $v : \mathbb{R}^d \longrightarrow \mathbb{R}$. We require that $-\nabla v =: V$ is Lebesgue-integrable (i.e., $\in \mathcal{L}^1$), essentialy bounded (i.e., $\in \mathcal{L}^\infty$), $\mathcal{C}^\infty$ and globally Lipschitz. One could think of $-v$ as a d-dimensional Gaussian to keep in mind something more concrete.
4. Each amoeba A^p spawns cAMP molecules $C^{p,g}$ (p=parent, g=generation) with a Poisson process M^p of parameter λ independent of p.
5. Each cAMP molecule is born at the position of its parent and is subjected from that time on to its own Brownian motion with constant diffusion coefficient σ_C.
6. All the Brownian motions for each particle are independent from each others.

Let us see all of this in equations. Let $N \in \mathbb{N}$ be the number of amoebas, and fix N independent Poisson processes $(M^p)_{i=1}^N$ of parameter λ. Define for each parent $p \in \{1, \dots, N\}$ and each generation $g \in \mathbb{N}$, the stopping time

$$T^{p,g} := \inf\{t : M_t^p \geq g\} \tag{2.1}$$

i.e., the $\text{Gamma}_{g,\lambda}$ distributed *spawning* or *birth time* of the $C^{p,g}$ cAMP molecule. Consider a countable collection of independent Brownian motions

$$\{B^{A,p}, B^{C,p,g}\}_{1 \leq p \leq N, g \in \mathbb{N}} =: \{B^p, B^{p,g}\}_{p,g} \tag{2.2}$$

i.e., N of them for the amoebas and a countable amount for the cAMP molecules. Now we need an important passage to accurately define the Brownian movement of the cAMP particles. In fact each of these molucules moves only after spawning, and thus such movement must be calibrated for starting after this moment, and not before.

Definition 2.1.1. Let T be a stopping time and B a Brownian motion. Consider the process

$$B_t^{(T)} := B_{T+t} - B_T \tag{2.3}$$

which is a B.m. (see for example [10, Theorem 2.20]). Then the *Quasi Brownian Motion* $Q_t^{B,T}$ is defined as

$$Q_t^{B,T} := \begin{cases} 0 & t < T \\ B_{t-T}^{(T)} = B_t - B_T & t \geq T \end{cases} \tag{2.4}$$

which can informally be written as "$B_{t-T}^{(T)} \mathbb{1}_{t-T\geq 0}$". Then in our case, since $\forall p, g$ a unique B.m. and birth time is identified, we can write

$$Q_t^{p,g} := Q_t^{B^{p,g},T^{p,g}} = "B_{t-T^{p,g}}^{p,g,(T^{p,g})} \mathbb{1}_{t-T^{p,g}\geq 0}" \tag{2.5}$$

which is lighter on notation. With these independent processes we can now finally write down the model as the set of Poisson processes and stochastic differential equations

$$\begin{cases} (M_t^p)_{p=1}^N \text{ i.i.d. and } \sim \text{Poisson}_{\lambda t} & \\ \quad T^{p,g} := \inf\{t : M_t^p \geq g\} & \\ \quad A_t^p = a_0 - \displaystyle\int_0^t \frac{1}{N} \sum_{q=1}^{N} \sum_{g=1}^{M_s^q} V(A_s^p - C_s^{q,g}) ds + \overbrace{\sigma^A B_t^p}^{\int_0^t \sigma_A dB_s^p} & \\ \quad C_t^{p,g} \quad \text{does not exist} & \text{if} \quad t < T^{p,g} \\ \quad C_t^{p,g} = A_{T^{p,g}}^p + \underbrace{\sigma_C Q_t^{p,g}}_{\int_0^t \sigma_C Q_s^{p,g}} & \text{if} \quad t \geq T^{p,g} \end{cases} \tag{2.6}$$

with $V \in \mathcal{L}^1(\mathbb{R}^d) \cap \mathcal{L}^\infty(\mathbb{R}^d) \cap \mathcal{C}^\infty(\mathbb{R}^d)$, globally Lipschitz and $a_0 \in \mathcal{L}^1(\mathbb{R}^d) \cup \mathcal{L}^\infty(\mathbb{R}^d)$.

Observations One can immediately notice that not even the dimension of the system 2.6 is deterministic: the particles $C_t^{p,g}$ are spawning at the stochastic time $T^{p,g}$, increasing at each of these instants the number of particles involved (i.e., the system is piecewise defined in time, according to stochastic times). This, together with the fact that the Poisson processes M^p cannot be represented with a SDE and that the $Q_t^{p,g}$ are not proper Brownian motions, makes this system formally not a system of SDEs, even though the equations involved are indeed differential and stochastic.

2.1.1. Alternative Formulation

We can build a mathematically handier version of the previous model. Since the Poisson processes M^ps determining the spawning times of particles are independent for each amoeba, we have that

$$M_t := \sum_{p=1}^{N} M_t^p \tag{2.7}$$

is Poisson$(Nt\lambda)$-distributed. Therefore the correspondent birth times

$$T_m := \inf\{t : M_t \geq m\} \tag{2.8}$$

are Gamma$(m, N\lambda)$-distributed. Consider these as spawning times and re-indicise the cAMP particles consequently (i.e., C^m is the m-th spawned molecule, born at time T_m).

Then which amoeba A^p is spawning C^m?

Definition 2.1.2. Define the random function $P : \mathbb{N} \longrightarrow \{1, \dots, N\}$ as

$$P(m) := \{p \ : \ A^p \text{ is the parent of } C^m\} \tag{2.9}$$

This function is called *parenting function*, and for each $m \in \mathbb{N}$ the random variable $P(m)$ is uniformly distirbuted on the amoebas $\{1, \dots, N\}$.

Proof.

$$\begin{aligned}
\mathbb{P}(P(m) = p) = \ \mathbb{P}(\exists g \in \mathbb{N}. T_m = T^{p,g}) \ &= \sum_{g=1}^{\infty} \mathbb{P}(T_m = T^{p,g}) \ = \sum_{g=1}^{\infty} \mathbb{P}(M_{T^{p,g}} = m) \\
&= \sum_{g=1}^{\infty} \mathbb{P}(M^p_{T^{p,g}} = g, \sum_{k \neq p} M^k_{T^{p,g}} = m - g) \\
&= \sum_{g=1}^{\infty} \mathbb{P}(g = g, \sum_{k \neq p} M^k_{T^{p,g}} = m - g) \\
&= \sum_{g=1}^{\infty} \int \mathbb{P}\left(\sum_{k \neq p} M^k_{T^{p,g}} = m - g \ \middle| \ T^{p,g} = s \right) d\mathbb{P}\left(T^{p,g} = s \right) \\
&= \sum_{g=1}^{\infty} \int \text{Poisson}_{(N-1)s\lambda}(\{m-g\}) \mathbb{1}_{m-g \geq 0} \underbrace{d\mathbb{P}(T^{p,g} = s)}_{:= \text{Ж}(p,g,s)ds}
\end{aligned}$$

And since $\text{Ж}(p, g, s)ds = d\text{Gamma}_{g,\lambda}(s)$, it is independent from p, we have that $\mathbb{P}(P(m) = p) = \text{K}_m$ and thus uniformly distributed on $\{1, \dots, N\}$. □

Then we can reformulate the model 2.6 as:

$$\begin{cases}
(M^p_t)_{p=1}^N \text{ i.i.d. and } \sim \text{Poisson}_{\lambda t}, \ M_t := \sum_{p=1}^{N} M^p_t & \\
\quad T_m := \inf\{t : M_t \geq m\} & \\
\quad A^p_t = a_0 - \int_0^t \frac{1}{N} \sum_{m=1}^{M_s} V(A^p_s - C^m_s) ds + \overbrace{\sigma^A B^p_t}^{\int_0^t \sigma_A dB^p_s} & \\
C^m_t \quad \text{does not exist} & if \quad t < T^m \\
C^m_t = A^{P(m)}_{T^m} + \underbrace{\sigma_C Q^m_t}_{\int_0^t \sigma_C Q^m_s} & if \quad t \geq T^m
\end{cases} \tag{2.10}$$

This makes the equations simpler and let us arrange the spawning time chronologically, a feature which is important to show that the system has a solution.

2.2. Existence and Uniqueness

The best chance we have to prove existence and uniqueness of solutions for the system 2.6 is to find a way to see it (or parts of it) as proper SDEs and use the theorems of such framework to reach the goal.

2.2.1. The SDEs Framework

An SDE is an expression of the form:

$$\begin{cases} dX_t = b(t, X_t)dt + \sigma(t, X_t)dB_t \\ X_0 \sim \rho_0 \end{cases} \tag{2.11}$$

which is a more conceptual notation to formally mean

$$X_t = \rho_0 + \int_0^t b(t, X_t)\, dt + \int_0^t \sigma(t, X_t)\, dB_t \tag{2.12}$$

where $b \in \mathbb{R}^n$, $\sigma \in \mathbb{R}^{n\times m}$ are given vectorial and matricial functions respectively (the given initial distribution ρ_0 is optional). They are called *drift* and *diffusion* (or *volatility*) respectively. We sign such SDE as a $E(b, \sigma)$.

Definition 2.2.1. A solution for $E(b, \sigma)$ consists of

- a complete filtered space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_t, \mathbb{P})$
- an m-dimensional $(\mathcal{F}_t)_t$-Brownian motion $B = (B^1, \dots, B^m)$
- an $(\mathcal{F}_t)_t$-adapted n-dimensional process X_t with $\mathcal{C}^0$ sample paths

such that they satisfy the equation $E(b, \sigma)$. If furthermore $X_0 = x$ we say that X solves $E_x(b, \sigma)$.

To ensure existence (strong existence, meaning that we can find a solution on each filtered probability space endowed with a Brownian motion) and uniqueness (path uniqueness, meaning that having two solutions X_t, Y_t with the same initial conditions, they are indistinguishable, i.e., $\mathbb{P}(\,\forall t. X_t = Y_t\,) = 1$) of solutions for $E(b, \sigma)$ we need to require conditions on b, σ. There are two main results:

- [10, Thm 8.3]: If the two coefficients are Lipschitz (eventually only locally) in space and continuous in time, then there is existence and uniqueness of solutions for $E(b, \sigma)$:

$$\begin{cases} \forall t.(|b(t,x) - b(t,y)|_\infty < L_b|x-y|_\infty & \text{and } |\sigma(t,x) - \sigma(t,y)|_\infty < L_\sigma|x-y|_\infty) \\ b \in \mathcal{C}^0 & \text{and } \sigma \in \mathcal{C}^0 \end{cases}$$

- [3, Thm 9.4]: If the two coefficients are Lipschitz in space (as before) and have sublinear growth in space, then there is existence and uniqueness of solutions for $E(b, \sigma)$:

$$\begin{cases} \forall t.(|b(t,x) - b(t,y)|_\infty < L_b|x-y|_\infty & \text{and } |\sigma(t,x) - \sigma(t,y)|_\infty < L_\sigma|x-y|_\infty) \\ \forall t.|b(t,x)|_\infty < M_b(1+|x|_\infty) & \text{and } |\sigma(t,x)|_\infty < M_\sigma(1+|x|_\infty) \end{cases}$$

2.2.2. Reducing the Model to the SDE Framework

Heuristic

The system 2.6 is not a proper SDE because of technical oddities:

- An hypothetical vector $X_t := (A_t^1, \dots, A_t^N, C_t^{p,g}$ already spawned at time $t)$ would have a dimension changing (growing, in particular) as the time flows. And it grows up to infinity.

- Not only it would be changing with the time, but it would be a *stochastic* dimension.
- If we were to write the equations for X in a similar way to the one in 2.2.1 we would have a drift term $b = b(t, X_t, (M_t^p)_p)$ depending also on the Poisson-distributed processes $(M_t^p)_p$, making it "too much stochastical" to be treated as in 2.2.1.

All these oddities originate from the fact that our system depends on N sequences $(T^{p,g})_g$ of *random* birth times (or equivalently from the Poisson processes generating them, i.e., $(M_t^p)_t$). We can notice that, while the movement of amoebas A^p and cAMP molecules $C^{p,g}$ depends on them, such sequences do not depend on $A^p, C^{p,g}$. Then heuristically we can imagine to fix these sequences to an arbitrary value $(\tau^{p,g})_{p,g} = (T^{p,g})_{p,g}$ (obtaining an equivalent $(\mu_t^p)_{p,t} = (M_t^p)_{t,p}$), and inquire if this system has a solution for it:

$$\begin{cases} A_t^p = A_0^p + \int_0^t \overbrace{-\frac{1}{N}\sum_{p=1}^{N}\sum_{g=1}^{\mu_s^p} V(A_s^p - C_s^{p,g})\, ds}^{=:b_p(s,X_s,(\mu_s^p)_p)=:b_p(s,X_s)} + \overbrace{\sigma^A}^{=:\sigma_p(s,X_s)} B_t^p & \\ C_t^{p,g} \quad \text{does not exist} & if \quad t < \tau^{p,g} \\ C_t^{p,g} = A_{\tau^{p,g}}^p + \underbrace{\sigma_C}_{=:\sigma_{p,g}(s,X_s)} Q_t^{p,g} & if \quad t \geq \tau^{p,g} \end{cases} \tag{2.13}$$

This solves the problem because $(\tau^{p,g})_{p,g}$ is deterministic. Since the value is arbitrary, if we find a solution $(X, B)_{(\tau^{p,g})_{p,g}}$ for 2.2.2, then we can assemble all these components into a global solution (X, B).

This is formally explained by defining defining a solution for the system 2.6 as a triple $((M^p)_p, X = (A, C), B)$ such that under the conditional probability measure

$$\overline{\mathbb{P}} := \mathbb{P}(\ \cdot \mid\ =\ (T^{p,g})_{p,g}\ ) \quad \mathbb{P}\big(\ \cdot \mid\ (M^p)_{1\leq p\leq N, t\in\mathbb{R}_+}\ \big) \tag{2.14}$$

the couple (X, B) solves 2.6. But we can notice that even in this way we need to operate piecewise in time to avoid a non-constant dimension (dictated by the number of particles, which is growing), i.e., define the solution as one which on each interval where the dimension is constant, solves the system locally.

Local Solutions

Consider the reformulation 2.10 and its birth times, which are chronologically ordered

$$T_0 := 0 < T_1 < T_2 < \ldots < T_m < \ldots \tag{2.15}$$

Define the correspondent intervals

$$I_0 := [0, T_1], I_1 := [T_1, T_2], \ldots, I_m := [T_1, T_{m+1}], \ldots \tag{2.16}$$

On each interval I_m the system can be described as this:

$$\begin{cases} t \in I_m = [T_m, T_{m+1}] & \\ A_t^p = A_{T_m}^p - \int_{T_m}^t \frac{1}{N}\sum_{p=1}^{N}\sum_{g=1}^{M_s^p} V(A_s^p - C_s^{p,g})ds + \sigma^A B_t^p & \\ C_t^{p,g} = C_{T^{p,g}}^{p,g} + \sigma_C Q_t^{p,g} & \text{if} \quad T^{p,g} < T_m \\ C_t^{p,g} = A_{T^{p,g}}^p + \sigma_C Q_t^{p,g} & \text{when} \quad T^{p,g} = T_m \end{cases} \tag{2.17}$$

Remark 2.2.1. We need to observe that only in $I_m \setminus \{T_{m+1}\}$ the number of particles (and thus the dimension) is constant, so why can we claim that the model 2.6 is described efficiently by 2.2.2 efficiently on I_m? Since the $\{T_{m+1}\}$ has Lebesgue-measure 0 the trajectory of the particles already spawned in $I_m \setminus \{T_{m+1}\}$ is not influenced in I_m by C^{m+1}. Then we can ignore its existence to determine the trajectories of the other particles in I_m and have thus a constant dimension on such interval.

Now the only passage that we need in order to describe 2.2.2 as a proper SDE is that of transforming the quasi-Brownian motions $Q^{p,g}$ into proper Brownian motions. This is easily done by shifting the system in time from I_m to $I_m - T_m = [0, T_{m+1} - T_m]$, obtaining:

$$\begin{cases} t \in I_m - T_m = [0, T_{m+1} - T_m] \\ A^p_t = A^p_{T_m} + \int_0^t \overbrace{-\frac{1}{N}\sum_{p=1}^{N}\sum_{g=1}^{M^p_s} V(A^p_s - C^{p,g}_s)}^{b_p}\, ds + \overbrace{\sigma^A}^{\sigma_p} \overbrace{(B^p_{t+T_m} - B^p_{T_m})}^{=:\overline{B}^p_t \text{ (B.m. in } t)} \\ \\ C^{p,g}_t = C^{p,g}_{T^{p,g}} + \sigma_C Q^{p,g}_{T_m+t} = C^{p,g}_{T^{p,g}} + \sigma_C \overbrace{(Q^{p,g}_{t+T_m} - Q^{p,g}_t)}^{\begin{array}{l}=: \overline{B}^{p,g}_t \text{ (B.m. in } t) \\ = (B^{p,g}_{t+T_m} - B^{p,g}_{T_m})\end{array}} \qquad \text{if} \quad T^{p,g} < T_m \\ C^{p,g}_t = A^p_{T^{p,g}} + \sigma_C Q^{p,g}_{T_m+t} = A^p_{T^{p,g}} + \underbrace{\sigma_C}_{\sigma_{p,g}} + \underbrace{B^{p,g}_t}_{=:\overline{B}^{p,g}_t} \qquad \text{when} \quad T^{p,g} = T_m \end{cases} \tag{2.18}$$

Suppose that the initial conditions $A^p_{T_m}, C^{p,g}_{T_m}$ are given and that such equations in 2.2.2 have a solution $({}^mY_t, \overline{B}_t)$ on $I_m - T_m$. Define then on I_m the processes

$${}^mX_t := {}^mY_{t-T_m} \qquad B^\star_t := \overline{B}^\star_{t-T_m}$$

Then it is straightforward to verify that $({}^mX_t, B^\star_t)$ fullfilles the equations on I_m, i.e., 2.2.2, with the correspondent initial conditions.

Global Solutions

Definition 2.2.2. A solution to 2.6 is a triple $((M^p)_{1\le p\le N}, X = (A, C), B)$ such that:

- $(M^p)_{1\le p\le N}$ are Poisson processes of parameter λ and do not depend on (X, B)
- Considering the random measure $\mathbb{P}(\ \cdot \mid\ (T^{p,g})_{p,g}\ )$ and the interval $I_m := [T_m, T_{m+1}]$ we have that the couple $(X = (A, C), B)$ on $I_m \setminus \{T_{m+1}\} = [T_m, T_{m+1})$ takes the form $({}^mX := (A, C^1, \dots, C^m), {}^mB := (B^A, Q^1, \dots, Q^m))$ and such couple $({}^mX, {}^mB)$ solves the local system 2.2.2 on the whole interval I_m.

In order for this to be a valid definition we should specify a global codomain for the processes X, B and we should verify that $\mathbb{P}(\ \cdot \mid\ (T^{p,g})_{p,g}\ )$ is a *regular* random probability measure, so that the SDEs considered remain valid. This is hinted in the appendix at A.3.1.

Remark 2.2.2. 1. In the interval I_m the number of particles is

$$D_t := N + \sum_{p=1}^{N} M^p_t = \overbrace{N}^{\#\text{amoebas}} + \overbrace{M_t}^{\#\text{cAMP}} \tag{2.19}$$

thus the dimension of the system is $D_t \cdot d$, where d is the dimension in which we are considering the particles (2 or 3).

2. Since by definition the particle C^{m+1} spawns on the point $A^p_{T_{m+1}}$ for some p, the solution ${}^{m+1}X$ on the interval I_{m+1} is glued with continuity with the solution ${}^{m}X$ on I_m for all the already existing particles. Furthermore, this continuity property remains true even regarding the spawning position of C^{m+1}, meaning that such particle spawns on $A^{P(m+1)}_{T_{m+1}}$. This can be visualised as the property that at each birth time T_m the path of one of the amoebas bifurcates.

3. Glueing the local solutions we might ask if for each particle its diffusion is *globally* still a B.m. , or a quasi-B.m. (under the measure $\mathbb{P}(\ \ |\ \ (T^{p,g})_{p,g}\ )$). The question depends essentially on this: ¿given two independent B.m. B^1 and B^2 and a stopping time T, is it true that the process

$$B^{T,1,2}_t := B^1_{t\wedge T} + \mathbb{1}_{t\geq T}(B^2_t - B^2_T)$$

is also Brownian motion? It is easy to verify that the answer is affirmative (use e.g., [10, Proposition 2.3] and discard the non-continuous sample-paths), and that thus for each particle we have a global B.m. or quasi-B.m. as a diffusion, i.e., the expressions B^p and $Q^{p,g}$ still make sense globally.

Let us sign the coordinates of space as

$$\begin{aligned} x &:= (x^1, \dots, x^N, x^{1,1}, x^{1,2}, \dots, x^{p,1}, \dots, x^{p,g}, x^{p,g+1}, \dots, x^{N,1}, \dots) \\ &= (a^1, \dots, a^N, c^{1,1}, \dots, c^{p,g}, \dots) \end{aligned}$$

Then we can notice that all the systems considered (the model 2.6, its local version 2.2.2 and its local shifted-in-time version 2.2.2) share the same expression for the functions which would represent the drift and the diffusion in an SDE

$$\begin{cases} b_k(t,x,(M^k_t)_k) := b_k(t,x) := -\dfrac{1}{N}\displaystyle\sum_{p=1}^{N}\sum_{g=1}^{M^p_t} V(x^k - x^{p,g}) & 1 \leq k \leq N \\ \qquad b_{p,g}(t,x) := 0 & \text{if } \exists (p,g) \\ \qquad \sigma_p(t,x) := \sigma_A & 1 \leq p \leq N \\ \qquad \sigma_{p,g}(t,x) := \sigma_C & \text{if } \exists (p,g) \end{cases} \tag{2.20}$$

At first sight it would seem that these expressions would require to be manipulated in order to become the drift and the diffusion of 2.2.2, because of the time translation of T_m. But the time dependence is given only by the random value M^p_s, which is fixed under $\mathbb{P}(\ \cdot\ |\ \ (T^{p,g})_{p,g}\ )$, giving us thus the exact same definition for the drift and diffusion throughout all systems.

Remark 2.2.3. Notice that $b_p(t,x),\ b_{p,g}(t,x) \in \mathbb{R}^d$. To avoid ambiguity between such d-dimensional components of the drift and its 1 dimensional ones, we sign as

$$b_{p;i}(t,x),\ b_{p,g;i}(t,x)$$

the i-th coordinate of the d-dimensional component for the functions $b_p(t,x), b_{p,g}(t,x) \in \mathbb{R}^d$ respectively.

2.2.3. Existence and Uniqueness for the Model

Now that we have reduced our model to a collection of conditional, piecewise-in-time defined SDE, it is enough to find a solution inductively for each I_m under $\mathbb{P}(\ \cdot\ |\ (T^{p,g})_{p,g}\ )$, i.e., a unique solution for 2.2.2 under such random probability measure. Since the case $m = 1$ has essentialy the same expression for the drift and diffusion as the arbitrary case m, we consider only the latter one.

In order to prove local existence and uniqueness we must watch at the drift and diffusion of the system (2.2.2). The diffusion coefficients are constant, and thus respect all the conditions required in 2.2.1. The same is true for the drift of the cAMP. This is not the case for the drift in the case of amoebas though: since the processes M_t^p are càdlàg[1], then also the coordinate functions $b_p(t, x)$ are càdlàg, i.e., not $\mathcal{C}^0$ in time. We need thus to use the result 2.2.1.

Lemma 2.2.1. *The drift function $b(t, x, (M^p)_p)$ defined coordinate-wise in 2.2.2 as*

$$b(t,x) := b(t,x,(M_t^p)_t) := \begin{cases} b_k(t,x) := -\dfrac{1}{N}\displaystyle\sum_{p=1}^{N}\sum_{g=1}^{M_t^p} V(x^k - x^{p,g}) \\ b_{p,g}(t,x) := 0 \end{cases}$$

is such that:

1. *it is Lipschitz continuous in space with fixed time:*

$$\forall t \in \mathbb{R}_+.\ |b(t,x) - b(t,y)|_\infty \leq K_V \frac{M_t}{N}\ |x-y|_\infty \tag{2.21}$$

   where K_V is a constant depending on the Lipschitz constant of V.

2. *There is a constant $K_V > 0$ such that*

$$||b(t,\cdot)||_\infty\ \leq\ K_V\ \frac{M_t}{N}$$

   i.e., it is bounded, hence in particular it grows sublinearly in space.

Remark 2.2.4. Since we work in a finite Banach space all norms are equivalent, thus for these properties we could choose any of them. Still, the Lipschitz constant of b with the infinity norm will play a more relevant role later, so for that instance we decided to specify it.

Proof. Fix t, x, y and let l be the Lipschitz constant of V. Then

[1] From the French expression "*continue à droite, limite à gauche*", meaning "right continuous with left limits".

$$
\begin{aligned}
|b(t,x)-b(t,y)|_\infty &= \max_{k\in\{1,\dots,N\}} |b_k(t,x)-b_k(t,y)|_\infty \\
&= \max_{k\in\{1,\dots,N\}} \left| \frac{1}{N}\sum_{p=1}^{N}\sum_{g=1}^{M_t^p} V(x^k-x^{p,g}) - \frac{1}{N}\sum_{p=1}^{N}\sum_{g=1}^{M_t^p} V(y^k-y^{p,g}) \right|_\infty \\
&\le \frac{1}{N} \max_{k\in\{1,\dots,N\}} \sum_{p=1}^{N}\sum_{g=1}^{M_t^p} \underbrace{\left| V(x^k-x^{p,g}) - V(y^k-y^{p,g}) \right|_\infty}_{\le l\left|(x^k-x^{p,g})-(y^k-y^{p,g})\right|_\infty} \\
&\le \frac{l}{N} \max_{k\in\{1,\dots,N\}} \sum_{p=1}^{N}\sum_{g=1}^{M_t^p} \left(\left|x^k-y^k\right|_\infty + |x^{p,g}-y^{p,g}|_\infty \right) \\
&= \frac{l}{N} \sum_{p=1}^{N}\sum_{g=1}^{M_t^p} \left(\max_{k\in\{1,\dots,N\}} \left|x^k-y^k\right|_\infty + |x^{p,g}-y^{p,g}|_\infty \right) \\
&\le \frac{l}{N} \sum_{p=1}^{N}\sum_{g=1}^{M_t^p} \left(|x-y|_\infty + |x-y|_\infty \right) \\
&= \frac{2l}{N} \sum_{p=1}^{N} M_t^p |x-y|_\infty \;=\; \frac{\mathrm{K}_V}{N} \sum_{p=1}^{N} M_t^p |x-y|_\infty
\end{aligned}
$$

where $\mathrm{K}_V := 2l$. Thus the Lipschitzianity has been proved. The drift b is bounded:

$$
\begin{aligned}
||b(t,x)||_\infty &\le \frac{l}{N} \max_{k\in\{1,\dots,N\}} \sum_{p=1}^{N}\sum_{g=1}^{M_t^p} \underbrace{\left|\left| V(x^k-x^{p,g}) \right|\right|_\infty}_{\le \cdot\, ||V||_\infty} \\
&\le \frac{l\sum_{p=1}^{N} M_t^p ||V||_\infty}{N} = \mathrm{K}_V \; \frac{M_t}{N}
\end{aligned}
$$

□

Therefore the conditions for the pathwise uniqueness and the strong existence of solutions of 2.2.1 are fullfilled, and thus the system 2.2.2 has the properties of pathwise uniqueness and strong existence of solutions, i.e., under $\mathbb{P}(\ \cdot \mid\ (T^{p,g})_{p,g}\ )$ on each interval $I_m = [T_m, T_{m+1}]$ there is a unique solution on each filtered probability space endowed with a B.m. . To be precise, the solution is unique for each initial condition x_m for the local solution ${}^mX$, but since $\forall m \ge 2$ all the initial conditions are determined by the previous local solution ${}^{m+1}X$, then there is a global solution to 2.6 which is unique for each initial condition $(a_1, \dots, a_N)$.

3. Macroscopic and Mesoscopic Descriptions

In this chapter we follow a similar path to the one traced by [11].

3.1. Introduction

In the previous chapter we have defined a microscopic system, i.e., a system where we track the movement of each individual. Consider $X_t^{(N)} = (A_t^{(N)}, C_t^{(N)})$, the solution to 2.6 with N amoebas, where $A_t^{(N)}$ represents the N amoebas and $C_t^{(N)}$ the cAMP molecules. Then such solution can be expressed in terms of its empirical measures:

$$\mu_t^{A,(N)} := \frac{1}{N}\sum_{p=1}^{N} \delta_{A_t^p}$$

$$\mu_t^{C,(N)} := \text{"}\mathbb{1}_{M_t \neq 0}\frac{1}{M_t}\sum_{m=1}^{M_t} \delta_{C_t^m}\text{"}$$

A macroscopic description of this model for the amoebas' part is a density $a(t,\cdot)$ independent form N such that, in some sense

$$\mu_t^{A,(N)} \xrightarrow{N\longrightarrow+\infty} a(t,\cdot)$$

This description is macroscopic, because we loose the information about each individual, but we still describe the movement of the individuals as a whole. In this chapter we craft a candidate for $a(t,\cdot)$ and we explain the condition above and how to prove it in a more handy way. Specifically we build a mesoscopic model, i.e., a model between the macroscopic and the microscopic one, which serves as a bridge. Finally we hint at the idea of finding a similar macroscopic description for the cAMP part, namely a density $c(t,\cdot)$ such that

$$\mu_t^{C,(N)} \xrightarrow{N\longrightarrow+\infty} c(t,\cdot)$$

3.2. Convergence of Empirical Measures

We follow [7, Section 11.4] and [42, Part 1, Ch. 6]. In general, having a sequence of probability measures $(\mu_N)_N$ and a probability measure μ on $\mathbb{R}^d$, the sequence is converging (weakly) to μ iff

$$\forall f \in \mathcal{C}_b(\mathbb{R}^d). \qquad \mu_N(f) := \int f d\mu_N \xrightarrow{N} \mu(f) = \int f d\mu$$

i.e., we see the space of probability measures $\mathcal{P}(\mathbb{R}^d)$ as the dual space of $\mathcal{C}_b(\mathbb{R}^d)$ endowed with the weak-* topology, where $\mathcal{C}_b(\mathbb{R}^d)$ is the space of bounded continuous functions $f : E \longrightarrow \mathbb{R}$ endowed with the norm $||\cdot||_\infty$. This definition cannot be directly applied to our case, because the empirical measures are *random*, i.e., are r.v. mapping into the space of probability measures. For this to be a valid definition we need to have a σ-algebra on $\mathcal{P}(\mathbb{R}^d)$:

Definition 3.2.1. The *weak Borel σ-algebra on $\mathcal{P}(\mathbb{R}^d)$* is the σ algebra generated by the weak-* topology on $\mathcal{P}(\mathbb{R}^d) = (\mathcal{C}_b(\mathbb{R}^d))^\star$, i.e., the coearset σ-algebra such that

$$\forall f \in \mathrm{BL}(\mathbb{R}^d). \qquad \phi_f : \mathcal{P}(\mathbb{R}^d) \longrightarrow \mathbb{R}_+, \qquad \phi_f(\mu) := \int f d\mu \quad \text{is measurable}$$

Then the condition that we actually need to show is the convergence in law of the r.v. $\mu_t^{(A),(N)}$ to the (constant) r.v. a_t:

$$\mu_t^{A,(N)} \xrightarrow{\text{in law in } \mathcal{P}(\mathbb{R}^d)} a_t \qquad \iff \quad \forall F \in \mathcal{C}_b(\mathcal{P}(\mathbb{R}^d)). \quad \mathbb{E}\left[\, F(\mu_t^{A,(N)}) \,\right] \xrightarrow{N\to\infty} \mathbb{E}\left[\, F(a_t) \,\right] \tag{3.1}$$

This condition is cumbersome, and we can do better than it. Thanks to the theorem [37, Proposition 2.2] we can find a less scary condition:

Proposition 3.2.1. *Let $A^{(N)} := (A^1, \dots, A^N)$ be exchangeable[1] r.v. in $\mathbb{R}^{dN}$, let $\Psi^N \in \mathcal{P}(\mathbb{R}^{dN})$ be its law and $\Psi^{N,(k)}$ the k-th marginal, i.e.,*

$$\Psi^{N,(k)}(x) := \int_{\mathbb{R}^{d(N-k)}} \Psi^N dx_{k+1} \dots dx_N$$

Let $\mu^{(A),N}$ its empirical measure. Then for a measure $a \in \mathcal{P}(\mathbb{R}^d)$ are equivalent the following statements:

- *The r.v. $\mu^{(A),N}$ with values in $\mathcal{P}(\mathbb{R}^d)$ converges in law to the constant r.v. $a \in \mathcal{P}(\mathbb{R}^d)$.*
- *For all $k \geq 1$ the marginal $\Psi^{N,(k)}$ converges weakly to $\otimes_{p=1}^N a$ in N.*
- *The 2-marginal $\Psi^{N,(2)}$ converges weakly to $a \otimes a$ in N.*

In this way we must "only" show the weak convergence of the k-marginals $\Psi^{N,(k)}$ to $\otimes_{p=1}^N a$. Then we will need another instrument: a suitable metric for the topology of the weak convergence of measures (that, as we have seen, is the weak-* topology on $\mathcal{C}_b(\mathbb{R}^d)$). If we restrict the space of allowed measures we can choose the *p-Wasserstein distance*, also known as *p-Kantorovich–Rubinstein*. Another popular name is *Earth-mover distance* (the latter one can be generalized to finite measures) because of the heuristic that we will see later. To see its definition we first we need the concept of *coupling* of measures.

Definition 3.2.2. Consider the space of Borel probability measures on $\mathbb{R}^u$

$$\mathcal{BP} := \mathcal{BP}(\mathbb{R}^u) := \{\mu : \mu \text{ is a } \textit{Borel} \text{ prob. meas. on } \mathbb{R}^u\} \tag{3.2}$$

and let $\mu, \nu \in \mathcal{P}$. A *coupling of μ and ν* is a measure $\gamma \in \mathcal{P}(\mathbb{R}^u \times \mathbb{R}^u)$ such that for all Borel-measurable and bounded functions ϕ we have

$$\int_{\mathbb{R}^u \times \mathbb{R}^u} \phi(x)\, d\gamma(x,y) = \int_{\mathbb{R}^u} \phi(x)\, d\mu(x) \quad \text{and} \quad \int_{\mathbb{R}^u \times \mathbb{R}^u} \phi(y)\, d\gamma(x,y) = \int_{\mathbb{R}^u} \phi(y)\, d\nu(x) \tag{3.3}$$

We sign as $\Gamma(\mu,\nu)$ the set of couplings of μ and ν.

Remark 3.2.1. The set $\Gamma(\mu,\nu)$ is never empty because the product measure $\mu \otimes \nu$ is always in it.

[1] Meaning that under each permutation π the $(A^{\pi(1)}, \dots, A^{\pi(N)})$ has the same law as $A^(N) := (A^1, \dots, A^N)$, a condition fullfilled by the system 2.6 for every t.

A way to visualise a coupling is as a transformation of μ into ν. Assume, for the sake of simplicity, that μ, ν are measures on $\mathbb{R}$ and are absolute continuous with respect to the Lebesgue measure Leb. Consider the densities $\frac{d\mu}{d\text{Leb}}(x)$, $\frac{d\nu}{d\text{Leb}}(x)$ and think of their graphs as piles of earth. A plan to move earth in $\mathbb{R}$ can be thought as a function $\overline{\gamma} : \mathbb{R} \times \mathbb{R} \longrightarrow \mathbb{R}$ which with the value $\overline{\gamma}(x, y) =: \overline{\gamma}(x \to y)$ tells how much earth we should move from the point x to the point y. In order for this to be a plan which precisely transforms $\frac{d\mu}{d\text{Leb}}$ into $\frac{d\nu}{d\text{Leb}}$, it must respect two symmetrical conditions:

1. For each $x \in \mathbb{R}$ the total earth (mass of) moved from it by the plan must be exactly equal to the earth present in x at the beginning, namely:

$$\int \overline{\gamma}(x, y)\, dy = \frac{d\mu}{d\text{Leb}}(x) \tag{3.4}$$

2. For each $y \in \mathbb{R}$ the earth (the mass) moved to it must be exactly equal to the earth present in y at the end, namely:

$$\int \overline{\gamma}(x, y)\, dx = \frac{d\nu}{d\text{Leb}}(y) \tag{3.5}$$

This is equivalent to 3.2.2, with $\overline{\gamma}$ playing the role of the density of the coupling γ. With this example in mind we could ask ourselves which is the best among such transformations of μ into ν. To do that we need to associate to each coupling $\gamma \in \Gamma(\mu, nu)$ a value, which indicates how "good" a plan γ might be. If we have a notion of distance *dist* on our space $\mathbb{R}^u$, then a plan can be evaluated with how much net distance we cover to transport all the earth around, i.e., with the expression

$$\int dist(x, y)\, d\gamma(x, y) \tag{3.6}$$

In this way we can say that two measures μ, ν are near to each others if we can find a coupling $\gamma \in \Gamma(\mu, \nu)$ such that its evaluation is very low. With this idea in mind, choosing the distance induced by the normalised 1-norm on $\mathbb{R}^u$ (meaning $\frac{||\cdot||_1}{u}$), one reaches the definition of the 1-Wassertein distance (with respect to the normalised 1-norm).

Definition 3.2.3. Consider the space of Borel probability measures with finite 1-moments on $\mathbb{R}^u$

$$\mathcal{BP}_1 := \mathcal{BP}_1(\mathbb{R}^u) := \{\mu \in \mathcal{BP} : \int ||x||_1 d\mu(x) < \infty\} \subseteq \mathcal{BP} \tag{3.7}$$

Then the 1-Wasserstein distance between two measures $\mu, \nu \in \mathcal{BP}_1$ is defined as

$$W_1(\mu, \nu) := \inf_{\gamma \in \Gamma(\mu,\nu)} \int_{\mathbb{R}^u \times \mathbb{R}^u} \frac{1}{u} \sum_{i=1}^{u} |x_i - y_i|\, d\gamma(x, y) \tag{3.8}$$

To see the proof that W_1 is a distance take a look at the proof after [42, Example 6.3]. The reason why we cover this concept is this result (from [42, Corollary 6.13]):

Theorem 3.2.2. *If (X, d) is a Polish space, and $\overline{d}$ is a bounded distance inducing the same topology on X as d, then the convergence of measures in the 1-Wasserstein sense for $\overline{d}$ is equivalent to the usual weak convergence of probability measures in $\mathcal{BP}_1(\mathbb{R}^u)$.*

Then we have the subsequent statement:

Lemma 3.2.3. *Let $\mu_t^{A,(N)}, a_t, \Psi_t^{(N,k)}, \Psi_t^{(N)}$ be as in 3.2.1 and let $\forall k \geq 1. \Psi_t^{(N,k)}, \Psi_t^{(N)} \in \mathcal{BP}_1$. Then the condition*

$$\sup_{0\leq t\leq T} W_1\big(\Psi_t^{(N,k)}, \bigotimes_{p=1}^{k} a_t\big), \quad \sup_{0\leq t\leq T} W_1\big(\Psi_t^{(N)}, \bigotimes_{p=1}^{N} a_t\big), \xrightarrow{N\to\infty} 0 \tag{3.9}$$

implies that the r.v. $\mu_t^{A,(N)}$ with values in $\mathcal{P}(\mathbb{R}^d)$ converges in law to the constant r.v. $a_t \in \mathcal{P}(\mathbb{R}^d)$.

Remark 3.2.2. Notice that the condition that $\forall k \geq 1. \Psi_t^{(N,k)}, \Psi_t^{(N)} \in \mathcal{BP}_1$ is not particularly strict, it translates to requiring that the amoebas do not have the possibility to have an infinite velocity.

Proof. By 3.2.1 we have that the condition

$$\mu_t^{A,(N)} \xrightarrow{\text{in law in } \mathcal{P}(\mathbb{R}^d)} a_t$$

is equivalent to ask that for all $k \geq 1$, the marginals $\Psi^{N,(k)}$ converge weakly to $\otimes_{p=1}^{N} a_t$ in N. Since for every $k \geq 1$ we have that $\Psi_t^{(N,k)} \in \mathcal{BP}_1$, and since $\mathcal{BP}_1$ is a Polish space, then also $\lim_N \Psi_t^{(N,k)} = \bigotimes_{p=1}^{k} a_t$ is an element of $\mathcal{BP}_1$. Since the 1-Wasserstein distance W_1 is a metric for the weak-convergence of measures, this implies that for every $k \geq 1$ we have that $\Psi_t^{(N,k)}$ converges weakly to $\bigotimes_{p=1}^{k} a_t$. By the proposition 3.2.1, this implies that the r.v. $\mu_t^{A,(N)}$ converges in law to a_t. □

3.3. Effective Equations and Mean-Field Particles

We want to find a good candidate $a_t := a(t, \cdot)$, but we have no analytical description of the solution of 2.6, we have only its evolution in time. Effective equations are description of the evolution of the macroscopic description, and this is what realistically we are searching for, i.e., a partial differential equation

$$\partial_t a_t = (\text{some function})(t, a_t)$$

satisfied by a_t, the macroscopic description of our system. More precisely, we expect this to happen up to the imposition of the same initial conditions, namely when we also impose that $a(0, \cdot) = a_0$, where a_0 is the density of A_0^p imposed in the model 2.6. This might work, but we do not know yet how to find a candidate, and even if we had one, it would still be cumbersome to work with it: since we have no analytical solution for the model 2.6, we cannot work directly with the marginals Ψ_t^N, and we are thus stuck with the empirical measure

$$\mu_t^{A,(N)} := \frac{1}{N} \sum_{p=1}^{N} \delta_{A_t^p}$$

which is a discrete object. On the other side we have a_t, which is a continuous guy. Then we need a bridge from the PDE-world to the SDE-world: the Fokker-Planck equation (refer for example to [30, Theorem 8.1.1]). Given an SDE

$$\overline{A}_t = a_0 + \int_0^t \overline{b}(t, \overline{A}_t)\, dt + \int_0^t \overline{\sigma}(t, X_t)\, dB_t \tag{3.10}$$

if this has a solution $\overline{A}_t$, then its density $a(t, \cdot)$ satifies the PDE

$$\begin{cases} \partial_t a(t,x) = -\sum_{i=1}^{d} \partial_{x_i}[\overline{b}^i(t,x)a(t,x)] + \sum_{i=1}^{d}\sum_{j=1}^{d} \frac{\partial^2}{\partial_{x_i}\partial_{x_j}}[D_{ij}(t,x)a(t,x)] \\ a(0,x) = a_0 \\ D_{ij} := \frac{1}{2}\sum_k \overline{\sigma}^{ik}\overline{\sigma}^{jk} = \frac{1}{2}(\overline{\sigma}\overline{\sigma}^T)_{ij} \end{cases}$$

The viceversa is also true: if we encounter a PDE of the form 3.3 with solution a_t, then we can observe the system 3.10 knowing that its solution has a_t as density. The idea is, in simple words, to exploit this to build somehow a system of i.i.d. N particles

$$\overline{A}_t^{(N)} = (\overline{A}_t^p)_{p=1}^N \tag{3.11}$$

such that each $\overline{A}_t^p$ solves a system of the form 3.10 and which for every t are with a high probability particularly close (in space) to the solutions $A_t^{(N)} = (A_t^p)_{p=1}^N$ of the system 2.6. In this way, if $\overline{A}_t^{(N)} = (\overline{A}_t^p)_{p=1}^N$ are i.i.d. , the marginals of $\overline{A}_t^{(N)}$ are exactly $\bigotimes_{p=1}^N a_t$ and we then just need to show that this closeness in space for every time t implies a closeness of the marginals of the two groups of particles, fullfilling 3.2.3 (all this is precisely stated and proven at 3.4.1). Since this new system $\overline{A}_t^{(N)}$ would be between the microscopic description and the macroscopic one, it is often referred to as *mesoscopic description/model*. More intuitively, if we have such particles $\overline{A}_t^{(N)}$ remaining close enough to the original ones $A_t^{(N)}$, we should expect their respective empirical densities

$$\mu_t^{\overline{A},(N)} = \frac{1}{N}\sum_{p=1}^N \delta_{\overline{A}_t^p}, \qquad \mu_t^{A,(N)} := \frac{1}{N}\sum_{p=1}^N \delta_{A_t^p}$$

to be close, and thus tend to the same limit in N, which we would expect to be the same one given by the Fokker-Planck equation. The same concept will be roughly applied also to the cAMP, meaning that, under the conditional probability measure $\mathbb{P}(\ \cdot\ |\ (T^{p,g})_{p,g}\ )$, we will search also for M_t independent particles $\overline{C}_t^{(N)}$ remaining close to the original $C_t^{(N)}$ (and obtaining then also a PDE for the cAMP part). This not only because it is conceptually more useful to think in two entire systems of both amoebas and cAMP being close to each others, but also because we hope that it is possible to show, in further work, that this implies also the convergence of measures for the cAMP part, i.e., that there is a density c_t easily derivable (with Fokker-Planck and other tools) from $\overline{C}_t^{(N)}$ such that

$$\mu_t^{C,(N)}, \mu_t^{\overline{C},(N)} \xrightarrow{N\longrightarrow+\infty} c(t,\cdot)$$

where $\mu_t^{\overline{C},(N)}$ is the empirical measure of $\overline{C}_t^{(N)}$. Then, practically, we are trying to craft a system of $N + M_t$ particles $\overline{X}_t^{(N)} = (\overline{A}_t^{(N)}, \overline{C}_t^{(N)})$ with these characteristics:

1. The particles $\overline{X}_t^{(N)} = (\overline{A}_t^{(N)}, \overline{C}_t^{(N)})$ should be coupled with their counterparts $X_t^{(N)} = (A_t^{(N)}, C_t^{(N)})$ (i.e., they should have the same B.m. in their stochastic evolution), in order to control more easily the distance between the two. Indeed in such case the distance process $X_t^{(N)} - \overline{X}_t^{(N)}$ is a finite variation process and can be thus more easily treated, depending only on the two drifts, and not on the diffusions.

2. The single particles $\overline{A}_t^1, \dots, \overline{A}_t^N$ must be i.i.d. so that on each one of them can be described by the same SDE and thus we can use Fokker-Planck separately to have N identical densities $a_t^p = a_t$, and have thus that the marginals of this system are

$\bigotimes_{p=1}^{k} a_t$.

3. The distance process $X_t^{(N)} - \overline{X}_t^{(N)}$ should be well-defined, and thus under the conditional probability $\mathbb{P}(\ \cdot \mid\ (T^{p,g})_{p,g}\ )$ the processes $X_t^{(N)}$ and $\overline{X}_t^{(N)}$ must have the same dimension, meaning each having at every time the same number of cAMP molecules spawned.

4. It would be nice that under the conditional probability $\mathbb{P}(\ \cdot \mid\ (T^{p,g})_{p,g}\ )$ the particles $\overline{C}_t^1, \dots, \overline{C}_t^{M_t}$ would be at least independent. It would be desirable to have them even i.i.d. to easily apply Fokker-Planck also in this case, but this is not possible. In fact, if we want them to be coupled with their counterparts $C_t^{p,g}$, we must necessarily have that they move of quasi-B.m. spawning at different random times, which automatically makes them differently distributed. As we will see in the paragraph 3.3.1, we will not be able even to obtain the independence.

Till now we found conditions on the mesoscopic system, but we still have not pointed out at the main way to provide our smart guess: the idea is that the macroscopic behaviour is dictated by the *average* dynamic of the system, more precisely, by the *mean* field felt by each amoeba A_t^p and cAMP molecule $C_t^{p,g}$):

$$\begin{aligned}
\overline{b}_p(t,x) := \overline{b}_p^{(N)}(t,x) &:= \mathbb{E}\left[\ b_p^{(N)}(t, \overline{X}_t^{(N)}) \ \middle|\ \overline{X}_t^{(N),p} = x,\ (T^{p,g})_{p,g}\ \right] \\
\overline{b}_{p,g}(t,x) := \overline{b}_{p,g}^{(N)}(t,x) &:= \mathbb{E}\left[\ b_{p,g}^{(N)}(t, \overline{X}_t^{(N)}) \ \middle|\ \overline{X}_t^{(N),p,g} = x,\ (T^{p,g})_{p,g}\ \right]
\end{aligned}$$

This in fact guarantees that the force felt by a particle p is dependent only on itself, and thus the independence. Besides then the SDE becomes the same for each amoeba, guaranteeing identical distribution for the amoebas' part. Then, imposing the same initial condition as for 2.6, the candidate for our system would be something of the form

$$\begin{cases}
\overline{A}_t^p = a_0 + \displaystyle\int_0^t \overline{b}_p^{(N)}(s, \overline{A}_s^k)ds + \sigma_A B_t^{A,k} \\
\overline{C}_t^{p,g} = \ (\text{some spawning position } P_{T^{p,g}}^{p,g})) + \displaystyle\int_0^t \overline{b}_{p,g}^{(N)}(s, \overline{C}_s^{p,g})ds + \sigma_C Q_t^{p,g} \qquad \text{from time } T^{p,g}
\end{cases} \tag{3.12}$$

We call such particles the *mean-field particles*, for obvious reasons.

3.3.1. Calculate the Mean-Field

First we notice that since the drift of the cAMP molecules is $\equiv 0$, their average drift field is also 0, i.e.,

$$\begin{aligned}
\forall p,g.\quad \overline{b}_{p,g}(t, \overline{X}_t^{p,g}) := \mathbb{E}\left[\ b_{p,g}(t, \overline{X}_t) \mid \overline{X}_t^{p,g}\ \right] &= \mathbb{E}\left[\ \ \mathbb{E}\left[\ b_{p,g}(t, \overline{X}_t) \mid \overline{X}_t^{p,g},\ (T^{p,g})_{p,g}\ \right] \ \middle|\ \overline{X}_t^{p,g}\ \right] \\
&= \mathbb{E}\left[\, \mathbb{E}\left[\ 0 \mid\ (T^{p,g})_{p,g}\ \right]\, \right] = 0
\end{aligned}$$

Then, with respect to the measure $\mathbb{P}(\ \cdot \mid\ (T^{p,g})_{p,g}\ )$, the Fokker-Planck equation (3.3) for such particle will be

$$\begin{cases} c^{T^{p,g}}(0,x) = 0 & \\ \partial_t c^{T^{p,g}}(t,x) = 0 & \text{if } t < T^{p,g} \\ c^{T^{p,g}}(T^{p,g},x) = \text{ some spawning position related to } a(T^{p,g},\cdot) & \\ \partial_t c^{T^{p,g}}(t,x) = D_C \triangle c^{T^{p,g}}(t,x) & \text{if } t \geq T^{p,g} \\ D_C := \frac{1}{2}\sum_{k=1}^{d} \sigma_C^2 = \frac{d\sigma_C^2}{2} & \end{cases} \tag{3.13}$$

We could expect to describe the initial density at the spawning $c^{T^{p,g}}(T^{p,g},\cdot)$ as simply $a(T^{p,g},\cdot)$, since the latter one should represent the spatial distribution of the amoebas. Then taking considering the arbitrary birth time as a parameter $\tau = T^{p,g}$ we can write

$$\begin{cases} c^\tau(t,x) \equiv 0 & \text{if } t < \tau \\ c^\tau(\tau,\cdot) = a(\tau,\cdot) & \\ \partial_t c^\tau = D_C \triangle c^\tau & \text{if } t \geq \tau \\ D_C := \frac{1}{2}\sum_{k=1}^{d} \sigma_C^2 = \frac{d\sigma_C^2}{2} & \end{cases} \tag{3.14}$$

This is simply a heat-equation starting with initial condition $a(\tau)$ at time τ. Therefore, considering the d-dimensional heat-kernel $\Phi_t^{D_C}$ (modified so that it is $\equiv 0$ for negative values of t) we can express it with the analytical solution

$$\begin{cases} c^\tau(\tau,x) = a(\tau) & \\ c^\tau(t,x) = \left(\Phi_{t-\tau}^{D_C} \star a(\tau,\cdot)\right)(x) & \\ \Phi_s^{D_C}(x) := \dfrac{1}{\left(4D_C\pi s\right)^{\frac{d}{2}}} \exp\left(-\dfrac{x\cdot x}{4D_C s}\right) \mathbb{1}_{s\geq 0} & \\ D_C := \frac{1}{2}\sum_{k=1}^{d} \sigma_C^2 = \frac{d\sigma_C^2}{2} & \end{cases} \tag{3.15}$$

where $\star$ is the convolution in space. Let us calculate now the case for the amoebas, using our candidate for the effective equations of the cAMP particles:

$$\begin{aligned} \overline{b}_p^{(N)}(t,x) :&= \mathbb{E}\left[\ b_p^{(N)}(t,\overline{X}_t^{(N)}) \ \middle|\ \overline{X}_t^{(N),p}\ \right](x) \ = \mathbb{E}\left[\ \mathbb{E}\left[\ b_p^{(N)}(t,\overline{X}_t^{(N)}) \ \middle|\ \overline{A}_t^p = x,\ (T^{p,g})_{p,g}\ \right] \ \middle|\ \overline{A}_t^p = x\ \right] \\ &= -\mathbb{E}\left[\ \ \mathbb{E}\left[\ \frac{1}{N}\sum_{p=1}^{N}\sum_{g=1}^{M_t^p} V(\overline{A}_t^p - \overline{C}_t^{p,g}) \ \middle|\ \overline{A}_t^p = x,\ (T^{p,g})_{p,g}\ \right] \ \middle|\ \overline{A}_t^p = x\ \right] \\ &= -\mathbb{E}\left[\ \frac{1}{N}\sum_{p=1}^{N}\sum_{g=1}^{M_t^p} \mathbb{E}\left[\ V(x-\overline{C}_t^{p,g}) \ \middle|\ (T^{p,g})_{p,g}\ \right] \ \right] \\ &= -\frac{1}{N}\sum_{p=1}^{N} \mathbb{E}\left[\ \sum_{g=1}^{M_t^p} \int V(x-y) d\mathbb{P}\left(\ \overline{C}_t^{p,g} = y \ \middle|\ (T^{p,g})_{p,g}\ \right) \right] \\ &= -\frac{1}{N}\sum_{p=1}^{N} \mathbb{E}\left[\ \sum_{g=1}^{M_t^p} \int V(x-y) c_t^{T^{p,g}}(y) dy \ \right] \ = -\frac{1}{N}\sum_{p=1}^{N} \mathbb{E}\left[\ \sum_{g=1}^{M_t^p} \left(V \star c_t^{T^{p,g}}\right)(x) \ \right] = \end{aligned}$$

Now recall that $M^p \sim M^1$ and thus also $T^{p,g} \sim T^{1,g}$, therefore

$$
\begin{aligned}
&= -\frac{1}{N}\sum_{p=1}^{N}\mathbb{E}\left[\ \sum_{g=1}^{M_t^1}\left(V \star c_t^{T^{1,g}}\right)(x)\ \right] \ = \ -\mathbb{E}\left[\ \sum_{g=1}^{M_t^1}\left(V \star c_t^{T^{1,g}}\right)(x)\ \right] \\
&= -\mathbb{E}\left[\ \sum_{g=1}^{\infty}\underbrace{\overbrace{\mathbb{1}_{T^{1,g}\leq t}}\mathbb{1}_{g\leq M_t^1}\left(V \star c_t^{T^{1,g}}\right)(x)}_{(\text{coordinatewise})\geq 0}\ \right]
\end{aligned}
$$

Since the argument in the expectation value is coordinate-wise positive, we can freely extract the infinite sum:

$$
\begin{aligned}
&= -\sum_{g=1}^{\infty}\mathbb{E}\left[\ \mathbb{1}_{T^{1,g}\leq t}\left(V \star c_t^{T^{1,g}}\right)(x)\ \right] = -\sum_{g=1}^{\infty}\int \mathbb{1}_{\tau\leq t}\left(V \star c_t^{\tau}\right)(x) d\underbrace{\mathbb{P}(T^{1,g}=\tau)}_{\text{Gamma}_{\lambda,g}} \\
&= -\sum_{g=1}^{\infty}\int_0^t \left(V \star c_t^{\tau}\right)(x)\frac{\lambda^g}{(g-1)!}x^{g-1}\, e^{-\lambda x}d\tau = -\int_0^t \left(V \star c_t^{\tau}\right)(x)\lambda\underbrace{\sum_{g=1}^{\infty}\frac{(\lambda x)^{g-1}}{(g-1)!}}_{e^{\lambda x}} e^{-\lambda x}d\tau \\
&= -\lambda\int_0^t \left(V \star c_t^{\tau}\right)(x)d\tau = -\left(V \star \int_0^t \lambda c_t^{\tau}d\tau\right)(x) \quad =: \overline{b}_p^{(N)}(t,x) \equiv \overline{b}_p(t,x)
\end{aligned}
$$

With these calculations we have that the mean-field amoebas are the solution of the system

$$
\begin{cases}
\overline{A}_t^0 \sim a_0 \\
\overline{A}_t^p = \overline{A}_t^0 - \displaystyle\int_0^t \left(V \star \int_0^s \lambda c_s^{\tau}d\tau\right)(\overline{A}_s^p)ds + \sigma_A B_t^p
\end{cases}
\tag{3.16}
$$

and a is therefore given by the Fokker-Planck equation (3.3):

$$
\begin{aligned}
\partial_t a(t,x)^1 &= -\sum_{i=1}^{d}\partial_{x_i}\left[\overline{b}_{p;i}(t,x)a(t,x)\right] + \frac{d\sigma_A^2}{2}\triangle a = \sum_{i=1}^{d}\partial_{x_i}\left[(\underbrace{V_i}_{-\partial_{x_i}v} \star \int_0^t \lambda c_t^{\tau}d\tau)a\right] + \frac{d\sigma_A^2}{2}\triangle a \\
&= \sum_{i=1}^{d}\left[\partial_{x_i}\left(-\partial_{x_i}v \star \int_0^t \lambda c_t^{\tau}d\tau\right)a + \left(-\partial_{x_i}v \star \int_0^t \lambda c_t^{\tau}d\tau\right)\partial_{x_i}a\right] + \frac{d\sigma_A^2}{2}\triangle a \\
&= \left(-\triangle v \star \int_0^t \lambda c_t^{\tau}d\tau\right)a + \left(-\nabla v \star \int_0^t \lambda c_t^{\tau}d\tau\right)\nabla a + \frac{d\sigma_A^2}{2}\triangle a \\
&= \nabla\left[\left(V \star \int_0^t \lambda c_t^{\tau}d\tau\right)a\right] + \frac{d\sigma_A^2}{2}\triangle a
\end{aligned}
\tag{3.17}
$$

Expressing such equations with the analytical solution of c (3.15) this becomes

[1]Recall the notation for the 1-dimensional components of the drift and diffusion: A.1.

$$= \nabla \left[\left(V \star \int_0^t \lambda \Phi_{t-\tau}^{D_C} \star a(t,\cdot) d\tau \right) a \right] + \frac{d\sigma_A^2}{2} \triangle a = \lambda \nabla \left[\left(\int_0^t V \star \Phi_{t-\tau}^{D_C} \star a(t,\cdot) d\tau \right) a \right] + \frac{d\sigma_A^2}{2} \triangle a$$

The cAMP Spawning Positions We require that the spawning position $P_{T^{p,g}}^{p,g}$ of $\overline{C}_t^{p,g}$ is $a_{T^{p,g}}$ distributed. Then $P_t^{p,g}$ fullfills the same SDE 3.16 as the mean-field amoebas do. Furthermore we need that the distance process

$$C_t^{p,g} - \overline{C}_t^{p,g} = A_{T^{p,g}}^p + \sigma_C Q_t^{p,g} - (P_{T^{p,g}}^p + \sigma_C Q_t^{p,g}) = A_{T^{p,g}}^p - P_{T^{p,g}}^p$$

is a finite variation process to easily treat it in the section 4.3.1. Then P_t^p must be coupled with A_t^p, and therefore it must necessarily be that $P_t^p = \overline{A}_t^p$, i.e., the cAMP particles fullfill the SDE

$$\overline{C}_t^{p,g} = \overline{A}_{T^{p,g}}^p + \sigma_C Q_t^{p,g} \qquad \text{from time } T^{p,g}$$

In other words we are forced to give up the independence for the particles $(\overline{C}_t^{p,g})_g$ (with a fixed p), but we still have that (for a fixed g) the particles $(\overline{C}_t^{p,g})_p$ are i.i.d. . Then our whole system of mean-field particles is given by

$$\begin{cases} \overline{A}_t^p = a_0 - \displaystyle\int_0^t \left(V \star \int_0^s \lambda c_s^\tau d\tau \right) (\overline{A}_s^p) ds + \sigma_A B_t^p \\ \overline{C}_t^{p,g} = \overline{A}_{T^{p,g}}^p + \sigma_C Q_t^{p,g} \qquad \text{from time } T^{p,g} \end{cases} \tag{3.18}$$

where c_t^τ is defined by the correspondent effective equations:

$$\begin{cases} a(0,\cdot) = a_0 \\ \quad \partial_t a = \nabla \left[\left(V \star \displaystyle\int_0^t \lambda c_t^\tau d\tau \right) a \right] + \frac{d\sigma_A^2}{2} \triangle a \\ c^\tau(t,\cdot) \equiv 0 \qquad \text{if } t < \tau \\ c^\tau(\tau,\cdot) = a(\tau,\cdot) \\ \quad \partial_t c^\tau = \dfrac{d\sigma_C^2}{2} \triangle c^\tau \qquad \text{if } t \geq \tau \end{cases} \tag{3.19}$$

The system 3.19 can be also expressed by substituting the density of cAMP molecules with its analytical expression (3.15) to obtain

$$\begin{cases} a(0,\cdot) = a_0 \\ \quad \partial_t a = \lambda \nabla \left[\left(\displaystyle\int_0^t V \star \Phi_{t-\tau}^{\frac{d\sigma_C^2}{2}} \star a(\tau,\cdot) d\tau \right) a \right] + \dfrac{d\sigma_A^2}{2} \triangle a \end{cases} \tag{3.20}$$

3.3.2. Solution of the Mean-Field Systems

The system 3.20 is a non-linear partial integro-differential equation (non-linear PIDE). In other words, it is a pretty wild beast. The best instrument to prove that it does possess a solution is the Schauder fixed point theorem (see [6, Exercise 6.26], which does not guarantee unicity:

Theorem 3.3.1. *Let E be a Banach space and let $C \subseteq E$ be a non-empty closed convex subset of E. Let $F : C \longrightarrow C$ be a $\mathcal{C}^0$ map such that $F(C) \subseteq K$, where $K \subseteq C$ is a compact subset of C. Then F has a fixed point in K.*

The usual procedure to apply the theorem is to separate the linear part of the differential operator from the non-linear one, to invert the linear part with the correspondent green function, and then to verify that, with K as the space of functions with the characteristics that we desire for our solution, the operator respects the hypothesis of the theorem 3.3.1. This is not proved in this master thesis, but it is assumed that the solution exist. Notice furthermore that the existence of a sufficiently regular solution a_t to the system 3.20 is equivalent to require the existence of solutions a_t and c_t^τ for each $\tau \in \mathbb{R}_+$ to 3.19.

With this view on the problem we make this assumptions regarding the solutions a_t, c_t of the system 3.19:

1. There are solution $a_t, c_t^\tau \in \mathcal{L}^1(\mathbb{R}^d)$ (for every $\tau \geq 0$) for 3.20 and it is valid that $\int a_t dx = 1 = \int c_t^\tau dx$, meaning that $\forall t$ they are probability distributions.

2. The solutions a_t, c_t^τ are not exploding to infinity in finite time, meaning that there is an increasing, positive and continuous function $f^\uparrow(T)$ in T such that

$$\sup_{t \leq T} ||a_t||_\infty, \ \sup_{t,\tau \leq T} ||c_t^\tau||_\infty \leq f^\uparrow(T)$$

   This condition will be necessary to guarantee the existence of solutions for the mean-field particle system 3.18 and to prove the law of large numbers at 5.4.1.

Lemma 3.3.2. *The mean-field drift $\bar{b}(t,x)$ defined in 3.18 coordinate-wise as*

$$\bar{b}_p(t,x) = -\left(V \star \int_0^t \lambda c_t^\tau d\tau\right)(x)$$

is Lipschitz and bounded in space with bound

$$\left|\left|\bar{b}(t,\cdot)\right|\right|_\infty \leq K_{V,\lambda} t$$

for a constant $K_{V,\lambda} > 0$ depending on V, λ.

Remark 3.3.1. This lemma obviously imply that the mean-field drift is sublinearly growing in space with a fixed time, and thus that by the theorem 2.2.1 the existence and uniqueness of solution for the system 3.18 is guaranteed.

Proof.

$$\begin{aligned}
\left|\bar{b}(t,x) - \bar{b}(t,y)\right|_\infty &= \max_{1\leq p\leq N} \max_{1\leq i\leq d} \left\{ \left| \left(V_i \star \int_0^t \lambda c_t^\tau d\tau\right)(x^p) - \left(V_i \star \int_0^t \lambda c_t^\tau d\tau\right)(y^p) \right| \right\} \\
&= \max_{1\leq p\leq N} \max_{1\leq i\leq d} \left\{ \lambda \left| \int V_i(x^p - z) \left(\int_0^t c_t^\tau(z) d\tau\right) dz - \int V_i(y^p - z) \left(\int_0^t c_t^\tau(z) d\tau\right) dz \right| \right\} \\
&= \max_{1\leq p\leq N} \max_{1\leq i\leq d} \left\{ \lambda \left| \int (V_i(x^p - z) - V_i(y^p - z)) \left(\int_0^t c_t^\tau(z) d\tau\right) dz \right| \right\} \\
&\leq \max_{1\leq p\leq N} \max_{1\leq i\leq d} \left\{ \lambda \int |V_i(x^p - z) - V_i(y^p - z)| \left(\int_0^t c_t^\tau(z) d\tau\right) dz \right\} \leq
\end{aligned}$$

Let l be the Lipschitz constant of V. Then we have that

$$\begin{aligned}
&\leq \max_{1\leq p\leq N} \max_{1\leq i\leq d} \left\{ \lambda \int l \ |(x^p - z) - (y^p - z)|_\infty \left(\int_0^t c_t^\tau(z) d\tau\right) dz \right\} \\
&= \ l \max_{1\leq p\leq N} |x^p - y^p|_\infty \int_0^t \overbrace{\int c_t^\tau(z) dz}^{=1} d\tau \ = tl|x-y|_\infty
\end{aligned}$$

Regarding the bound we have that

$$\begin{aligned}\left|\left|\bar{b}(t,x)\right|\right|_\infty &= \lambda\left|\left|V \star \int_0^t c_t^\tau d\tau\right|\right|_{\mathcal{L}^\infty(\mathbb{R}^d)} \leq \lambda||V||_\infty\left|\left|\int_0^t c_t^\tau d\tau\right|\right|_{\mathcal{L}^1(\mathbb{R}^d)} \\ &= \lambda||V||_\infty \int_0^t \int c_t^\tau dx d\tau = \lambda||V||_\infty t\end{aligned}$$

□

Continuity It is important to underline that also the mesoscopic particles $\overline{X}_t^{(N)} = (\overline{A}_t^{(N)}, \overline{C}_t^{(N)})$ have continuous paths in the sense of 2, because the mean-field particles $\overline{C}_t^m$ are spawning on the position of the mean-field amoebas $\overline{A}_t^{P(m)}$. This will be relevant to inquire about the properties of the distance process $\left|X_t - \overline{X}_t\right|_\infty$.

3.4. The Best Condition to Obtain the Convergence

Now that the mean-field particles are well-defined we can finally treat the condition we are gonna use to prove the convergence of the empirical measures of the amoebas of the original system 2.6 towards the distribution a_t.

Lemma 3.4.1. *Let $\overline{A}_t^{(N)} = (\overline{A}_t^p)_{p=1}^N$ be the amoebas part of the solution of 3.18. Suppose that the marginals $\Psi^{(N,k)}$ of $\Psi^{(N)} = \mathcal{L}(A_t^{(N)})$ are elements of $\mathcal{BP}_1$. Suppose that there exists parameters $0 < \delta < \xi < \frac{1}{2}$ and a constant $K > 0$ independent form N such that*

$$\mathbb{P}\left(\max_{1\leq p\leq N}\left|A_t^p - \overline{A}_t^p\right|_\infty > N^{-\delta}\right) \leq K\ N^{-(\xi-\delta)} \tag{3.21}$$

Then we have that

$$\mu_t^{A,(N)} \xrightarrow{\text{in law in } \mathcal{P}(\mathbb{R}^d)} a_t$$

Proof. By the lemma 3.2.3 it is sufficient to show that the condition 3.21 implies

$$\sup_{0\leq t\leq T} W_1\big(\Psi_t^{(N,k)}, \bigotimes_{p=1}^k a_t\big), \quad \sup_{0\leq t\leq T} W_1\big(\Psi_t^{(N)}, \bigotimes_{p=1}^N a_t\big), \xrightarrow{N\to\infty} 0$$

Here the proof will be shown for $\Psi_t^{(N)}$, the case is analogous for $\Psi_t^{(N,k)}$.

$$\begin{aligned}W_1\big(\Psi_t^{(N)}, \bigotimes_{p=1}^N a_t\big) &= \inf_{\gamma\in\Gamma(\Psi_t^{(N)}, \bigotimes_{p=1}^N a_t)} \int_{\mathbb{R}^{dN}\times\mathbb{R}^{dN}} \frac{1}{dN} \sum_{i=1}^d \sum_{p=1}^N \left|x^{p;i} - y^{p;i}\right| d\gamma(x,y) \\ &\leq \inf_{\gamma\in\Gamma(\Psi_t^{(N)}, \bigotimes_{p=1}^N a_t)} \int_{\mathbb{R}^{dN}\times\mathbb{R}^{dN}} |x-y|_\infty\, d\gamma(x,y) \leq\end{aligned}$$

Since $\overline{A}_t^{(N)} = (\overline{A}_t^p)_{p=1}^N$ are i.i.d. each with law a_t, we can take the specific coupling

$$\overline{\gamma} := \Psi_t^{(N)} \otimes (\bigotimes_{p=1}^N a_t) = \mathcal{L}(A_t^{(N)}) \otimes \mathcal{L}(\overline{A}_t^{(N)})$$

and obtain

$$\leq \int_{\mathbb{R}^{dN}\times\mathbb{R}^{dN}} |x-y|_\infty \; d\mathcal{L}(A_t^{(N)})(x) \; d\mathcal{L}(\overline{A}_t^{(N)})(y) = \mathbb{E}\left[\; \left|A_t^{(N)} - \overline{A}_t^{(N)}\right|_\infty \;\right] =$$

Then, if we define the set

$$\Xi_t := \left\{ \left|A_t^{(N)} - \overline{A}_t^{(N)}\right|_\infty \geq N^{-\delta} \right\}$$

by the condition 3.21 we have that $\mathbb{P}\,(\,\Xi_t\,) \leq KN^{-(\xi-\delta)}$ and therefore

$$\begin{aligned}
&= \mathbb{E}\left[\; \left|A_t^{(N)} - \overline{A}_t^{(N)}\right|_\infty \mathbb{1}_{\Xi_t} \;\right] + \mathbb{E}\left[\; \left|A_t^{(N)} - \overline{A}_t^{(N)}\right|_\infty \mathbb{1}_{\Xi_t^c} \;\right] \leq \mathbb{E}\left[\; \left|A_t^{(N)} - \overline{A}_t^{(N)}\right|_\infty \mathbb{1}_{\Xi_t} \;\right] + N^{-\delta} \\
&= \mathbb{E}\left[\; \left| a_0 + \int_0^t b(s, \overline{A}_s^p)ds + \sigma_A B_t^p - \left(a_0 + \int_0^t \overline{b}(s, \overline{A}_s^p)ds + \sigma_A B_t^p \right) \right|_\infty \mathbb{1}_{\Xi_t} \;\right] + N^{-\delta} \\
&= \mathbb{E}\left[\; \left| \int_0^t b(s, \overline{A}_s^p) - \overline{b}(s, \overline{A}_s^p)ds \right|_\infty \mathbb{1}_{\Xi_t} \;\right] + N^{-\delta} \;\leq\; \mathbb{E}\left[\; \int_0^t ||b(s,\cdot)||_\infty + \left|\left|\overline{b}(s,\cdot)\right|\right|_\infty ds \; \mathbb{1}_{\Xi_t} \;\right] + N^{-\delta}
\end{aligned}$$

Recalling now the bound for the drift (2.2.1) and the mean-field drift (3.3.2) we can proceed as following

$$\begin{aligned}
&\leq \mathbb{E}\left[\; \int_0^t \mathrm{K}_V \; \frac{M_s}{N} + \mathrm{K}_{V,\lambda} s \; ds \; \mathbb{1}_{\Xi_t} \;\right] + N^{-\delta} \\
&\leq\; \mathbb{E}\left[\; t\mathrm{K}_V \; \frac{M_t}{N} \; \mathbb{1}_{\Xi_t} \;\right] + \mathbb{E}\left[\; \mathrm{K}_{V,\lambda} \frac{t^2}{2} \; \mathbb{1}_{\Xi_t} \;\right] + N^{-\delta} \\
&= \frac{t\mathrm{K}_V}{N} \; \mathbb{E}\,[\; M_t \mathbb{1}_{\Xi_t} \;] + \mathrm{K}_{V,\lambda} \frac{t^2}{2} \; \mathbb{E}\,[\; \mathbb{1}_{\Xi_t} \;] + N^{-\delta} \leq
\end{aligned}$$

Now we can use Hölder on $\mathbb{E}\,[\; M_t \mathbb{1}_{\Xi_t} \;]$ for some $p, q \in (1, +\infty)$ such that $\frac{1}{p} + \frac{1}{q} = 1$ together with the bound on the measure of Ξ_t to obtain that

$$\begin{aligned}
&\leq\; \frac{t\mathrm{K}_V}{N} \; (\mathbb{E}\,[\; (M_t)^q \;])^{\frac{1}{q}} \mathbb{E}\left[\; \mathbb{1}_{\Xi_t}^p \;\right]^{\frac{1}{p}} + \mathrm{K}_{V,\lambda} t^2 \; \mathbb{E}\,[\; \mathbb{1}_{\Xi_t} \;] + N^{-\delta} \\
&\leq\; \frac{t\mathrm{K}_V}{N} \; (\mathbb{E}\,[\; (M_t)^q \;])^{\frac{1}{q}} \; K^{\frac{1}{p}} N^{\frac{-(\xi-\delta)}{p}} \; + \mathrm{K}_{V,\lambda} t^2 \; K \; N^{-(\xi-\delta)} + N^{-\delta} \\
&=\; \frac{t \; \mathrm{K}_{V,p}}{N} \; (\mathbb{E}\,[\; (M_t)^q \;])^{\frac{1}{q}} \; N^{\frac{-(\xi-\delta)}{p}} \; + \mathrm{K}_{V,\lambda} t^2 \; N^{-(\xi-\delta)} + N^{-\delta}
\end{aligned}$$

Then, applying the estimate of Poisson processes moments from A.2.1 we obtain that

$$\leq \frac{t \; \mathrm{K}_{V,p}}{N} \; (f^{\uparrow}(q)^{\frac{1}{q}} \; (\lambda N t)^{q\frac{1}{q}} \vee 1) \; N^{\frac{-(\xi-\delta)}{p}} \; + \mathrm{K}_{V,\lambda} t^2 \; N^{-(\xi-\delta)} + N^{-\delta}$$

Supposing that $\lambda N t \geq 1$ (which is true for N sufficiently big if $t > 0$) we obtain that

$$\begin{aligned}
&= \frac{t \; \mathrm{K}_{V,p,q,\lambda}}{N} \; (Nt) N^{\frac{-(\xi-\delta)}{p}} \; + \mathrm{K}_{V,\lambda} t^2 \; N^{-(\xi-\delta)} + N^{-\delta} \\
&\leq \mathrm{K}_{V,p,q,\lambda} \; t^2 \; N^{\frac{-(\xi-\delta)}{p}} \; + \mathrm{K}_{V,\lambda} t^2 \; N^{-(\xi-\delta)} + N^{-\delta}
\end{aligned}$$

Resuming, then, we have that

$$W_1\big(\Psi_t^{(N)}, \bigotimes_{p=1}^{N} a_t\big) \leq \mathrm{K}_{V,\lambda} t^2 N^{-(\xi-\delta)} + t^2\ \mathrm{K}_{V,\lambda,q,p}\ N^{\frac{-(\xi-\delta)}{p}} + N^{-\delta}$$

and thus

$$\sup_{t\in[0,T]} W_1\big(\Psi_t^{(N)}, \bigotimes_{p=1}^{N} a_t\big) \leq \mathrm{K}_{V,\lambda} T^2 N^{-(\xi-\delta)} + T^2\ \mathrm{K}_{V,\lambda,q,p}\ N^{\frac{-(\xi-\delta)}{p}} + N^{-\delta} \xrightarrow{N\to+\infty} 0$$

where the limit goes to 0 because $0 < \delta < \xi < \frac{1}{2}$. □

3.5. A Macroscopic Description for the cAMP Molecules?

The cAMP molecules are created at a rate λ, and each molecule created at a random time τ has a density fullfilling

$$\begin{cases} c^\tau(t,\cdot) \equiv 0 & \text{if } t < \tau \\ c^\tau(\tau,\cdot) = a(\tau,\cdot) & \\ \quad \partial_t c^\tau = \dfrac{d\sigma_C^2}{2} \triangle c^\tau & \text{if } t \geq \tau \end{cases}$$

This, together with the form of the effective equations 3.19, where the a_t should follow the cAMP by the convolution, seems to suggest that

$$\text{"}\mathbb{1}_{M_t\neq 0}\frac{1}{M_t}\sum_{m=1}^{M_t}\delta_{C_t^m}\text{"} =: \mu_t^{C,(N)} \xrightarrow{N\longrightarrow+\infty} c(t,\cdot) := \frac{\int_0^t \lambda c_t^\tau d\tau}{\int\int_0^t \lambda c_t^\tau d\tau dx} = \frac{1}{t}\int_0^t c_t^\tau d\tau$$

The problem is that the cAMP molecules, both in the microscopic and in the mesoscopic model, are *not* exchangeable, and thus the proposition of Sznitman 3.2.1 cannot be used, and one might have to manually use the condition 3.1. There are some properties of the cAMP particles which show somehow symmetry, namely, under the conditional probability $\mathbb{P}(\ \cdot \mid\ (T^{p,g})_{p,g}\ )$:

- The spawning position of each $C_t^{p,g}$ is given by A_t^p, and the amoebas are exchangeable. Even better, $\overline{C}_t^{p,g}$ are spawning on $\overline{A}_t^p$, and the latter ones are i.i.d. .
- Each $C_t^{p,g}$ moves of a quasi-B.m. starting at $T^{p,g}$, and these birth times are i.i.d. with respect to the parameter p. Since the mesoscopic molecules $\overline{C}_t^{p,g}$ are coupled with the microscopic ones, the same happens for them

Because of these facts, for each generation g, we have that $(C^{p,g})_p$ are exchangeable, which might come in handy. Some nuances come from the fact that the the number of molecules spawned is stochastically distributed with the r.v. $(M_t^p)_p$, and thus also for the W_1-convergence a bit work would be necessary (maybe even approximating first with the conditional probability $\mathbb{P}(\ \cdot \mid\ (T^{p,g})_{p,g}\ )$ and then with $\mathbb{P}$). A possible strategy could be to follow a path analogous to the one Sznitman followed in the proof in 3.2.1, namely to prove that the sequence $(\mu_t^{C,(N)})_N$ is tight and that converges finitely to c, which exploits [4, Theorem 5.1, Corollary] to obtain the convergence. Since the key question of this work is the macroscopic description for the amoebas, not a lot of effort has been further spent on this topic, leaving it for eventual future works.

4. Preliminary: the Dini Derivation

In this chapter the objective is to find a good way to handle the Dini-derivative of the distance process

$$\sup_{0\le t\le T}\left|X_t-\overline{X}_t\right|_\infty=\sup_{0\le t\le T}\left|X_t^{(N)}-\overline{X}_t^{(N)}\right|_\infty$$

Such tool will be fundamental for the proof of 6.0.1.

4.1. Dini Derivatives

We follow [12] and [11].

Definition 4.1.1 (Dini derivatives)**.** For any function $f:\mathbb{R}\longrightarrow\mathbb{R}$ we define the *right upper, right lower, left upper* and *left lower Dini derivatives* in t as:

$$\overline{D}^+f(t)=\limsup_{h\to0^+}\frac{f(t+h)-f(t)}{h}\qquad \underline{D}^+f(t)=\liminf_{h\to0^+}\frac{f(t+h)-f(t)}{h}$$
$$\overline{D}^-f(t)=\limsup_{h\to0^-}\frac{f(t+h)-f(t)}{h}\qquad \underline{D}^-f(t)=\liminf_{h\to0^-}\frac{f(t+h)-f(t)}{h}$$

For any function $g:\mathbb{R}^{\mathrm{Ч}}\longrightarrow\mathbb{R}$ and any direction (i.e., unit vector) $v\in\mathbb{R}^{\mathrm{Ч}}$ we define the *upper* and *lower dini derivative in the direction* v as:

$$\overline{D}_vg(x)=\limsup_{h\to0^+}\frac{g(x+hv)-g(x)}{h}\qquad \underline{D}_vg(x)=\liminf_{h\to0^+}\frac{g(x+hv)-g(x)}{h}$$

Remark 4.1.1. The right upper Dini derivative $\overline{D}^+f(t)$ might be signed also as $\overline{D}^+_tf(t)$, and analogous for the others. The Dini derivative of the multivariate case has been added for the sake of completeness, but it will never be used in this work.

Remark 4.1.2. Notice that the Dini derivatives are always defined if we allow them to take also values in $[-\infty,\infty]$. Furthermore it is easy to see that

- If f is locally Lipschitz, then its dini derivatives are all finite.
- If f is right-differentiable $\partial^+f(t)=\overline{D}^+f(t)=\underline{D}^+f(t)$. Analogously happens if f is left-differentiable.
- If f is differentiable then $\partial f(t)=\partial^+f(t)=\partial^-f(t)$ and all the dini derivatives coincide.

The proof of the first two points of the next lemma is easily achieved exploiting the properties of lim sup, while the third one is found in the quoted article.

Lemma 4.1.1 (Properties of Dini derivatives)**.** *Let $f,h:\mathbb{R}\longrightarrow\mathbb{R}$ be $\mathcal{C}^0$. Then*

- $\overline{D}^+(f+h)\le\overline{D}^+f+\overline{D}^+h$
- *If $f>0$. then* $\overline{D}^+(fh)\le(\overline{D}^+f)h+f(\overline{D}^+h)$

- *([12, Theorem 11]) If f has* finite *Dini derivative $\overline{D}^+$ at every point in $t \in \mathbb{R}$, and if $\overline{D}^+ f$ is Lebesgue integrable on $[a, b]$, then*

$$f(b) - f(a) = \int_a^b \overline{D}^+ f(t) dt. \tag{4.1}$$

With this background in mind, we state the next result, of which the proof will be achieved by the end of the section (4.4).

Lemma 4.1.2. *Let $X^{(N)} = X$ be the solution to the model 2.6 and $\overline{X}^{(N)} = \overline{X}$ its coupled mean-field particles. Let $b^{(N)} = b$ and $\overline{b}^{(N)} = \overline{b}$ be the correspondent drifts for the two. Then we have that with probability 1, for all $T > 0$*

$$\overline{D}_T^+ \sup_{0 \le t \le T} \left| X_t - \overline{X}_t \right|_\infty \le d \cdot \left| b(T, X_T) - \overline{b}(T, \overline{X}_T) \right|_\infty \tag{4.2}$$

4.2. Proceed Beyond the sup

In this part we want to show this lemma.

Lemma 4.2.1. *For all $T > 0$*

$$\overline{D}_T^+ \sup_{0 \le t \le T} \left| X_t - \overline{X}_t \right|_\infty \le \max_{s \in \{0,T\}} \left\{ \overline{D}_s^+ \left| X_s - \overline{X}_s \right|_\infty \right\} \tag{4.3}$$

Even though the next lemma (4.2.2) show that the distance function $\left| X_t - \overline{X}_t \right|_\infty$ has continuous paths, we will show that the lemma 4.2.1 is valid also under weaker conditions. In fact, defining the random function

$$Ч(t) := \left| X_t - \overline{X}_t \right|_\infty \tag{4.4}$$

and considering its paths $Ч(t)$, we have that these are continuous in between the birth times $(T_m)_m$, but the behaviour at such instants depends on the spawning position of the cAMP. For a general mesoscopic model $\overline{X}_t$ which spawn new molecules C_t^m in random position, and not necessarily along the path of other already existing particles A_t^p, we would have a this more general class of paths:

Definition 4.2.1. Let $\mathbb{D}_F$ be the set of functions $g : \mathbb{R}_+ \longrightarrow \mathbb{R}_+$ such that:

- $g(0) = 0$
- g is càdlàg
- For each $T > 0$, g has finite amount of discontinuity in $[0, T]$

Furthermore, considering $g(t-) := \lim_{s \to t^-} g(s)$ and $g(t+) := \lim_{s \to t^+} g(s)$, we define the set $\mathbb{D}_F^{\hat{\wedge}}$ as the subset of elements $g \in \mathbb{D}_F$ such that if g is discontinuous in t_i then $g(t_i-) < g(t_i)$.

Remark 4.2.1. Let g be a càdlàgfunction. Then the condition

$$\text{if } g \text{ is discontinuous in } t_i \text{ then } g(t_i-) < g(t_i)$$

is equivalent to

$$\forall t. g(t-) \le g(t+) = g(t)$$

Notice that such property does not necessarily imply that g is increasing, it only means that if it has a jump, it's necessarily a jump up.

Lemma 4.2.2. *The paths of* $\Psi(t) = \left|X_t - \overline{X}_t\right|_\infty$ *are continuous. In particular are elements of* $\mathbb{D}_F^{\bar{\wedge}}$.

Proof. Since both $X_t = X_t^{(N)}$ and $\overline{X}_t = \overline{X}_t^{(N)}$ are solutions to SDEs on $I_m \setminus \{T_{m+1}\} = [T_m, T_{m+1})$, they are continuous on such interval. This, together with the fact that $|\cdot|_\infty$ is $\mathcal{C}^0$ and that $\forall\omega$ the number of birth times T_m in $[0,T]$ is finite, means that the discontinuouties in $[0,T]$ is finite. Furthermore we have that $[T_m, T_{m+1})$ is closed on the left, thus the paths are right-continuous. By definition we have also that

$$\lim_{t\to T_{m+1}} X_t^{(N)} = (A^1_{T_{m+1}}, \dots, A^N_{T_{m+1}}, C^1_{T_{m+1}}, \dots, C^m_{T_{m+1}}) \tag{4.5}$$

and analogously for $\overline{X}_t^{(N)}$. Therefore the paths of $\left|X_t - \overline{X}_t\right|_\infty$ are càdlàg.

Let us now prove that $\Psi(T_m-) \le \Psi(T_m)$ for every discontinuouty T_m in $[0,T]$. We have that

$$\begin{aligned}
\Psi(T_m-) &= \lim_{t\to T_{m+1}^-} \Psi(t) = \lim_{t\to T_{m+1}^-} \left|X_t^{(N)} - \overline{X}_t^{(N)}\right|_\infty \\
&= \lim_{t\to T_{m+1}^-} \max\left\{ \max_{p\in\{1,\dots,N\}} \left|A_t^p - \overline{A}_t^p\right|_\infty, \max_{1\le i\le m} \left|C_t^i - \overline{C}_t^i\right|_\infty \right\} \\
&= \max\left\{ \max_{p\in\{1,\dots,N\}} \left|A^p_{T_{m+1}} - \overline{A}^p_{T_{m+1}}\right|_\infty, \max_{1\le i\le m} \left|C^i_{T_{m+1}} - \overline{C}^i_{T_{m+1}}\right|_\infty \right\} \\
&\le \max\left\{ \max_{p\in\{1,\dots,N\}} \left|A^p_{T_{m+1}} - \overline{A}^p_{T_{m+1}}\right|_\infty, \max_{1\le i\le m} \left|C^i_{T_{m+1}} - \overline{C}^i_{T_{m+1}}\right|_\infty, \left|C^{m+1}_{T_{m+1}} - \overline{C}^{m+1}_{T_{m+1}}\right|_\infty \right\} \\
&= \left|X^{(N)}_{T_{m+1}} - \overline{X}^{(N)}_{T_{m+1}}\right|_\infty = \Psi(T_{m+1})
\end{aligned} \tag{4.6}$$

In particular, since the cAMP molecules are spawning on the positions of some amooebas, i.e.,

$$C^{m+1}_{T_{m+1}} - \overline{C}^{m+1}_{T_{m+1}} = A^{P(m+1)}_{T_{m+1}} - \overline{A}^{P(m+1)}_{T_{m+1}}$$

then the inequality above becomes an equality, and thus the paths are continuous. □

Remark 4.2.2. Notice that if the mean-field cAMP would not be born on some mean-field amoeba, i.e., there is no bifurcation of paths at birth times, at each birth time T_m it would be possible to have that the equation 4.6 becomes a strict inequality.

Now we need to work on how to Dini-derive the sup operator of functions in $\mathbb{D}_F^{\bar{\wedge}}$.

Lemma 4.2.3. *Let* $g \in \mathbb{D}_F^{\bar{\wedge}}$ *and consider* $g_\star(t) := \sup_{0\le s\le t} g(s)$. *Then*

$$g_\star(t) = \max_{0\le s\le t} g(s) \tag{4.7}$$

and $g_\star \in \mathbb{D}_F^{\bar{\wedge}}$.

Proof. Let $\{\tau_1, \dots, \tau_m\}$ be the set of discontinuities of g in $[0,t]$, and define $\tau_0 := 0, \tau_{m+1} := t$. Define then, for each $i \in \{1,\dots,m\}$, the function

$$g_i(s) := \mathbb{1}_{[\tau_{i-1},\tau_i)}(s) g(s) + \mathbb{1}_{\{\tau_i\}}(s) g(\tau_i-) = \begin{cases} g(s) & s \in [\tau_{i-1}, \tau_i) \\ g(\tau_i-) & s = \tau_i \end{cases}$$

The functions g_i are obviously continuous, and since $[\tau_{i-1}, \tau_i]$ is a closed interval, we have that $\sup_s g_i(s) = \max_s g_i(s)$. We have two possible cases: either τ_i is a maximal point for g_i, or it is not. If τ_i is a maximal point for it we have that

$$g_i(s) \leq g_i(\tau_i) = g(\tau_i-) \leq g(\tau_i) \quad \Rightarrow \max g_i \leq g(\tau_i) \tag{4.8}$$

Otherwise if τ_i is not a maximal point for g_i, we have that $\exists s_i \in [\tau_{i-1}, \tau_i)$ such that $\forall s.\ g_i(s) \leq g(s_i) = g_i(s_i)$. Since $g_i = g$ on all points but on τ_i, and since $(g_i)_i$ is a sequence with a finite number of elements, we can affirm that in all cases

$$\sup_{s\in[\tau_{i-1},\tau_i]} g(s) = \max\left\{\max_{s\in[\tau_{i-1},\tau_i]} g_i(s), g(\tau_i)\right\} = \max_{s\in[\tau_{i-1},\tau_i]} g(s) \tag{4.9}$$

Hence it is obvious that

$$\sup_{s\in[0,t]} g(s) = \max_i \left\{\sup_{s\in[\tau_{i-1},\tau_i]} g(s)\right\} = \max_i \left\{\max_{s\in[\tau_{i-1},\tau_i]} g(s)\right\} \tag{4.10}$$

$$= \max_{s\in[0,t]} g(s) \tag{4.11}$$

Therefore we have that $g_\star(t) = \sup_{[0,t]} g = \max_{[0,t]} g$. Let us now prove that $g_\star$ is an element of $\mathbb{D}^{\overline{\wedge}}_F$.

Claim 4.2.1. *The set of discontinuities of $g_\star$ is contained in $\{\tau_1, \ldots, \tau_m\}$, the set of discontinuities of g.*

Proof. Let t be a point where g is continuous. We show that also $g_\star$ is continuous in t.

We have two possibilities: either $g(t) = g_\star(t)$, or $g(t) < g_\star(t)$.

- If $g(t) < g_\star(t)$, since g is $\mathcal{C}^0$ in t, there is a neighbourhood U_t of t such that $g|_{U_t} < g_\star(t)$, and since $g_\star(t) = \max_{[0,t]} g$, we have that $(g_\star)|_{U_t} \equiv g_\star(t)$, and thus that $g_\star$ is continuous in t.

- If $g(t) = g_\star(t)$, consider an $\epsilon > 0$ small enough so that g is monotonous on the left and right intervals $U^-_\epsilon := (t-\epsilon, t)$ and $U^+_\epsilon := (t, t+\epsilon)$. This can be done otherwise the function g would oscillate infinite times in a neighbourhood, and we could construct a sequence $(t_n)_n$ such that $(g(t_n))_n$ does not converge to $g(t)$, contraddicting the continuouity in t. Let us show that $g_\star$ is continuous on both left and right neighbourhoods, being thus continuous.

  $\left(U^-_\epsilon\right)$ g can be either constant or strictly increasing on U^-_ϵ (otherwise if g would be strictly decreasing, we would have a point $s < t$ such that $g(s) > g(t) = g_\star(t) = \max_{[0,t]} g$, contraddiction). If g is constant on this left neighbourhood, then so is $g_\star$ and thus it is left continuous. If g is stricly increasing on U^-_ϵ we have either that the two functions coincide on such domain, or that $g_\star$ is constant on it, and in both case we have that $g_\star$ is left continuous.

  $\left(U^+_\epsilon\right)$ Consider now the case on $U^+_\epsilon = (t, t+\epsilon)$: if g is non-increasing on it, then $(g_\star)|_{U^+_\epsilon} \equiv g_\star(t)$. If instead g is increasing on it, the two functions coincide on it. In either case, $g_\star$ is right-$\mathcal{C}^0$.

Therefore $g_\star$ is continuous on t. □

Claim 4.2.2. *$g_\star$ is right-$\mathcal{C}^0$, i.e., $g_\star(t) = g_\star(t+)$.*

Proof of claim.

$$
\begin{aligned}
g_\star(t+) &= \lim_{h\to 0^+} g_\star(t+h) = \lim_{h\to 0^+} \sup_{s\le t+h} g_\star(s) \\
&= \lim_{h\to 0^+} \max\Big\{ \underbrace{\sup_{s\in[0,t)} g(s)}_{\le \sup_{s\in[0,t]} g(s) = g_\star(t)} \quad , \quad \underbrace{\sup_{s\in[t,t+h]} g(s)}_{\sup_{\delta\in[0,h]} g(t+\delta)} \Big\} \\
&\le \max\{g_\star(t), \lim_{h\to 0^+} \sup_{\delta\in[0,h]} g(t+\delta)\} \\
&= \max\{g_\star(t), \limsup_{s\to t^+} g(s)\} \qquad\qquad = \max\{g_\star(t), g(t)\} = g_\star(t)
\end{aligned}
$$

Thus $g_\star(t+) \le g_\star(t)$, but since $g_\star$ is increasing by definition we also have that $g_\star(t+) \ge g_\star(t)$, therefore $g_\star(t+) = g_\star(t)$. □

Claim 4.2.3. *For each t we have that $\exists g_\star(t-)$ and $g_\star(t-) \le g_\star(t)$.*

Proof of claim. If $g_\star$ is $\mathcal{C}^0$ in t the claim is obvious, let us thus assume that $g_\star$ is discontinuous in t. Notice that since $g_\star$ is right-continuous and increasing, if the left limit $g_\star(t-)$ exists, then it must necessarily be that $g_\star(t-) < g_\star(t)$. Therefore we need only to show that such left limit exists.

Let $(t_n)_n$ be a sequence of points such that $\forall n.\, t_n \le t_{n+1} < t$ and $t_n \longrightarrow t$. Since the set of discontinuities of g on $[0,T]$ is finite, then it is discrete, thus there is an $\epsilon > 0$ such that g (and hence $g_\star$) are $\mathcal{C}^0$ on $U_\epsilon^- := (t-\epsilon, t)$ and on $U_\epsilon^+ := (t, t+\epsilon)$. We then observe, that since (t_n) is a non-decreasing sequence and $g_\star$ is a non-decreasing function, we have that

$$g_\star(t_n) \le g_\star(t_{n+1}) \tag{4.12}$$

i.e., that the sequence $(g_\star(t_n))_n$ is non-decreasing. Furthermore we have that $(g_\star(t_n))_n$ is bounded by $g_\star(t)$ (because of the definition of $g_\star$). Then, being the sequence $(g_\star(t_n))_n$ bounded and non-decreasing, it has a limit l. But being such sequence arbitrary, we have that

$$l = \lim_n g_\star(t_n) = \lim_{s\to t^-} g(s) = g(t-) \tag{4.13}$$

□

And thus $g_\star \in \bar{\mathbb{D}}_F^\uparrow$ and the proof is concluded. □

Lemma 4.2.4. *Let g be a function in $\bar{\mathbb{D}}_F^\uparrow$. Then*

$$\overline{D}_t^+ g_\star(t) = \overline{D}_t^+ \sup_{0\le s\le t} g(s) \le \left(\overline{D}_0^+ g(0)\right) \vee \left(\overline{D}_t^+ g(t)\right) = \max_{s\in\{0,T\}} \overline{D}_s^+ g(s) \tag{4.14}$$

with $s_t \in \{0, t\}$

Proof. Observe that since $g \ge 0$ and since $g_\star$ is non-decreasing, we have that

$$\text{L.H.S. (4.14)} = \lim_{h\to 0^+} \sup_{\delta\in(0,h]} \frac{g_\star(t+\delta) - g_\star(t)}{\delta} \ge 0 \tag{4.15}$$

Thus we have only two possibilities:

$$\text{L.H.S. (4.14)} = 0 \qquad \text{or} \qquad \text{L.H.S. (4.14)} > 0 \tag{4.16}$$

Consider the case L.H.S. (4.14) $= 0$. Then it is sufficient to show that R.H.S. (4.14) ≥ 0 to obtain the result. Since $g \ge 0$, we trivially have that $g(t) \ge g(0)$, and thus

$$\forall \delta > 0 \quad \frac{g(\delta) - g(0)}{\delta} \geq 0 \qquad \overline{D}_0^+ g(0) \geq 0 \tag{4.17}$$

And thus in such case we have that

$$\text{L.H.S. } (4.14) = 0 \leq \overline{D}_0^+ g(0) \leq \left(\overline{D}_0^+ g(0)\right) \vee \left(\overline{D}_t^+ g(t)\right) = \text{R.H.S. } (4.14) \tag{4.18}$$

And so this case is covered.
Let us then assume that L.H.S. $(4.14) > 0$. We have that

$$\begin{aligned} \text{L.H.S. } (4.14) > 0 &\iff \forall h \geq 0. \exists 0 < \delta \leq h. \quad \frac{g(t+\delta) - g(t)}{\delta} > 0 \\ &\iff \forall h \geq 0. \exists 0 < \delta \leq h. \quad g(t+\delta) - g(t) > 0 \\ &\iff \forall n \in \mathbb{N}. \exists 0 < \delta_n \leq \frac{1}{n}. \quad g(t+\delta_n) - g(t) > 0 \end{aligned} \tag{4.19}$$

The strategy is to show that under the condition 4.19 in t, we can approximate the Dini derivative of $g_\star$ with the one of g, obtaining thus:

$$\text{L.H.S. } = \overline{D}_t^+ g_\star(t) \simeq \overline{D}_t^+ g(t) \leq \sup_{s \in [0,t]} \overline{D}_s^+ g(s) = \text{R.H.S.}$$

This will require some preliminary claims to be proved.

Claim 4.2.1. *Under the condition 4.19 we have that $g(t) = g_\star(t)$.*

Proof. By 4.19 we have that $\forall n. \exists 0 < \delta_n \leq \frac{1}{n}$ such that

$$\max_{[0,t+\delta_n]} g = g_\star(t+\delta_n) >^{4.19} g_\star(t) = \max_{[0,t]} g \tag{4.20}$$

Then $\forall n. \exists r_n \in (0, \delta_n] \subseteq (0, \frac{1}{n})$ such that $g(t+r_n) = g_\star(t+\delta_n) > g_\star(t)$. This, together with the fact that $r_n \xrightarrow{n} 0$ and that the functions involved are right-continuous, let us infer that

$$g(t) \leq g_\star(t) \leq \lim_n g(t+r_n) = g(t+) = g(t) \tag{4.21}$$

And then $g(t) = g_\star(t)$. □

Claim 4.2.2. *Under the condition 4.19 in t consider $\delta > 0$ such that $g, g_\star$ are $\mathcal{C}^0$ on $U_\delta^+ := [t, t+\delta)$. Define*

$$\xi_{t,\delta} := \xi_\delta := \xi := \inf\{\nu \geq 0 \ : \ g_\star(t+\nu) = g_\star(t+\delta)\} \tag{4.22}$$

Then ξ is strictly positive, is $\leq \delta$ and is such that $g(t+\xi) = g_\star(t+\xi) = g_\star(t+\delta)$.

Proof. We have that $\xi \leq \delta$ is trivially true by definition.

(¿ $\xi > 0$?) Since $g_\star$ is $\mathcal{C}^0$ on $[t, t+\delta)$, we have that

$$\begin{aligned} A :&= \{\nu \geq 0 \ : \ g_\star(t+\nu) = g_\star(t+\delta)\} = (g_\star)|^{-1}_{[t,t+\delta)} (g_\star(t+\delta)) - t \\ &= \Big(\underbrace{(g_\star)|^{-1}_{[t,t+\delta)} - t}_{\mathcal{C}^0 \text{function}} \Big) \Big(\underbrace{\{g_\star(t+\delta)\}}_{\text{closed set}} \Big) \end{aligned}$$

and thus A is closed. Therefore we have $\xi := \inf A = \min A$. Furthermore, because of 4.19, we have that $g_\star(t) < g_\star(t+\delta)$, and thus $0 \notin A$. Hence $\xi > 0$.

(¿ $g(t+\xi) = g_\star(t+\xi)$?) Suppose ad absurdum that $g(t+\xi) < g_\star(t+\xi)$. Because of 4.19, and since $\xi > 0$, we have that $g_\star(t) < g_\star(t+\xi)$. Then $\exists 0 < \nu < \xi$ such that

$$g(t+\nu) = g_\star(t+\xi) = \max_{[0,t+\nu]} g = g_\star(t+\nu) \tag{4.23}$$

And this is a contraddiction, because $\nu < \xi = \min A$, but $\nu \in A$.
Finally, it is true by definition that $g_\star(t+\xi) = g_\star(t+\delta)$. □

Now we have finally all the instrument to prove the final result.

$$\begin{aligned}
\text{L.H.S. (4.14)} &= \overline{D}_t^+ g_\star(t) \\
&= \lim_{h\to 0^+} \sup_{\delta\in(0,h]} \frac{g_\star(t+\delta) - g_\star(t)}{\delta} \\
&= \lim_{h\to 0^+} \sup_{\delta\in(0,h]} \frac{g_\star(t+\delta) - g(t)}{\delta} && \text{by 4.2.1} \\
&= \lim_{h\to 0^+} \sup_{\delta\in(0,h]} \frac{g(t+\xi_\delta) - g(t)}{\delta} && \text{by 4.2.2}
\end{aligned}$$

Then, since $0 < \xi_\delta \le \delta$ (by 4.2.2), we have that $\frac{1}{\delta} \le \frac{1}{\xi_\delta}$ and thus

$$\begin{aligned}
&\le \lim_{h\to 0^+} \sup_{\delta\in(0,h]} \frac{g(t+\xi_\delta) - g(t)}{\xi_\delta} && \text{by 4.2.2} \\
&\le \lim_{h\to 0^+} \sup_{\delta\in(0,h]} \sup_{\nu\in(0,\delta]} \frac{g(t+\nu) - g(t)}{\nu} \\
&= \lim_{h\to 0^+} \sup_{\delta\in(0,h]} \frac{g(t+\delta) - g(t)}{\delta} \\
&= \overline{D}_t^+ g(t) \le \left(\overline{D}_0^+ g(0)\right) \vee \left(\overline{D}_t^+ g(t)\right) = \text{R.H.S. (4.14)}
\end{aligned}$$

And thus the result stated above is valid. □

Thus we proved the statement 4.2.1:

$$\overline{D}_T^+ \sup_{0\le t\le T} \left|X_t - \overline{X}_t\right|_\infty \le \max_{s\in\{0,T\}} \left\{\overline{D}_s^+ \left|X_s - \overline{X}_s\right|_\infty\right\} \tag{4.24}$$

4.3. Get Rid of the Integral Sign

Here the result we need to show is this:

Lemma 4.3.1.

$$\overline{D}_t^+ \left|X_t - \overline{X}_t\right|_\infty \le d\left|b(t, X_t) - \bar{b}(t, \overline{X}_t)\right|_\infty$$

Notice that the mean-field particles are coupled with the system 2.6, and the initial conditions a_0 are the same for all the amoebas. Then we have

$$\begin{aligned}\left|X_t^{(N)} - \overline{X}_t^{(N)}\right|_\infty &= \max\left\{\left|A_t^{(N)} - \overline{A}_t^{(N)}\right|_\infty, \left|C_t^{(N)} - \overline{C}_t^{(N)}\right|_\infty\right\} \\ &= \max\left\{\left|A_0^{(N)} + \int_0^t b^{(N)}(s, X_s^{(N)})ds + \sigma B_t^N - \left(\overline{A}_0^{(N)} + \int_0^t b^{(N)}(s, \overline{X}_s^{(N)})ds + \sigma B_t^N\right)\right|_\infty,\right. \\ &\qquad \left.\max_{1\le m\le M_t}\left|A_{T_m}^{(N),P(m)} + Q_t^m - (\overline{A}_{T_m}^{(N),P(m)} + Q_t^m)\right|_\infty\right\} \\ &= \max\left\{\left|\int_0^t b(s, X_s) - \overline{b}(s, \overline{X}_s)ds\right|_\infty, \max_{1\le m\le M_t}\left|A_{T_m}^{(N),P(m)} - \overline{A}_{T_m}^{(N),P(m)}\right|_\infty\right\}\end{aligned}$$

Notice that $A_{T_m}^{(N)} - \overline{A}_{T_m}^{(N)}$ is not dependent on t and it is therefore removed with the derivation $\overline{D}_t^+$, unless $t = T_m$. Since the latter case happens with probability 0 we simply remove the paths for which this manifests, i.e., we assume that $t \neq T_m$. Therefore, if we try to Dini-derive such quantity we might sketch this:

$$\begin{aligned}\overline{D}_t^+ \left|X_t - \overline{X}_t\right|_\infty &= \overline{D}_t^+ \left|\underbrace{\int_0^t b(s, X_s) - \overline{b}(s, \overline{X}_s)ds}_{:= I_t = (I_t^{p;i})_{p,i}}\right|_\infty \\ &\qquad \Big(\text{if } \forall p, i.\ I_t^{p;i} \text{ is right-differentiable}\Big) \\ &= \partial_t^+ \left|\int_0^t b(s, X_s) - \overline{b}(s, \overline{X}_s)ds\right|_\infty \\ &= \underbrace{[\nabla_x |x|_\infty]_{x=I_t}}_{?} \Big(\underbrace{\partial_t^+ \int_0^t b_{p;i}(s, X_s) - \overline{b}_{p;i}(s, \overline{X}_s)ds}_{¿\quad = b_{p;i}(t,X_t) - \overline{b}_{p;i}(t,\overline{X}_t)\quad ?}\Big)_{p,i}\end{aligned}$$

The first is an easy computation.

Lemma 4.3.2. *Almost everywhere we have that*

$$\partial_i |x|_\infty = sgn(x_i)\mathbb{1}_{|x_i| = |x|_\infty} \tag{4.25}$$

Proof. Let us suppose that $|x|_\infty = |x_i|$. Then

$$\partial_i |x|_\infty = \partial_i |x_i| = sgn(x_i)$$

If instead there is $j \neq i$ such that $|x|_\infty = |x_j| > |x_i|$, then

$$\partial_i |x|_\infty = \partial_i |x_j| = 0$$

Hence $\partial_i |x|_\infty = sgn(x_i)\mathbb{1}_{|x_i| = max_j |x_j|}$. □

The second one is a confirmation of an intuitive fact:

Lemma 4.3.3. *The quantity*

$$\int_0^t b(s, X_s) - \overline{b}(s, \overline{X}_s)ds \tag{4.26}$$

is right-differentiable in each of its coordinates and is such that

$$\partial_t^+ \int_0^t b_{p;i}(s, X_s) - \overline{b}_{p;i}(s, \overline{X}_s) ds = b_{p;i}(t, X_t) - \overline{b}_{p;i}(t, \overline{X}_t) \tag{4.27}$$

almost everywhere.

Proof. Just apply the Lebesgue differentiation theorem (see [9, Theorem 3.21]). □

Then we can write

$$\begin{aligned}
\overline{D}_t^+ \left| X_t - \overline{X}_t \right|_\infty &= \partial_t^+ \left| \left(\underbrace{\int_0^t b_{p;i}(s, X_s) - \overline{b}_{p;i}(s, \overline{X}_s) ds}_{:= I_t^{p;i}} \right)_{p,i} \right|_\infty \\
&= \sum_{p=1}^N \sum_{i=1}^d \left(\partial_{p;i} |x|_\infty \right)_{x_{p;i} = I_t^{p;i}} \left(\partial_t^+ \int_0^t b_{p;i}(s, X_s) - \overline{b}_{p;i}(s, \overline{X}_s) ds \right) \\
&= \sum_{p=1}^N \sum_{i=1}^d \left(\operatorname{sgn}(I_t^{p;i}) \mathbb{1}_{|I_t^{p;i}| = |I_t|_\infty} \right) \left(b_{p;i}(t, X_t) - \overline{b}_{p;i}(t, \overline{X}_t) \right) \\
&\leq \sum_{p=1}^N \sum_{i=1}^d \left(\mathbb{1}_{|I_t^{p;i}| = |I_t|_\infty} \right) \left| b_{p;i}(t, X_t) - \overline{b}_{p;i}(t, \overline{X}_t) \right|
\end{aligned}$$

We want to get rid of the sum in N, and in order to do that we need to prove that $\left| I_t^{p;i} \right|$ reaches $|I_t|_\infty$ only for one p at a time. Or more weakly, that for two different amoebas $p \neq q$, the probability that $\left| I_t^{p;i} \right| = \left| I_t^{q;j} \right| = |I_t|_\infty$ for some coordinate i, j is 0. This might seem obvious, because corresponds roughly to the statement that different particles will happen to be in the same point with probability 0, but it is still not obvious from a mathematical point of view.

Lemma 4.3.4. *For all $t > 0$ and for all $p \neq q$ we have that $\mathbb{P}\left(\exists i, j. \ \left| I_t^{p;i} \right| = \left| I_t^{q;j} \right| \right) = 0$*

Proof. We just need to prove that this happens conditionally with respect to the birth times.

$$\begin{aligned}
&\mathbb{P}\left(\ \exists i, j. \ \left| I_t^{p;i} \right| - \left| I_t^{q;j} \right| = 0 \ \middle| \ (T^{p,g})_{p,g} \right) \leq \sum_{i=1,j=1}^{d,d} \mathbb{P}\left(\ \left| I_t^{p;i} \right| - \left| I_t^{q;j} \right| = 0 \ \middle| \ (T^{p,g})_{p,g} \right) = \\
&= \sum_{i=1,j=1}^{d,d} \mathbb{E}\left[\ \mathbb{1}_0(\left| I_t^{p;i} \right| - \left| I_t^{q;j} \right|) \ \middle| \ (T^{p,g})_{p,g} \right] = \\
&= \sum_{i=1,j=1}^{d,d} \int \mathbb{1}_{\{0\}} \Bigg(\underbrace{\left| \int_0^t b_{p;i}(s, x) - \overline{b}_{p;i}(s, y) ds \right| - \left| \int_0^t b_{q;j}(s, x) - \overline{b}_{q;j}(s, y) ds \right|}_{=: \mathcal{K}_t(x,y)} \Bigg) \cdot \\
&\qquad \cdot d(X_s, \overline{X}_s)_\star \mathbb{P}\left(\ x, y \mid \ (T^{p,g})_{p,g} \right)
\end{aligned}$$

where in the last inequality we treat all the birth times as deterministic. If we we prove that the set $\{(x, y) \ : \ \mathcal{K}_t(x, y) = 0\}$ is such that

$$\mathrm{Leb}(\{(x,y) \in \mathbb{R}^{dD_t} \times \mathbb{R}^{dD_t} \ : \ \text{Ж}_t(x,y) = 0\}) = 0$$

then the integral above is 0 (since the laws of the stochastic processes involved are absolutely continuous with respect to the Lebesgue measure), yielding the thesis. We will show this exploiting the implicit function theorem, proving that the set above is locally the graph of a function $f : \mathbb{R}^{dD_t} \times \mathbb{R}^{dD_t - 1} \longrightarrow \mathbb{R}$ and has thus Lebesgue measure 0 in $\mathbb{R}^{dD_t} \times \mathbb{R}^{dD_t}$. First we can write

$$\begin{aligned}
\text{Ж}_t(x,y) &= \left| \int_0^t b_{p;i}(s,x) - \overline{b}_{p;i}(s,y) ds \right| - \left| \int_0^t b_{q;j}(s,x) - \overline{b}_{q;j}(s,y) ds \right| \\
&= \left| \int_0^t b_{p;i}(s,x) - \lambda \left(V_i \star \int_0^s c_s^\tau d\tau \right)(y^p) ds \right| - \left| \int_0^t b_{q;j}(s,x) - \lambda \left(V_j \star \int_0^s c_s^\tau d\tau \right)(y^q) ds \right|
\end{aligned}$$

where $y^p, y^q \in \mathbb{R}^d$.

Let us see that at least one of its derivatives is always different from 0, in order to fullfill the conditions of the implicit function theorem. Since all functions involved are integrable and their derivative are integrable and dominated, we can derive under the integral with respect to the coordinates $y^{p;i}$ and obtain almost everywhere:

$$\begin{aligned}
\partial_{y^{p;i}} \text{Ж}_t(x,y) = \mathrm{sgn}(I_t^{p;i}) \ & \int_0^t \partial_{y^{p;i}} b_{p;i}(s,x) - \lambda \partial_{y^{p;i}} \left(V_i \star \int_0^s c_s^\tau d\tau \right)(y^p) ds \\
& - \partial_{y^{p;i}} \ \left| \int_0^t b_{q;j}(s,x) - \lambda \left(V_j \star \int_0^s c_s^\tau d\tau \right)(y^q) ds \right| \\
= \mathrm{sgn}(I_t^{p;i}) \ & \int_0^t \lambda \partial_{y^{p;i}} \left(V_i \star \int_0^s c_s^\tau d\tau \right)(y^p) ds =
\end{aligned}$$

Now notice that both V_i and $\partial_i V_i$ are bounded and integrable, and that by hypothesis also c_s^τ is bounded and integrable, therefore we can transport the derivative $\partial_{y^{p;i}}$ inside the convolution (see e.g., [9, Theorem 2.27]) and obtain:

$$\begin{aligned}
&= \mathrm{sgn}(I_t^{p;i}) \ \int_0^t \lambda \left(\partial_i V_i \star \int_0^s c_s^\tau d\tau \right)(y^p) ds \\
&= \mathrm{sgn}(I_t^{p;i}) \ \lambda \ \left(\partial_i V_i \star \int_0^t \int_0^s c_s^\tau d\tau ds \right)(y^p)
\end{aligned}$$

Define now $g_t(x) := \left(\int_0^t \int_0^s c_s^\tau d\tau ds \right)(x)$ and observe that since $c_s^\tau(x)$ has a constant mass when $s \geq \tau$ (after the correspondent birth time), then for all $t > 0$ we have that $g_t(x)$ is a positive function with positive mass. Since also $\partial_i V_i$ is a function always $\neq 0$, we have necessarily, whenever $\mathrm{sgn}(I_t^{p;i}) \neq 0$, that $\partial_{y^{p;i}} \text{Ж}_t(x,y) \neq 0$, and thus that the correspondent set can be described locally as the graph of some function $f_{\text{Ж}}$.

If instead $\mathrm{sgn}(I_t^{p;i}) = 0$, i.e., $I_t^{p;i} = 0$, we can act in a similar fashion, but with a different function:

$$\text{Ч}_t(x,y) := I_t^{p;i} = \int_0^t b_{p;i}(s,x) - \lambda \left(V_i \star \int_0^s c_s^\tau d\tau \right)(y^p) ds$$

And now we can show that the set $\left\{ (x,y) \ : \ I_t^{p;i} = 0 \right\} = \{\text{Ч}_t(x,y) = 0\}$ is locally the graph of some function with the implicit function theorem, obtaining the same result. In

fact, proceeding as before, we have that

$$\partial_{y^{p;i}} \text{Ч}_t(x,y) = \lambda \left(\partial_i V_i \star \int_0^t \int_0^s c_s^\tau d\tau ds \right)(y^p) > 0$$

And thus also in this case we have an implicit function $f_{\text{Ч}}$ which locally represents the set $\{\text{Ч}_t(x,y) = 0\}$. Resuming, we can write:

$$\begin{aligned} \text{Leb}\left(\{\text{Ж}_t(x,y) = 0\}\right) = & \ \text{Leb}\left(\left\{ \text{Ж}_t(x,y) = 0,\ I_t^{p;i} \neq 0 \right\} \cup \{\text{Ч}_t(x,y) = 0\} \right) \\ = & \ \text{Leb}\Big(\left\{ \text{Graph of } f_{\text{Ж}} : \mathbb{R}^{dD_t} \times \mathbb{R}^{dD_t - 1} \longrightarrow \mathbb{R} \right\} \cup \\ & \cup \left\{ \text{Graph of } f_{\text{Ч}} : \mathbb{R}^{dD_t} \times \mathbb{R}^{dD_t - 1} \longrightarrow \mathbb{R} \right\} \Big) = 0 \end{aligned}$$

And thus the thesis is reached. □

This statement allows as to say that with probability 1 there exist some index $k \in \{1, \dots, N\}$ such that

$$\begin{aligned} \overline{D}_t^+ \left| X_t - \overline{X}_t \right|_\infty &\leq \sum_{p=1}^N \sum_{i=1}^d \left(\mathbb{1}_{|I_t^{p;i}| = |I_t|_\infty} \right) \left| b_{p;i}(t, X_t) - \overline{b}_{p;i}(t, \overline{X}_t) \right| \\ &= \sum_{i=1}^d \left(\mathbb{1}_{\left|I_t^{k;i}\right| = |I_t|_\infty} \right) \left| b_{k;i}(t, X_t) - \overline{b}_{p;i}(t, \overline{X}_t) \right| \\ &\leq \sum_{i=1}^d 1 \cdot \left| b(t, X_t) - \overline{b}(t, \overline{X}_t) \right|_\infty \leq d \left| b(t, X_t) - \overline{b}(t, \overline{X}_t) \right|_\infty \end{aligned}$$

And thus the result 4.3.1 is proved.

4.4. Assembling the Proof of Lemma 4.1.2

For $T > 0$ we have that

$$\begin{aligned} \overline{D}_T^+ \sup_{0 \leq t \leq T} \left| X_t - \overline{X}_t \right|_\infty &\leq \max_{s \in \{0,T\}} \left\{ \overline{D}_s^+ \left| X_s - \overline{X}_s \right|_\infty \right\} && 4.2.1 \\ &\leq \max_{s \in \{0,T\}} \left\{ d \left| b(s, X_s) - \overline{b}(s, \overline{X}_s) \right|_\infty \right\} && 4.3.1 \end{aligned}$$

Now notice that since for every amoeba p $M_0^p = 0$ with probability 1, then we have that

$$b_p(0, X_0) = \frac{1}{N} \sum_{q=1}^N \sum_{g=1}^{M_0^q} V(A_s^p - C_s^{q,g}) = 0$$

and the same is valid for the mean-field drift:

$$\overline{b}(0, \overline{X}_0)_\infty = \lambda \left(V \star \int_0^0 c_0^\tau d\tau \right)(y^{p;i}) = 0$$

Then we have that

$$\max_{s \in \{0,T\}} \left\{ d \left| b(s, X_s) - \overline{b}(s, \overline{X}_s) \right|_\infty \right\} = d \left| b(T, X_T) - \overline{b}(T, \overline{X}_T) \right|_\infty$$

And the proof is concluded.

5. Preliminary: the Law of Large Numbers

The following chapter focuses on showing, in some sense, that the difference between the real and the average drift, namely the quantity

$$\left|b(t,\overline{X}_t)-\overline{b}(t,\overline{X}_t)\right|_\infty=\left|b^{(N)}(t,\overline{X}_t^{(N)})-\overline{b}^{(N)}(t,\overline{X}_t^{(N)})\right|_\infty \tag{5.1}$$

is very unlikely to be big as N (the number of amoebas) grows. We prove it in a slightly more general settings than the one that we will use in the result 6.0.1: the attraction taken in consideration has also an added Coulomb-type singularity. More formally we would like to prove that, chosen some $t\leq T$ and some $0<\xi<\frac{1}{2}$ then

$$\mathbb{P}\left(\left|b^{(N)}(t,\overline{X}_t^{(N)})-\overline{b}^{(N)}(t,\overline{X}_t^{(N)})\right|_\infty\geq N^{-\xi}\right)\leq f_n^{\uparrow}(T)\ N^{-\mathrm{K}_\xi n+1} \tag{5.2}$$

for all $n\in\mathbb{N}$, where K_ξ is a constant dependent only on ξ and $f_n^{\uparrow}$ is an increasing continuous function dependent on n and independent from N. In order to prove it the idea is to notice that for the k-th particle

$$\begin{aligned}
N^{-\xi}\ &\leq\left|b_k(t,\overline{X}_t)-\overline{b}_k(t,\overline{X}_t)\right|_\infty\\
&=\left|b_k(t,\overline{X}_t)-\mathbb{E}\left[\ \ b_k(t,\overline{X}_t)\ \middle|\ \overline{A}_t\ \right]\right|_\infty\\
&\leq\left|b_k(t,\overline{X}_t)-\mathbb{E}\left[\ \ b_k(t,\overline{X}_t)\ \middle|\ \overline{A}_t,\ \ (T^{p,g})_{p,g}\ \right]\right|_\infty+\\
&\qquad+\left|\mathbb{E}\left[\ \ b_k(t,\overline{X}_t)\ \middle|\ \overline{A}_t,\ \ (T^{p,g})_{p,g}\ \right]-\mathbb{E}\left[\ \ b_k(t,\overline{X}_t)\ \middle|\ \overline{A}_t\ \right]\right|_\infty
\end{aligned}$$

and therefore, using this knowledge together with A.2.3, we can obtain

$$\begin{aligned}
&\mathbb{P}\left(\left|b(t,\overline{X}_t)-\overline{b}(t,\overline{X}_t)\right|_\infty\geq N^{-\xi}\right)=\mathbb{P}\left(\sup_{1\leq k\leq N}\left|b_k(t,\overline{X}_t)-\overline{b}_k(t,\overline{X}_t)\right|_\infty\geq N^{-\xi}\ \right)\leq\\
&\leq\mathbb{P}\Bigg(\sup_{1\leq k\leq N}\left|b_k(t,\overline{X}_t)-\mathbb{E}\left[\ \ b_k(t,\overline{X}_t)\ \middle|\ \overline{A}_t,\ \ (T^{p,g})_{p,g}\ \right]\right|_\infty+\\
&\qquad+\sup_{1\leq k\leq N}\left|\mathbb{E}\left[\ \ b_k(t,\overline{X}_t)\ \middle|\ \overline{A}_t,\ \ (T^{p,g})_{p,g}\ \right]-\mathbb{E}\left[\ \ b_k(t,\overline{X}_t)\ \middle|\ \overline{A}_t\ \right]\right|_\infty\geq N^{-\xi}\ +N^{-\xi}\Bigg)\\
&\leq\overbrace{\mathbb{P}\left(\sup_{1\leq k\leq N}\left|b_k(t,\overline{X}_t)-\mathbb{E}\left[\ \ b_k(t,\overline{X}_t)\ \middle|\ \overline{A}_t,\ \ (T^{p,g})_{p,g}\ \right]\right|_\infty\geq N^{-\xi}\right)}^{\text{Term 1}}+\\
&\qquad\underbrace{+\mathbb{P}\left(\sup_{1\leq k\leq N}\left|\mathbb{E}\left[\ \ b_k(t,\overline{X}_t)\ \middle|\ \overline{A}_t,\ \ (T^{p,g})_{p,g}\ \right]-\mathbb{E}\left[\ \ b_k(t,\overline{X}_t)\ \middle|\ \overline{A}_t\ \right]\right|_\infty\geq N^{-\xi}\right)}_{\text{Term 2}}
\end{aligned} \tag{5.3}$$

and consequently we can bound separately the two addends. This was the "divide" part of the usual strategy "divide et impera". The first section is an explains the generalization of the attraction V and shows some estimate about it, the two central sections treat

respectively the first and the second term, and the final fourth section composes all the pieces together in a ready-to-use tool.

5.1. Necessary premises

Only for this chapter we will use a more general case for the attraction: we will use an additional term which also admits a cut-off Coulomb-type of singularity.

Definition 5.1.1 (Coulomb-type attraction-potential, generalized attraction and drift)**.** Let $\alpha \geq 0$. Let $V_\lambda = V_\lambda(N, \alpha) : \mathbb{R}^d \longrightarrow \mathbb{R}^d$ be a function such that

$$0 \leq ||V_\lambda||_2 \leq \mathrm{K}_\lambda \left(\frac{1}{||x||_2} \wedge N^\alpha \right) = \begin{cases} \dfrac{\mathrm{K}_\lambda}{||x||_2} & x \in \{||x||_2 \geq N^{-\alpha}\} \\ \mathrm{K}_\lambda \, N^\alpha & \text{otherwise} \end{cases} \tag{5.4}$$

for some positive constant K_λ, i.e., such that it is less than a cut-off Coulomb-type singularity, with the cut-off of parameter N^α. Then we define the generalized attraction with cut-off of parameters α as $V_G := V_G(N, \alpha, \mathrm{K}_\lambda) := V + V_\lambda(N, \alpha)$, and the generalized drift as the drift of 2.2.2 with V substituted with V_G, i.e.,

$$\begin{cases} {}^{V_G}b_k^{(N)}(t, x, (M_t^k)_k) := b_p(t, x) := \dfrac{1}{N} \displaystyle\sum_{p=1}^{N} \sum_{g=1}^{M_t^p} V_G(x^k - x^{p,g}) & 1 \leq k \leq N \\ {}^{V_G}b_{p,g}^{(N)}(t, x) := 0 & \text{if } \exists (p, g) \end{cases} \tag{5.5}$$

Analogous is the case for the generalized mean-field drift:

$$\begin{cases} {}^{V_G}\overline{b}_k^{(N)}(t, x, (M_t^k)_k) := \overline{b}_p(t, x) := \lambda \displaystyle\int (V_G \star c_t^\tau)(x) d\,\tau & 1 \leq k \leq N \\ {}^{V_G}\overline{b}_{p,g}^{(N)}(t, x) := 0 & \text{if } \exists (p, g) \end{cases} \tag{5.6}$$

Therefore with the symbols $b, \overline{b}$, in this chapter, we will mean ${}^{V_G}b, {}^{V_G}\overline{b}$ respectively.

Remark 5.1.1. The usual attraction can be considered as a generalized attraction of parameter $\alpha = 0$.

Before proceeding to the law of large numbers we need to gather some preliminary lemmas. The first is a simple result of [5], that we simply report here.

Lemma 5.1.1. *For a function f define the eventually infinite quantities*

$$||f||_{1\wedge\infty} := \inf_{f_1 + f_\infty = f} (||f_1||_{\mathcal{L}^1} + ||f_\infty||_{\mathcal{L}^\infty}), \qquad ||f||_{1\vee\infty} := ||f||_{\mathcal{L}^1} + ||f||_{\mathcal{L}^\infty}$$

Then for arbitrary functions f, g it is true that

$$||f \star g||_\infty \leq ||f||_{1\wedge\infty} \cdot ||g||_{1\vee\infty} \tag{5.7}$$

Proof. Let us indicate the norms $||\cdot||_{\mathcal{L}^\infty}, ||\cdot||_{\mathcal{L}^1}$ with $||\cdot||_\infty, ||\cdot||_1$ respectively. Let $g_1 + g_\infty = g$ be a decomposition of g. Then

$$\begin{aligned} &||f \star g||_\infty \leq ||f \star g_1||_\infty + ||f \star g_1||_1 \leq \\ &\leq ||f||_\infty \, ||g_1||_1 + ||f||_1 \, ||g_\infty||_\infty \\ &\leq (||f||_\infty + ||f||_1)(||g_1||_1 + ||g_\infty||_\infty) = ||f||_{1\vee\infty} \, (||g_1||_1 + ||g_\infty||_\infty) \end{aligned}$$

Since this is valid for every decomposition $g_1 + g_\infty = g$, we have our result. □

The second lemma that we need is a simple evaluation of these quantities for the generalized attraction.

Lemma 5.1.2. *Let V_G be the generalized attraction, then we have that there exists a constant $K_{V,\lambda}$ depending only on V, K_λ such that*

$$||V_G||_{1\wedge\infty}\,,\ \left|\left|V_G^2\right|\right|_{1\wedge\infty} \leq K_{V,\lambda} \tag{5.8}$$

Proof. Let us first operate for the power 1. Choose the decomposition

$$V_G = V + V_\lambda = \underbrace{V + V_\lambda \mathbb{1}_{|x|\leq 1}}_{=:(V_G)_1} + \underbrace{V_\lambda \mathbb{1}_{|x|>1}}_{=:(V_G)_\infty} \tag{5.9}$$

Let us again sign the norms $||\cdot||_{\mathcal{L}^\infty}, ||\cdot||_{\mathcal{L}^1}$ with $||\cdot||_\infty, ||\cdot||_1$ respectively. Then we have that

$$\begin{aligned}
&||(V_G)_1||_1 \leq ||V||_1 + \left|\left|V_\lambda \mathbb{1}_{||x||_2\leq 1}\right|\right|_1 = \\
&= ||V||_1 + \left|\left|\mathrm{K}_\lambda N^\alpha \mathbb{1}_{0\leq||x||_2\leq N^{-\alpha}}\right|\right|_1 + \left|\left|\frac{\mathrm{K}_\lambda}{||x||_2}\mathbb{1}_{N^{-\alpha}\leq||x||_2\leq 1}\right|\right|_1 \\
&\overset{1}{=} ||V||_1 + \mathrm{K}_\lambda N^\alpha \cdot \mathrm{Leb}(\mathbb{B}^{d-1}[N^{-\alpha}]) + \int_{N^{-\alpha}}^1 \frac{\mathrm{K}_\lambda}{||x||_2}\,dx \\
&\leq ||V||_1 + \mathrm{K}_\lambda N^\alpha \max\left\{\pi(N^{-\alpha})^2, \frac{4}{3}\pi(N^{-\alpha})^3\right\} + \\
&\qquad + \mathrm{K}_\lambda \max\left\{\int_{N^{-\alpha}}^1 \frac{1}{r}r\,dr\,2\pi, \int_{N^{-\alpha}}^1 \frac{1}{r}r^2\,dr\,2\pi \int_0^\pi \sin(\theta)d\theta\right\} \\
&\leq ||V||_1 + \mathrm{K}_\lambda N^{\alpha-2\alpha}2\pi + \mathrm{K}_\lambda 2\pi \cdot 2\max\left\{\int_0^1 dr, \int_0^1 r\,dr\right\} \\
&\leq ||V||_1 + \mathrm{K}_\lambda 6\pi
\end{aligned}$$

And obviously we have that

$$\left|\left|V_\lambda \mathbb{1}_{||x||_2>1}\right|\right|_\infty \leq \left|\left|\frac{\mathrm{K}_\lambda}{||x||_2}\mathbb{1}_{||x||_2>1}\right|\right|_\infty = \mathrm{K}_\lambda$$

Then the first inequality is proven.
Let us proceed with the power 2. Choose this decomposition:

$$\begin{aligned}
V_G^2 &= (V + V_\lambda)^2 = V^2 + 2VV_\lambda + V_\lambda^2 \\
&= \underbrace{V^2 + 2VV_\lambda \mathbb{1}_{||x||_2\leq 1}}_{=:(V_G^2)_1} + \underbrace{VV_\lambda \mathbb{1}_{||x||_2>1} + V_\lambda^2 \mathbb{1}_{||x||_2>1}}_{=:(V_G^2)_\infty}
\end{aligned}$$

Then, proceeding as before, we have that

$$\left|\left|(V_G^2)_1\right|\right|_1 \leq \left|\left|V^2\right|\right|_1 + 2\,||V||_\infty \int |V_\lambda|\,\mathbb{1}_{|x|\leq 1} \leq \left|\left|V^2\right|\right|_1 + 2\,||V||_\infty \mathrm{K}_\lambda 6\pi$$

And regarding the other term:

[1] $\mathrm{Leb}(\mathbb{B}^{d-1}[N^{-\alpha}])$ indicates the Lebesgue measure of the $d-1$-ball of radius $N^{-\alpha}$

$$\left|\left|VV_\lambda \mathbb{1}_{|x|>1} + V_\lambda^2 \mathbb{1}_{||x||_2>1}\right|\right|_\infty \leq \left|\left|\frac{\mathrm{K}_\lambda V}{||x||_2}\mathbb{1}_{||x||_2>1}\right|\right|_\infty + \left|\left|\frac{\mathrm{K}_\lambda^2}{||x||_2^{\,2}}\mathbb{1}_{||x||_2>1}\right|\right|_\infty = \mathrm{K}_\lambda \, ||V||_\infty + \mathrm{K}_\lambda^2$$

□

5.2. The First Term

Now we can finally approach the first part of the law of large numbers. The proof is classical law of large numbers, following for example [11]. It uses a notation akin to the relabelled model defined in 2.10.

Proposition 5.2.1 (The first part of the law of large numbers)**.** *Let M_t be the spawning Poisson process of 2.10. Let $\overline{X}^{(N)}$ be a r.v. that under $\mathbb{P}(\ \cdot \mid\ (T^{p,g})_{p,g}\ )$ takes the form*

$$\overline{X}^{(N)} = (\overline{A}^1, \dots, \overline{A}^N, \overline{C}^1, \dots, \overline{C}^{M_t}) \in (\mathbb{R}^d)^{D_t} = (\mathbb{R}^d)^{N+M_t} \tag{5.10}$$

Let $b^{(N)}(t,\cdot)$ be the generalized *drift of parameter α defined in 5.1.1. Assume that:*

1. *All the coordinates of $\overline{X}^{(N)}$ are $\perp\!\!\!\perp$ under the p. measure $\mathbb{P}(\ \mid\ (T^{p,g})_{p,g}\ )$.*
2. *All the coordinates $\overline{A}^p$ of $\overline{X}^{(N)}$ are i.i.d. under the p. measure $\mathbb{P}(\ \mid\ (T^{p,g})_{p,g}\ )$.*
3. *The densities of all coordinates of $\overline{X}^{(N)}$ are uniformly bounded in $\mathcal{L}^\infty$, i.e.,*

$$\exists K_{dens} > 0.\ \forall p, m.\ \ ||c^m||_\infty, ||a^p||_\infty < K_{dens}$$

where a^p and c^m represent the laws of $\overline{A}^p$ and $\overline{C}^m$ under $\mathbb{P}(\ \cdot \mid\ (T^{p,g})_{p,g}\ )$ respectively.

Consider $\xi > 0$ be such that $0 < \xi + \alpha < \frac{1}{2}$ and define the sets

$$S_A^{N,\xi} := \left\{ \sup_{1\leq k\leq N} \left| b_k^{(N)}(t, \overline{X}^{(N)}) - \mathbb{E}\left[\ b_k^{(N)}(t, \overline{X}^{(N)}) \ \middle|\ \overline{A}^k,\ (T^{p,g})_{p,g}\ \right]\right|_\infty \geq N^{-\xi} \right\}$$

$$S^{N,\xi} := \left\{ \sup_{1\leq k\leq N} \left| b_k^{(N)}(t, \overline{X}^{(N)}) - \mathbb{E}\left[\ b_k^{(N)}(t, \overline{X}^{(N)}) \ \middle|\ (T^{p,g})_{p,g}\ \right]\right|_\infty \geq N^{-\xi} \right\}$$

Then, signing as $S_*^{N,\xi}$ either of the two sets, for $S_*^{N,\xi}$ is valid the laws of large numbers, i.e., $\exists \overline{N} = \overline{N}(\alpha, \lambda)$ such that for all K_ξ such that $0 < K_\xi \leq 1 - 2(\alpha + \xi)$ and for all $n \in \mathbb{N}$ exists $K_n = K_{V,\lambda,\lambda,n} > 0$ and an increasing positive continuous function $f_n^\uparrow$ (independent from N, α, ξ) so that it is valid:

$$\forall N > \overline{N}. \qquad \mathbb{P}\left(\ S_*^{N,\xi}\ \right) \leq K_n\ f_n^\uparrow(K_{dens})\ f_n^\uparrow(t)\ N^{-K_\xi n+1} \tag{5.11}$$

Remark 5.2.1. All along the subsequent proof it will be assumed that $t, \lambda \geq 1$. This is because in general for $0 < p < q$ we have that

$$t^p \leq (t \vee 1)^q$$

and this clutters the notation and creates only non-necessary headaches for a function of t which cannot explode (all the powers involved are positive integers).

We will now prove such result. For the sake of shortness, together with the convention at A.1, we will use the subsequent notation:

$$S_*^{N,\xi} := \text{ either of } S_A^{N,\xi} \text{ and } S^{N,\xi}$$

Notice that

$$\begin{aligned}
\mathbb{P}\left(S_*^{N,\xi} \right) &= \mathbb{P}\left(\sup_{1\le k\le N} \left| b_k^{(N)}(t,\overline{X}^{(N)}) - \widetilde{\mathbb{E}}_*\left[b_k^{(N)}(t,\overline{X}^{(N)}) \right] \right|_\infty \ge N^{-\xi} \right) \\
&\le \sum_{k=1}^{N}\sum_{i=1}^{d} \mathbb{P}\left(\left| b_{k;i}^{(N)}(t,\overline{X}^{(N)}) - \widetilde{\mathbb{E}}_*\left[b_{k;i}^{(N)}(t,\overline{X}^{(N)}) \right] \right| \ge N^{-\xi} \right) \\
&\le N\cdot 3\cdot \max_{k,i} \underbrace{\mathbb{P}\left(\left| b_{k;i}^{(N)}(t,\overline{X}^{(N)}) - \widetilde{\mathbb{E}}_*\left[b_{k;i}^{(N)}(t,\overline{X}^{(N)}) \right] \right| \ge N^{-\xi} \right)}_{\text{Ч}(k;i)}
\end{aligned}$$

where recall that $b_{k;i}$ indicates the i-th coordinate for the d-dimensional drift of the k-th particle. Since the expression has the same form for each coordinate $(k;i)$, we assume that the maximum of $\text{Ч}(k;i)$ is reached in $(1;1)$, and tame with a bound independent from such assumption.

We shall begin by applying Markov inequality with function $f(x) := |x|^{2n}$ for a generic $n \in \mathbb{N}$.

$$\begin{aligned}
&= 3N\cdot \mathbb{P}\left(\left| b_{1;1}^{(N)}(t,\overline{X}^{(N)}) - \widetilde{\mathbb{E}}_*\left[b_{1;1}^{(N)}(t,\overline{X}^{(N)}) \right] \right| \ge N^{-\xi} \right) \\
&\le \frac{3N}{(N^{-\xi})^{2n}} \mathbb{E}\left[\left| b_{1;1}^{(N)}(t,\overline{X}^{(N)}) - \widetilde{\mathbb{E}}_*\left[b_{1;1}^{(N)}(t,\overline{X}^{(N)}) \right] \right|^{2n} \right] \\
&\overset{1}{=} 3N^{\xi 2n+1}\mathbb{E}\left[\left| \frac{1}{N}\sum_{m=1}^{M_t} (V_G)_1(\overline{A}^1 - \overline{C}^m) - \frac{1}{N}\sum_{m=1}^{M_t} \widetilde{\mathbb{E}}_*\left[(V_G)_1(\overline{A}^1 - \overline{C}^m) \right] \right|^{2n} \right] \\
&= 3N^{\xi 2n+1}\frac{1}{N^{2n}}\mathbb{E}\left[\left| \sum_{m=1}^{M_t} \left[(V_G)_1(\overline{A}^1 - \overline{C}^m) - \widetilde{\mathbb{E}}_*\left[\left((V_G)_1 \star c^m \right)(\overline{A}^1) \right] \right] \right|^{2n} \right] \\
&= 3N^{(\xi-1)2n+1}\mathbb{E}\left[\left| \sum_{m=1}^{M_t} \left[\underbrace{(V_G)_1(\overline{A}^1 - \overline{C}^m)}_{:=\Psi_m} - \underbrace{\widetilde{\mathbb{E}}_*\left[\left((V_G)_1 \star c^m \right)(\overline{A}^1) \right]}_{=\widetilde{\mathbb{E}}_*[\,\Psi_m\,]=:\psi_m^*} \right] \right|^{2n} \right] =
\end{aligned}$$

Setting $\text{Ж}_m := |\Psi_m - \psi_m^*|$, we can apply the multinomial formula:

$$\left(\sum_{m=1}^{M} \text{Ж}_m \right)^{2n} = \sum_{\beta_1+\ldots+\beta_M=2n} \frac{(2n)!}{\beta_1!\ldots\beta_M!} \prod_{m=1}^{M} \text{Ж}_m^{\beta_m} \tag{5.12}$$

$$= \sum_{\beta_1+\cdots+\beta_M=2n} \frac{(2n)!}{(\beta)!} \prod_{m=1}^{M} \text{Ж}_m^{\beta_m} \tag{5.13}$$

where β is the multi-index $(\beta_1,\ldots,\beta_M)$ and $(\beta)! := \beta_1!\ldots\beta_M!$. Then we obtain

[1] Here $(V_G)_1$ indicates the first coordinate of V_G, and not a first part of any decomposition of V_G.

$$= 3N^{(\xi-1)2n+1}\mathbb{E}\left[\sum_{\beta_1+\cdots+\beta_{M_t}=2n}\frac{(2n)!}{(\beta)!}\prod_{m=1}^{M_t}|\Psi_m-\psi_m^*|^{\beta_m}\right]$$

$$= 3N^{(\xi-1)2n+1}\mathbb{E}\left[\widetilde{\mathbb{E}}_*\left[\sum_{\beta_1+\cdots+\beta_{M_t}=2n}\frac{(2n)!}{(\beta)!}\prod_{m=1}^{M_t}|\Psi_m-\psi_m^*|^{\beta_m}\right]\right]$$

$$= 3N^{(\xi-1)2n+1}\mathbb{E}\left[\sum_{\beta_1+\cdots+\beta_{M_t}=2n}\frac{(2n)!}{(\beta)!}\widetilde{\mathbb{E}}_*\left[\prod_{m=1}^{M_t}|\Psi_m-\psi_m^*|^{\beta_m}\right]\right]$$

$$= 3(2n)!\ N^{(\xi-1)2n+1}\mathbb{E}\left[\sum_{\beta_1+\cdots+\beta_{M_t}=2n}\widetilde{\mathbb{E}}_*\left[\prod_{m=1}^{M_t}|\Psi_m-\psi_m^*|^{\beta_m}\right]\right]=$$

Notice now that for each m the r.v.

$$\Psi_m-\psi_m^* \quad = \quad (V_G)_1(\overline{A}^1-\overline{C}^m)-\widetilde{\mathbb{E}}_*\left[\left((V_G)_1\star c^m\right)(\overline{A}^1)\right] \tag{5.14}$$

are $\perp\!\!\!\perp$ under $\widetilde{\mathbb{P}}_1(\cdot)=\mathbb{P}\left(\ \cdot\mid \overline{A}^1,\ (T^{p,g})_{p,g}\right)$, therefore

$$\begin{aligned}
&= 3(2n)!\ N^{(\xi-1)2n+1}\mathbb{E}\left[\sum_{\beta_1+\cdots+\beta_{M_t}=2n}\widetilde{\mathbb{E}}_*\left[\widetilde{\mathbb{E}}_1\left[\prod_{m=1}^{M_t}|\Psi_m-\psi_m^*|^{\beta_m}\right]\right]\right]\\
&= 3(2n)!\ N^{(\xi-1)2n+1}\mathbb{E}\left[\sum_{\beta_1+\cdots+\beta_{M_t}=2n}\prod_{m=1}^{M_t}\widetilde{\mathbb{E}}_*\left[\widetilde{\mathbb{E}}_1\left[|\Psi_m-\psi_m^*|^{\beta_m}\right]\right]\right] \qquad (5.15)\\
&= 3(2n)!\ N^{(\xi-1)2n+1}\mathbb{E}\left[\sum_{\beta_1+\cdots+\beta_{M_t}=2n}\prod_{m=1}^{M_t}\underbrace{\widetilde{\mathbb{E}}_*\left[|\Psi_m-\psi_m^*|^{\beta_m}\right]}_{\leq ?}\right]
\end{aligned}$$

Claim 5.2.1. *There exists a constant $K_{V,dens,\curlywedge}>0$ depending on V, K_{dens}, and there exists $\overline{N}$ depending on $K_{\curlywedge}, V, \alpha$ such that*

$$\forall N>\overline{N}. \qquad \widetilde{\mathbb{E}}_*\left[|\Psi_m-\psi_m^*|^{\beta_m}\right]\leq\begin{cases}1 & \text{if} \quad \beta_m=0\\ 0 & \text{if} \quad \beta_m=1\\ (K_{V,\curlywedge}f^{\uparrow}(K_{dens})N^{\alpha})^{\beta_m-2} & \text{otherwise}\end{cases} \tag{5.16}$$

Proof of the claim. If $\beta_m=0$ we have

$$\widetilde{\mathbb{E}}_*\left[|\Psi_m-\psi_m^*|^0\right]=\widetilde{\mathbb{E}}_*\left[\widetilde{\mathbb{E}}_1[\,1\,]\right]=1$$

If instead $\beta_m=1$ we have:

$$\widetilde{\mathbb{E}}_*\left[|\Psi_m-\psi_m^*|^1\right]=\widetilde{\mathbb{E}}_*\left[\widetilde{\mathbb{E}}_1[\,|\Psi_m-\psi_m^*|\,]\right]=\widetilde{\mathbb{E}}_*\left[\left|\Psi_m-\widetilde{\mathbb{E}}_*[\,\Psi_m\,]\right|\right]=0$$

We need then to take care only of the case $\beta_m\geq 2$. Here we have

$$\widetilde{\mathbb{E}}_*\left[\ |\Psi_m-\psi_m^*|^{\beta_m}\ \right]=\widetilde{\mathbb{E}}_*\left[\ |\Psi_m-\psi_m^*|^{\beta_m-2}\cdot|\Psi_m-\psi_m^*|^2\ \right]$$
$$\leq||\Psi_m-\psi_m^*||_\infty^{\beta_m-2}\cdot\widetilde{\mathbb{E}}_*\left[\ |\Psi_m-\psi_m^*|^2\ \right]$$

Then we can dominate the two terms separately.

$$\begin{aligned}||\Psi_m-\psi_m^*||_\infty&=\left|\left|(V_G)_1(\overline{A}^1-\overline{C}^m)-\widetilde{\mathbb{E}}_*\left[\ ((V_G)_1\star c^m)\,(\overline{A}^1)\ \right]\right|\right|_\infty\\&=||V_G||_\infty+||V_G\star c^m||_\infty\\&\leq||V_G||_\infty+||V_G||_\infty\,||c^m||_1\\&=2\,||V_G||_\infty=2(||V||_\infty+||V_\lambda||_\infty)\\&\leq2(2\,||V_\lambda||_\infty)=\mathrm{K}_\lambda 4N^\alpha\end{aligned}$$

where the last inequality is valid if N is big enough, i.e., $N>\overline{N}(\alpha,\lambda,V)$. Then

$$||\Psi_m-\psi_m^*||_\infty^{\beta_m-2}\leq(\mathrm{K}_\lambda 4)^{\beta_m-2}N^{\alpha(\beta_m-2)}=(\mathrm{K}_\lambda 4N^\alpha)^{\beta_m-2}$$

Let us now check the other term:

$$\begin{aligned}&\widetilde{\mathbb{E}}_*\left[\ |\Psi_m-\psi_m^*|^2\ \right]=\\&=\widetilde{\mathbb{E}}_*\left[\ \widetilde{\mathbb{E}}_1\left[\ \left|(V_G)_1(\overline{A}^1-\overline{C}^m)-\psi_m^*\right|^2\ \right]\ \right]\\&=\widetilde{\mathbb{E}}_*\left[\ \int\left((V_G)_1(\overline{A}^1-x)-\psi_m^*\right)^2\,c^m(x)\,dx\ \right]\\&=\widetilde{\mathbb{E}}_*\left[\ \int(V_G)_1^2(\overline{A}^1-x)c^m(x)dx-\int2(V_G)_1(\overline{A}^1-x)\psi_m^*c^m(x)dx+(\psi_m^*)^2\int c^m(x)dx\ \right]\\&=\widetilde{\mathbb{E}}_*\left[\ \left((V_G)_1^2\star c^m\right)(\overline{A}^1)-2\psi_m^*\left((V_G)_1\star c^m\right)(\overline{A}^1)+(\psi_m^*)^2\ \right]\\&\leq\widetilde{\mathbb{E}}_*\left[\ \left|\left|(V_G)_1^2\star c^m\right|\right|_\infty+2\psi_m^*\,||(V_G)_1\star c^m||_\infty+(\psi_m^*)^2\ \right]\\&=\left|\left|(V_G)_1^2\star c^m\right|\right|_\infty+2\psi_m^*\,||(V_G)_1\star c^m||_\infty+(\psi_m^*)^2\\&\leq\left|\left|(V_G)_1^2\star c^m\right|\right|_\infty+2\,||(V_G)_1\star c^m||_\infty^2+||(V_G)_1\star c^m||_\infty^2\\&=\left|\left|(V_G)_1^2\star c^m\right|\right|_\infty+3\,||(V_G)_1\star c^m||_\infty^2\\&\overset{1}{\leq}\left|\left|V_G^2\right|\right|_{1\wedge\infty}||c^m||_{1\vee\infty}+3\left(||V_G||_{1\wedge\infty}\,||c^m||_{1\vee\infty}\right)^2\\&=\mathrm{K}_{\lambda,V}(1+\mathrm{K}_{\mathrm{dens}})+3\mathrm{K}_{\lambda,V}^2(1+\mathrm{K}_{\mathrm{dens}})^2\\&\leq(\mathrm{K}_{\lambda,V}+3\mathrm{K}_{\lambda,V}^2)\ [1+\mathrm{K}_{\mathrm{dens}}+(1+\mathrm{K}_{\mathrm{dens}})^2]\\&=\mathrm{K}_{V,\lambda}\ f^\uparrow(\mathrm{K}_{\mathrm{dens}})\leq[\mathrm{K}_{V,\lambda}f^\uparrow(\mathrm{K}_{\mathrm{dens}})]^{\beta_m-2}\end{aligned}$$

Then we have

[1] By 5.1.1 and 5.1.2

$$\widetilde{\mathbb{E}}_*\left[\,|\Psi_m-\psi_m^*|^{\beta_m}\,\right]\le||\Psi_m-\psi_m^*||_\infty^{\beta_m-2}\cdot\widetilde{\mathbb{E}}_*\left[\,|\Psi_m-\psi_m^*|^2\,\right]$$
$$\le(\mathrm{K}_{\lightning}4N^\alpha)^{\beta_m-2}[\mathrm{K}_{V,\lightning}f^\uparrow(\mathrm{K}_{\mathrm{dens}})]^{\beta_m-2}\qquad=[\mathrm{K}_{V,\lightning}f^\uparrow(\mathrm{K}_{\mathrm{dens}})N^\alpha]^{\beta_m-2}$$

And this concludes the proof of the claim. □

Now we finally obtained a bound independent from the coordinate $(1,1)$ of the drift, thus, resuming with calculations from 5.15 and exploiting the claim 5.2.1, we have that

$$\mathbb{P}\left(\,S_*^{N,\xi}\,\right)=\mathbb{P}\left(\sup_{1\le k\le N}\left|b_k^{(N)}(t,\overline{X}^{(N)})-\widetilde{\mathbb{E}}_*\left[\,b_k^{(N)}(t,\overline{X}^{(N)})\,\right]\right|_\infty\ge N^{-\xi}\right)$$
$$\le N\cdot3\cdot\max_{k,i}\,\mathbb{P}\left(\,\left|b_{k;i}^{(N)}(t,\overline{X}^{(N)})-\widetilde{\mathbb{E}}_*\left[\,b_{k;i}^{(N)}(t,\overline{X}^{(N)})\,\right]\right|\ge N^{-\xi}\,\right)$$
$$\le3(2n)!\;N^{(\xi-1)2n+1}\mathbb{E}\left[\sum_{\beta_1+\cdots+\beta_{M_t}=2n}\prod_{m=1}^{M_t}\widetilde{\mathbb{E}}_*\left[\,|\Psi_m-\psi_m^*|^{\beta_m}\,\right]\right]$$
$$\le3(2n)!\;N^{(\xi-1)2n+1}\mathbb{E}\left[\sum_{\beta_1+\cdots+\beta_{M_t}=2n}\prod_{m=1}^{M_t}\left(\mathbb{1}_{\beta_m=0}+\mathbb{1}_{\beta_m\ge2}\underbrace{(\mathrm{K}_{V,\lightning}f^\uparrow(\mathrm{K}_{\mathrm{dens}})N^\alpha}_{=:\text{Ю}})^{\beta_m-2}\right)\right]$$

We can assume that $\text{Ю}\ge1$ (eventually modulate the value $\mathrm{K}_{V,\mathrm{dens},\lightning}$ accordingly).

Claim 5.2.2. *$\text{Ю}\ge1$ is independent from the multi-index β, therefore*

$$\sum_{\beta_1+\cdots+\beta_M=2n}\prod_{m=1}^{M}\left(\mathbb{1}_{\beta_m=0}+\mathbb{1}_{\beta_m\ge2}\text{Ю}^{\beta_m-2}\right)\le(2n)^n\;(M)^n\;\text{Ю}^{2n-2}\tag{5.17}$$

Proof of the claim. First of all notice that if the multi-index β has an entry $\beta_m=1$, then the whole product under $\prod_{m=1}^M$ goes to 0. Besides, if $\beta_m=0$, then the relative factor is 1 and therefore does not influence the product. In equations:

$$\prod_{m=1}^{M}\left(\mathbb{1}_{\beta_m=0}+\mathbb{1}_{\beta_m\ge2}\text{Ю}^{\beta_m-2}\right)=\prod_{\substack{m=1\\\beta_m\neq0}}^{M}\left(\mathbb{1}_{\beta_m\ge2}\text{Ю}^{\beta_m-2}\right)\;=\;\prod_{\substack{m=1\\\beta_m\neq0}}^{M}(\mathbb{1}_{\beta_m\neq1}\text{Ю}^{\beta_m-2})=$$
$$=\mathbb{1}_{\beta_m\neq1}\prod_{\substack{m=1\\\beta_m\neq0}}^{M}\text{Ю}^{\beta_m}\;\cdot\;\prod_{\substack{m=1\\\beta_m\neq0}}^{M}\text{Ю}^{-2}\;=\mathbb{1}_{\beta_m\neq1}\text{Ю}^{\sum_{\beta_m\neq0}\beta_m}\text{Ю}^{-2\sum_{\beta_m\neq0}1}$$

Therefore:

$$\sum_{\beta_1+\cdots+\beta_M=2n}\prod_{m=1}^{M}\left(\mathbb{1}_{\beta_m=0}+\mathbb{1}_{\beta_m\ge2}\text{Ю}^{\beta_m-2}\right)=\sum_{\beta_1+\cdots+\beta_M=2n}\mathbb{1}_{\beta_m\neq1}\text{Ю}^{\sum_{\beta_m\neq0}\beta_m}\text{Ю}^{-2\sum_{\beta_m\neq0}1}$$
$$=\sum_{\substack{\beta_1+\cdots+\beta_M=2n\\\forall m.\beta_m\neq1}}\text{Ю}^{\sum_{\beta_m\neq0}\beta_m}\text{Ю}^{-2\sum_{\beta_m\neq0}1}$$
$$=\sum_{\substack{\beta_1+\cdots+\beta_M=2n\\\forall m.\beta_m\neq1}}\text{Ю}^{2n-2(\sum_{\beta_m\neq0}1)}$$

We now need to take a closer look at the multi-index $\beta = (\beta_1, \dots, \beta_M)$ and its entries. For each β_m define the quantity

$$\#\{\beta_m \neq 0\} := \#\{i \ : \ \beta_i \text{ is non-zero entry of } \beta\} = \sum_{\beta_m \neq 0} 1$$

Observe that we are taking into consideration only multi-indices in this set:

$$\{\beta = (\beta_1, \dots, \beta_M) \ : \ |\beta| = 2n, \ \forall m. \beta_m \neq 1\}$$

Therefore, since the minimum value that we allow in an entry β_m is 2, the maximum amount of non-zero entries, for a multi-index $\beta = (\beta_1, \dots, \beta_M)$ such that $|\beta| = \beta_1 + \cdots + \beta_M = 2n$, will be n (n entries of value 2). Then we can write

$$\sum_{\substack{\beta_1+\cdots+\beta_M=2n \\ \forall m. \beta_m \neq 1}} \text{Ю}^{2n-2(\sum_{\beta_m \neq 0} 1)} = \sum_{\substack{\beta_1+\cdots+\beta_M=2n \\ \forall m. \beta_m \neq 1}} \text{Ю}^{2n-2(\#\{\beta_m \neq 0\})} =$$

$$= \sum_{k=1}^{n} \sum_{\substack{\beta_1+\cdots+\beta_M=2n \\ \forall m. \beta_m \neq 1 \\ \#\{\beta_m \neq 0\}=k}} \text{Ю}^{2n-2k} = \sum_{k=1}^{n} \text{Ю}^{2n-2k} \sum_{\substack{\beta_1+\cdots+\beta_M=2n \\ \forall m. \beta_m \neq 1 \\ \#\{\beta_m \neq 0\}=k}} 1 \quad {}^{1} \leq \text{Ю}^{2n-2} \sum_{k=1}^{n} \sum_{\substack{\beta_1+\cdots+\beta_M=2n \\ \forall m. \beta_m \neq 1 \\ \#\{\beta_m \neq 0\}=k}} 1$$

We can notice that the last term is simply counting:

$$\sum_{\substack{\beta_1+\cdots+\beta_M=2n \\ \forall m. \beta_m \neq 1 \\ \#\{\beta_m \neq 0\}=k}} 1 = \#\left\{\beta = (\beta_m)_{m=1}^{M} \ : \ \sum \beta_m = 2n, \ \forall m. \beta_m \neq 1, \ \#^{\widehat{0}}(\beta) = k\right\}$$

It can be evaluated with the classical stars and bars problem, with $k-1$ bars and n stars (technically there are $2n$ stars, but the condition $\beta_m \neq 1$ is equivalent to require that between each bars we have only even quantities):

$$\begin{aligned}
&\#\left\{\beta = (\beta_m)_{m=1}^{M} \ : \ \sum \beta_m = 2n, \ \forall m. \beta_m \neq 1, \ \#^{\widehat{0}}(\beta) = k\right\} \\
&= \#\left(\text{ways to choose } k \text{ entries in } \beta = (\beta_1, \dots, \beta_M)\right) \cdot \\
&\quad \cdot \#\left(\text{ways to arrange } k-1 \text{ bars between } n \text{ stars}\right) \\
&= \binom{M}{k} \binom{n+k-1}{k-1} \\
&= \frac{M \cdot (M-1) \cdot \cdots \cdot (M-k+1)}{k!} \cdot \frac{(n+k-1) \cdot (n+k-2) \cdot \cdots \cdot (n+1)}{(k-1)!} \\
&\leq \left[M \cdot (M-1) \cdot \cdots \cdot (M-k+1)\right] \left[(n+k-1) \cdot (n+k-2) \cdot \cdots \cdot (n+1)\right] \\
&= (M)_{(k)} (n+k-1)_{(k-1)} \\
&\leq (M)^k (n+k)^{k-1}
\end{aligned}$$

Then, resuming, we have

[1] $\text{Ю} \geq 1$

$$\sum_{\beta_1+\cdots+\beta_M=2n}\prod_{m=1}^{M}\left(\mathbb{1}_{\beta_m=0}+\mathbb{1}_{\beta_m\geq 2}\text{Ю}^{\beta_m-2}\right) \;\leq\; \text{Ю}^{2n-2}\sum_{k=1}^{n}\sum_{\substack{\beta_1+\cdots+\beta_M=2n\\ \forall m.\beta_m\neq 1\\ \#\{\beta_m\neq 0\}=k}} 1 \leq$$

$$\leq \text{Ю}^{2n-2}\sum_{k=1}^{n}(M)^k(n+k)^{k-1} \;\leq\; \text{Ю}^{2n-2}\,(M)^n\,(2n)^{n-1}\sum_{k=1}^{n}1 =$$

$$\leq (2n)^n\,(M)^n\,\text{Ю}^{2n-2}$$

And this concludes the proof of the claim. □

Resuming everything and applying the last claim we obtain:

$$\mathbb{P}\left(\,S_*^{N,\xi}\,\right) = \mathbb{P}\left(\sup_{1\leq k\leq N}\left|b_k^{(N)}(t,\overline{X}^{(N)}) - \widetilde{\mathbb{E}}_*\left[\,b_k^{(N)}(t,\overline{X}^{(N)})\,\right]\right|_\infty \geq N^{-\xi}\right)$$

$$\leq N\cdot 3\cdot\max_{k,i}\mathbb{P}\left(\left|b_{k;i}^{(N)}(t,\overline{X}^{(N)}) - \widetilde{\mathbb{E}}_*\left[\,b_{k;i}^{(N)}(t,\overline{X}^{(N)})\,\right]\right| \geq N^{-\xi}\right)$$

$$\leq 3(2n)!\;N^{(\xi-1)2n+1}\mathbb{E}\left[\sum_{\beta_1+\cdots+\beta_{M_t}=2n}\prod_{m=1}^{M_t}\left(\mathbb{1}_{\beta_m=0}+\mathbb{1}_{\beta_m\geq 2}(\text{K}_{V,\text{ʎ}}f^{\uparrow}(\text{K}_{\text{dens}})N^{\alpha})^{\beta_m-2}\right)\right]$$

$$\leq 3(2n)!\;N^{(\xi-1)2n+1}\mathbb{E}\left[\,(2n)^n\,(\text{K}_{V,\text{ʎ}}f^{\uparrow}(\text{K}_{\text{dens}})N^{\alpha})^{2n-2}\,(M_t)^n\,\right]$$

$$= (\text{K}_{V,\text{ʎ}})^{2n-2}\,[f^{\uparrow}(\text{K}_{\text{dens}})]^{2n-2}\,3(2n)!(2n)^n\,N^{(\xi+\alpha-1)2n+1-2\alpha}\,\mathbb{E}\left[\,(M_t)^n\,\right]$$

$$= \text{K}_{V,\text{ʎ},n}\,f_n^{\uparrow}(\text{K}_{\text{dens}})\,N^{(\xi+\alpha-1)2n+1-2\alpha}\,\mathbb{E}\left[\,(M_t)^n\,\right] \leq$$

Now we can use the result A.2.1 to evaluate the n-momenta of the Poisson process $M_t \sim \text{Poisson}(\lambda N t)$, and recall that by 5.2.1 we assume without loss of generality that the Poisson parameter is bigger than 1:

$$\leq \text{K}_{V,\text{ʎ},n}\,f_n^{\uparrow}(\text{K}_{\text{dens}})\,N^{(\xi+\alpha-1)2n+1-2\alpha}\,f^{\uparrow}(n)\,(Nt\lambda)^n$$

$$= \text{K}_{V,\text{ʎ},n,\lambda}\,f_n^{\uparrow}(\text{K}_{\text{dens}})\,t^n N^{(\xi+\alpha-1)2n+1-2\alpha}\,N^n$$

$$\leq \text{K}_{V,\text{ʎ},n,\lambda}\,f_n^{\uparrow}(\text{K}_{\text{dens}})\,f_n^{\uparrow}(t)\,N^{[2(\xi+\alpha)-2+1]n+1-2\alpha} \leq$$

Then, for every $K = \text{K}_\xi$ such that $0 < \text{K}_\xi \leq 1-2(\alpha+\xi)$ (it is guaranteed to exist because $\alpha+\xi<\frac{1}{2}$), we have

$$\leq\; \text{K}_{V,\text{ʎ},n,\lambda}\,f_n^{\uparrow}(\text{K}_{\text{dens}})\,f_n^{\uparrow}(t)\,N^{-\text{K}_\xi n+1-2\cdot 0} \;=\; \text{K}_{V,\text{ʎ},n,\lambda}\,f_n^{\uparrow}(\text{K}_{\text{dens}})\,f_n^{\uparrow}(t)\,N^{-\text{K}_\xi n+1}$$

And thus the proof of the proposition is finished.

5.3. The Second Term

To prove the second part of the law of large numbers it will be necessary to prove that a specific r.v. has finite q-moments, but in order to better drive the comprehension and the attention of the reader this will be done after this proposition.

Proposition 5.3.1 (The second part of the law of large numbers)**.** *Let $\overline{X}_t^{(N)}$ be the solution of the mean field system 3.18 with N amoebas with the attraction V substituted with V_G. Let $t \leq T$ and choose $0 < \xi \leq \frac{1}{2}$ and define the sets*

$$R_A^{N,\xi} := \left\{ \sup_{1\leq k\leq N} \left| \mathbb{E}\left[\ b_k^{(N)}(t,\overline{X}_t^{(N)}) \ \middle| \ \overline{A}^k,\ (T^{p,g})_{p,g} \ \right] - \mathbb{E}\left[\ b_k^{(N)}(t,\overline{X}_t^{(N)}) \ \middle| \ \overline{A}^k \ \right] \right|_\infty \geq N^{-\xi} \right\}$$

$$R^{N,\xi} := \left\{ \sup_{1\leq k\leq N} \left| \mathbb{E}\left[\ b_k^{(N)}(t,\overline{X}_t^{(N)}) \ \middle| \ (T^{p,g})_{p,g} \ \right] - \mathbb{E}\left[\ b_k^{(N)}(t,\overline{X}_t^{(N)}) \ \right] \right|_\infty \geq N^{-\xi} \right\}$$

Then, signing as

$$R_*^{N,\xi} := \text{ either of } R_A^{N,\xi} \text{ and } R^{N,\xi},$$

for $R_*^{N,\xi}$ is valid the law of large numbers, i.e., for all $K = K_\xi$ such that $0 < K_\xi \leq 1-2\xi$ and for all $n \in \mathbb{N}$ exists $K_n = K_{V,\lambda,n,\lambda} > 0$ and a positive increasing continuous function $f_n^\uparrow$ independent from N, ξ such that:

$$\mathbb{P}\left(\ R_*^{N,\xi} \ \right) \ \leq K_n \ f_n^\uparrow(T) \ N^{-K_\xi n+1} \tag{5.18}$$

Let us prove this result. As for the first law of large numbers at the remark 5.2.1, we assume here as well that $t, \lambda \geq 1$. As for the first part of the law of large numbers, together with $R_*^{N,\xi}$ we use the convention at A.1. Then we can write, as for the previous proof:

$$\begin{aligned}
\mathbb{P}\left(\ R_*^{N,\xi} \ \right) =& \leq \mathbb{P}\left(\sup_{1\leq k\leq N} \left| \widetilde{\mathbb{E}}_*\left[\ b_k^{(N)}(t,\overline{X}^{(N)}) \ \right] - \mathbb{E}_*\left[\ b_k^{(N)}(t,\overline{X}^{(N)}) \ \right] \right|_\infty \geq N^{-\xi} \right) \\
&\leq \sum_{i=1}^{d}\sum_{k=1}^{N} \max_{k,i} \ \mathbb{P}\left(\left| \widetilde{\mathbb{E}}_*\left[\ b_{k;i}^{(N)}(t,\overline{X}^{(N)}) \ \right] - \mathbb{E}_*\left[\ b_{k;i}^{(N)}(t,\overline{X}^{(N)}) \ \right] \right| \geq N^{-\xi} \right) =
\end{aligned}$$

Suppose, as in the proof of the previous LLN, that the maximum is reached in the coordinate $(k;i) = (1,1)$. Now we will use again the original notation 2.6.

$$\begin{aligned}
&= d \ N \ \mathbb{P}\left(\left| \widetilde{\mathbb{E}}_*\left[\ b_{1;1}^{(N)}(t,\overline{X}_t^{(N)}) \ \right] - \mathbb{E}_*\left[\ b_{1;1}^{(N)}(t,\overline{X}_t^{(N)}) \ \right] \right| \geq N^{-\xi} \right) \\
&\overset{1}{\leq} 3 \ N \ N^{\xi 2n} \ \mathbb{E}\left[\left| \widetilde{\mathbb{E}}_*\left[\ b_{1;1}^{(N)}(t,\overline{X}_t^{(N)}) \ \right] - \mathbb{E}_*\left[\ b_{1;1}^{(N)}(t,\overline{X}_t^{(N)}) \ \right] \right|^{2n} \right] \\
&= 3 \ N^{1+\xi 2n} \ \mathbb{E}\left[\left| \frac{1}{N} \sum_{p=1}^{N} \sum_{g=1}^{M_t^p} \widetilde{\mathbb{E}}_*\left[\ (V_G)_1(\overline{A}_t^1 - \overline{C}_t^{p,g}) \ \right] - \frac{1}{N} \sum_{p=1}^{N} \mathbb{E}_*\left[\sum_{g=1}^{M_t^p} (V_G)_1(\overline{A}_t^1 - \overline{C}_t^{p,g}) \right] \right|^{2n} \right] \\
&= 3 \ N^{1+(\xi-1)2n} \ \mathbb{E}\left[\left| \sum_{p=1}^{N} \left(\underbrace{\sum_{g=1}^{M_t^p} \widetilde{\mathbb{E}}_*\left[\ (V_G)_1(\overline{A}_t^1 - \overline{C}_t^{p,g}) \ \right]}_{=:\Phi_p^*} - \underbrace{\mathbb{E}_*\left[\sum_{g=1}^{M_t^p} (V_G)_1(\overline{A}_t^1 - \overline{C}_t^{p,g}) \right]}_{=\mathbb{E}_*\left[\ \Phi_p^* \ \right] =: \phi_p^*} \right) \right|^{2n} \right] =
\end{aligned}$$

[1] Markov inequality with $f(x) = |x|^{2n}$

As for the first part of the law of large numbers, we apply on $Ж_p := \left|\Phi_p^* - \phi_p^*\right|$ the multinomial formula

$$\left(\sum_{p=1}^{N} Ж_p\right)^{2n} = \sum_{\beta_1+\cdots+\beta_N=2n} \frac{(2n)!}{(\beta)!} \prod_{p=1}^{N} Ж_p^{\beta_p}$$

obtaining

$$= 3\ N^{1+(\xi-1)2n}\ \mathbb{E}\left[\ \sum_{\beta_1+\cdots+\beta_N=2n} \frac{(2n)!}{(\beta)!} \prod_{p=1}^{N} \left|\Phi_p^* - \phi_p^*\right|^{\beta_p}\ \right]$$

$$= 3\ N^{1+(\xi-1)2n} \sum_{\beta_1+\cdots+\beta_N=2n} \frac{(2n)!}{(\beta)!}\ \mathbb{E}\left[\ \prod_{p=1}^{N} \left|\Phi_p^* - \phi_p^*\right|^{\beta_p}\ \right] =$$

Now notice that

$$\Phi_p^* - \phi_p^* = \sum_{g=1}^{M_t^p} \widetilde{\mathbb{E}}_*\left[\ (V_G)_1(\overline{A}_t^1 - \overline{C}_t^{p,g})\ \right] - \mathbb{E}_*\left[\ \sum_{g=1}^{M_t^p} (V_G)_1(\overline{A}_t^1 - \overline{C}_t^{p,g})\ \right]$$

depends exclusively on $M_t^p, \overline{C}_t^{p,g}$ for a fixed p (and eventually on $\overline{A}_t^1$ if $* = 1$). Plus, for different p the quantities above are i.i.d. under $\mathbb{P}_*(\ )$ and thus we can go on with the calculations so:

$$= 3\ N^{1+(\xi-1)2n} \sum_{\beta_1+\cdots+\beta_N=2n} \frac{(2n)!}{(\beta)!}\ \mathbb{E}\left[\ \mathbb{E}_*\left[\ \prod_{p=1}^{N} \left|\Phi_p^* - \phi_p^*\right|^{\beta_p}\ \right]\ \right]$$

$$= 3\ N^{1+(\xi-1)2n} \sum_{\beta_1+\cdots+\beta_N=2n} \frac{(2n)!}{(\beta)!} \prod_{p=1}^{N} \mathbb{E}\left[\ \mathbb{E}_*\left[\ \left|\Phi_1^* - \phi_1^*\right|^{\beta_p}\ \right]\ \right]$$

$$= 3\ N^{1+(\xi-1)2n} \sum_{\beta_1+\cdots+\beta_N=2n} \frac{(2n)!}{(\beta)!} \prod_{p=1}^{N} \mathbb{E}\left[\ \left|\Phi_1^* - \phi_1^*\right|^{\beta_p}\ \right]$$

$$\leq 3\ (2n)!\ N^{1+(\xi-1)2n} \sum_{\beta_1+\cdots+\beta_N=2n} \prod_{p=1}^{N} \mathbb{E}\left[\ \left|\Phi_1^* - \phi_1^*\right|^{\beta_p}\ \right] =$$

Now, since $\phi_1^* = \mathbb{E}_*\left[\ \Phi_1^*\ \right]$, we can deduce, as for the first part of the law of large numbers:

$$= \mathrm{K}_n\ N^{1+(\xi-1)2n} \sum_{\substack{\beta_1+\cdots+\beta_N=2n \\ \forall p.\beta_p\neq 1}} \prod_{p=1}^{N} \left(\mathbb{1}_{\beta_p=0} + \mathbb{1}_{\beta_p\geq 2}\mathbb{E}\left[\ \left|\Phi_1^* - \phi_1^*\right|^{\beta_p}\ \right]\right)$$

Now we need to use the subsequent bound for $\mathbb{E}\left[\ \left|\Phi_1^* - \phi_1^*\right|^{\beta_p}\ \right]$, which will be proved in this proposition 5.3.2 (just put $q = \beta_p$):

$$\mathbb{E}\left[\ \left|\Phi_1^* - \phi_1^*\right|^{\beta_p}\ \right] \leq \mathrm{K}_{V,ʎ,\beta_p,\lambda}\ f^{\uparrow}(\beta_p, T)$$

for some positive constant $\mathrm{K}_{V,ʎ,q,\lambda}$ and some function $f^{\uparrow}$ increasing in both arguments

and continuous in T. Then, supposing without loss of generality that R.H.S. (5.3) ≥ 1, maximizing on β_p and elevating to β_p we have

$$\begin{aligned}\mathbb{E}\left[\ |\Phi_1^* - \phi_1^*|^{\beta_p}\ \right] &\leq \mathrm{K}_{V,\lambda,\beta_p,\lambda}\ f^{\uparrow}(\beta_p, T) \leq [\max_{\beta_p} \mathrm{K}_{V,\lambda,\beta_p,\lambda}\ f^{\uparrow}(\beta_p, T)]^{\beta_p} \\ &= [\max_{\beta_p} \mathrm{K}_{V,\lambda,\beta_p,\lambda}\ f^{\uparrow}(2n, T)]^{\beta_p} = [\mathrm{K}_{V,\lambda,n,\lambda}\ f_n^{\uparrow}(T)]^{\beta_p}\end{aligned}$$

loosing the dependence of the constant on β_p before elevating to the power. Applying this to our calculations, and resuming all since the beginning, we obtain that:

$$\mathbb{P}\left(\ R_*^{N,\xi}\ \right) \leq \sum_{i=1}^{d}\sum_{k=1}^{N} \max_{k,i}\ \mathbb{P}\left(\ \left|\widetilde{\mathbb{E}}_*\left[\ b_{k;i}^{(N)}(t, \overline{X}^{(N)})\ \right] - \mathbb{E}_*\left[\ b_{k;i}^{(N)}(t, \overline{X}^{(N)})\ \right]\right| \geq N^{-\xi}\ \right)$$

$$\leq 3\ N^{1+(\xi-1)2n}\ \mathbb{E}\left[\left|\sum_{p=1}^{N}\left(\underbrace{\sum_{g=1}^{M_t^p}\widetilde{\mathbb{E}}_*\left[\ (V_G)_1(\overline{A}_t^1 - \overline{C}_t^{p,g})\ \right]}_{=:\Phi_p^*} - \underbrace{\mathbb{E}_*\left[\ \sum_{g=1}^{M_t^p}(V_G)_1(\overline{A}_t^1 - \overline{C}_t^{p,g})\ \right]}_{=\mathbb{E}_*\left[\ \Phi_p^*\ \right]=:\phi_p^*}\right)\right|^{2n}\right]$$

$$\leq \mathrm{K}_n\ N^{1+(\xi-1)2n} \sum_{\substack{\beta_1+\cdots+\beta_N=2n \\ \forall p.\beta_p\neq 1}}\ \prod_{p=1}^{N}\ \left(\mathbb{1}_{\beta_p=0} + \mathbb{1}_{\beta_p\geq 2}\mathbb{E}\left[\ |\Phi_1^* - \phi_1^*|^{\beta_p}\ \right]\right)$$

$$\leq \mathrm{K}_n\ N^{1+(\xi-1)2n} \underbrace{\sum_{\substack{\beta_1+\cdots+\beta_N=2n \\ \forall p.\beta_p\neq 1}}\ \prod_{p=1}^{N}\ \left(\mathbb{1}_{\beta_p=0} + \mathbb{1}_{\beta_p\geq 2}\left(\mathrm{K}_{V,\lambda,n,\lambda}\ f_n^{\uparrow}(T)\right)^{\beta_p}\right)}_{\text{Б}} \leq$$

Now apply claim 5.2.2 from the proof of the first part of the law of large numbers 5.2.1 to bound Б:

$$\begin{aligned}&\leq \mathrm{K}_n\ N^{1+(\xi-1)2n}\ (2n)^n\ \left(\mathrm{K}_{V,\lambda,n,\lambda}\ f_n^{\uparrow}(T)\right)^{2n-2} \\ &= \mathrm{K}_{V,\lambda,n,\lambda}\ f_n^{\uparrow}(T)\ N^{1+(2\xi-2+1)n}\ = \mathrm{K}_{V,\lambda,n,\lambda}\ f_n^{\uparrow}(T)\ N^{1+(2\xi-1)n}\end{aligned}$$

Therefore, for every K_ξ such that $0 < \mathrm{K}_\xi \leq 1 - 2\xi$, we have that

$$1 + (2\xi - 1)n \leq 1 - \mathrm{K}_\xi n \tag{5.19}$$

and therefore we can conclude the proof:

$$\mathbb{P}\left(\ R_*^{N,\xi}\ \right) \leq \mathrm{K}_{V,\lambda,n,\lambda}\ f_n^{\uparrow}(T)\ N^{1+(2\xi-1)n}\ \leq \mathrm{K}_{V,\lambda,n,\lambda}\ f_n^{\uparrow}(T)\ N^{1-\mathrm{K}_\xi n}$$

5.3.1. Proof of the Auxiliary Bound 5.3

Notice that in 5.3 we contained from above the average attraction exercised by the cAMP particles released by a single amoeba. Let us show that such bound holds.

Proposition 5.3.2. *Let $t \leq T$ and consider the r.v.*

$$\Phi_p^*(k,i) \;:=\; \sum_{g=1}^{M_t^p} \widetilde{\mathbb{E}}_* \Big[\; (V_G)_i(\overline{A}_t^k - \overline{C}_t^{p,g}) \;\Big] \tag{5.20}$$

and its average $\phi_p^*(k,i) := \mathbb{E}_* \big[\; \Phi_p^*(k,i) \;\big]$*, where* $*$ *is either* k *or absent. Then exists* $K_{V,\lambda,q,\lambda} > 0$ *and an positive function* $f^{\uparrow}(q,T)$ *increasing in both arguments, continuous in* T *and bounded in* q *such that*

$$\forall q \geq 1.\ \forall p,k,i. \qquad \mathbb{E}\left[\; \left|\Phi_p^*(k,i) - \phi_p^*(k,i)\right|^q \;\right] \leq K_{V,\lambda,q,\lambda}\ f^{\uparrow}(q,T) \tag{5.21}$$

Remark 5.3.1. The r.v.

$$\Phi_p^*(k,i) \;:=\; \sum_{g=1}^{M_t^p} \widetilde{\mathbb{E}}_* \Big[\; (V_G)_i(\overline{A}_t^k - \overline{C}_t^{p,g}) \;\Big] \tag{5.22}$$

represents the attraction that the amoeba $\overline{A}_t^k$ feels at its coordinate i due only to the cAMP released by the amoeba $\overline{A}_t^p$ till the time t.

Let us prove this result. As for the first part of the law of large numbers at the remark 5.2.1, we assume here also that $t, \lambda \geq 1$. Observe that since the $(M_t^p)_p$ are i.i.d. , then we have that also $(\Phi_p^*(k,i))_p$ are i.i.d. , and thus we can operate directly with $\Phi_1^*(k,i)$. Then let us observe closely the object in question.

$$\begin{aligned}
\Phi_1^*(k,i) \;&=\; \sum_{g=1}^{M_t^1} \widetilde{\mathbb{E}}_* \Big[\; (V_G)_i(\overline{A}_t^k - \overline{C}_t^{1,g}) \;\Big] \\
&= \sum_{g=1}^{M_t^1} \widetilde{\mathbb{E}}_* \Big[\; \widetilde{\mathbb{E}}_k \Big[\; (V_G)_i(\overline{A}_t^k - \overline{C}_t^{1,g}) \;\Big] \;\Big] \;= \sum_{g=1}^{M_t^1} \widetilde{\mathbb{E}}_* \left[\; \int (V_G)_i(\overline{A}_t^k - x)\ c_t^{1,g}(x)dx \;\right] \\
&= \sum_{g=1}^{M_t^1} \widetilde{\mathbb{E}}_* \Big[\; \Big((V_G)_i \star c_t^{1,g}\Big)(\overline{A}_t^k) \;\Big] \;= \sum_{g=1}^{M_t^1} \widetilde{\mathbb{E}}_* \left[\; \int (V_G)_i(y)\ c_t^{1,g}(y - \overline{A}_t^k)dy \;\right] \\
&\overset{1}{=} \sum_{g=1}^{M_t^1} \int (V_G)_i(y)\ \widetilde{\mathbb{E}}_* \Big[\; c_t^{1,g}(y - \overline{A}_t^k) \;\Big]\, dy = \int (V_G)_i(y)\ \widetilde{\mathbb{E}}_* \left[\; \sum_{g=1}^{M_t^1} c_t^{1,g}(y - \overline{A}_t^k) \;\right] dy
\end{aligned}$$

Then in the same way we can observe that

$$\begin{aligned}
\phi_1^*(k,i) \;&=\; \mathbb{E}_* \left[\; \Phi_1^*(k,i) \;\right] \;= \mathbb{E}_* \left[\; \int (V_G)_i(y)\ \widetilde{\mathbb{E}}_* \left[\; \sum_{g=1}^{M_t^1} c_t^{1,g}(y - \overline{A}_t^k) \;\right] dy \;\right] \\
&= \int (V_G)_i(y)\ \mathbb{E}_* \left[\; \sum_{g=1}^{M_t^1} c_t^{1,g}(y - \overline{A}_t^k) \;\right] dy
\end{aligned}$$

Then we can write

[1] Fubini

$$
\begin{aligned}
&\left|\Phi_p^*(k,i) - \phi_p^*(k,i)\right| \; = \\
&= \left| \int (V_G)_i(y)\; \widetilde{\mathbb{E}}_* \left[\sum_{g=1}^{M_t^1} c_t^{1,g}(y - \overline{A}_t^k) \right] dy - \int (V_G)_i(y)\; \mathbb{E}_* \left[\sum_{g=1}^{M_t^1} c_t^{1,g}(y - \overline{A}_t^k) \right] dy \right| \\
&= \left| \int (V_G)_i(y) \; \left(\widetilde{\mathbb{E}}_* \left[\sum_{g=1}^{M_t^1} c_t^{1,g}(y - \overline{A}_t^k) \right] - \mathbb{E}_* \left[\sum_{g=1}^{M_t^1} c_t^{1,g}(y - \overline{A}_t^k) \right] \right) \; dy \right| \\
&\leq \int |(V_G)_i(y)| \; \cdot \left| \widetilde{\mathbb{E}}_* \left[\sum_{g=1}^{M_t^1} c_t^{1,g}(y - \overline{A}_t^k) \right] - \mathbb{E}_* \left[\sum_{g=1}^{M_t^1} c_t^{1,g}(y - \overline{A}_t^k) \right] \right| dy
\end{aligned}
\tag{5.23}
$$

We want to detach $(V_G)_i$ from the rest of the object. In order to do that we can use an argument very similar to the one illustrated in 5.1.1.

Claim 5.3.1. *Let f, g be functions. Then*

$$
||f \cdot g||_{\mathcal{L}^1} \leq ||f||_{1 \wedge \infty} \cdot ||g||_{1 \vee \infty} \tag{5.24}
$$

Proof. Choose a decomposition $f_1 + f_\infty = f$ and proceed with the Hölder-inequality, analogously to 5.1.1:

$$
||f \cdot g||_{\mathcal{L}^1} \leq ||f_1 \cdot g||_{\mathcal{L}^1} + ||f_\infty \cdot g||_{\mathcal{L}^1} \leq ||f_1||_{\mathcal{L}^1} ||g||_{\mathcal{L}^\infty} + ||f_\infty||_{\mathcal{L}^\infty} ||g||_{\mathcal{L}^1} \tag{5.25}
$$

Since this is valid for every decomposition we have the thesis, and this ends the proof of the claim. □

Exploiting this for the calculations of 5.23 we have:

$$
\begin{aligned}
\left|\Phi_p^*(k,i) - \phi_p^*(k,i)\right| &\leq \int |(V_G)_i(y)| \; \cdot \left| \widetilde{\mathbb{E}}_* \left[\sum_{g=1}^{M_t^1} c_t^{1,g}(y - \overline{A}_t^k) \right] - \mathbb{E}_* \left[\sum_{g=1}^{M_t^1} c_t^{1,g}(y - \overline{A}_t^k) \right] \right| dy \\
&\leq ||(V_G)_i||_{1 \wedge \infty} \; \cdot \left|\left| \widetilde{\mathbb{E}}_* \left[\sum_{g=1}^{M_t^1} c_t^{1,g}(y - \overline{A}_t^k) \right] - \mathbb{E}_* \left[\sum_{g=1}^{M_t^1} c_t^{1,g}(y - \overline{A}_t^k) \right] \right|\right|_{1 \vee \infty} \\
&\overset{1}{\leq} \mathrm{K}_{V,\lambda} \cdot \left|\left| \underbrace{\widetilde{\mathbb{E}}_* \left[\sum_{g=1}^{M_t^1} c_t^{1,g}(y - \overline{A}_t^k) \right]}_{=:\Theta_1} - \underbrace{\mathbb{E}_* \left[\sum_{g=1}^{M_t^1} c_t^{1,g}(y - \overline{A}_t^k) \right]}_{=:\theta_1^* = \mathbb{E}_*[\,\Theta_1^*\,]} \right|\right|_{1 \vee \infty}
\end{aligned}
\tag{5.26}
$$

Resuming all this we have that:

[1] 5.1.2

$$\begin{aligned}
\mathbb{E}\left[\ \left|\Phi_p^*(k,i)-\phi_p^*(k,i)\right|^q\ \right] &\leq \mathbb{E}\left[\ (\mathrm{K}_{V,\lambda})^q\,||\Theta_1-\theta_2||_{1\vee\infty}^q\ \right]\\
&= \mathrm{K}_{V,\lambda,q}\mathbb{E}\left[\ (||\Theta_1-\theta_2||_{\mathcal{L}^1}+||\Theta_1-\theta_2||_{\mathcal{L}^\infty})^q\ \right]\\
&\overset{2}{=} \mathrm{K}_{V,\lambda,q}\mathbb{E}\left[\ \sum_{j=0}^{q}\binom{q}{j}\ ||\Theta_1-\theta_1||_{\mathcal{L}^1}^{q-j}\ ||\Theta_1-\theta_1||_{\mathcal{L}^\infty}^{j}\ \right]\\
&= \mathrm{K}_{V,\lambda,q}\ \sum_{j=0}^{q}\binom{q}{j}\ \mathbb{E}\left[\ ||\Theta_1-\theta_1||_{\mathcal{L}^1}^{q-j}\ ||\Theta_1-\theta_1||_{\mathcal{L}^\infty}^{j}\ \right]
\end{aligned}$$

Let us start to find a bound for the infinity norm of $\Theta_1-\theta_1$:

$$\begin{aligned}
||\Theta_1-\theta_1||_{\mathcal{L}^\infty} &= \left|\left|\widetilde{\mathbb{E}}_*\left[\ \sum_{g=1}^{M_t^1}c_t^{1,g}(y-\overline{A}_t^k)\ \right]-\mathbb{E}_*\left[\ \sum_{g=1}^{M_t^1}c_t^{1,g}(y-\overline{A}_t^k)\ \right]\right|\right|_\infty\\
&\leq \left|\left|\widetilde{\mathbb{E}}_*\left[\ \sum_{g=1}^{M_t^1}c_t^{1,g}(y-\overline{A}_t^k)\ \right]\right|\right|_\infty+\left|\left|\mathbb{E}_*\left[\ \sum_{g=1}^{M_t^1}c_t^{1,g}(y-\overline{A}_t^k)\ \right]\right|\right|_\infty\\
&\leq \sum_{g=1}^{M_t^1}\widetilde{\mathbb{E}}_*\left[\ \left|\left|c_t^{1,g}\right|\right|_\infty\ \right]+\mathbb{E}_*\left[\ \sum_{g=1}^{M_t^1}\left|\left|c_t^{1,g}\right|\right|_\infty\ \right]
\end{aligned}$$

Recall now that by assumption 3.3.2, since $t\leq T$, we have that there exists a positive continuous increasing function $f^\uparrow$ such that:

$$\left|\left|c_t^{1,g}\right|\right|_{\mathcal{L}^\infty(\mathbb{R}^d)}\ \leq \sup_{t,\tau\leq T}||c_t^\tau||_{\mathcal{L}^\infty(\mathbb{R}^d)}\ \leq f^\uparrow(T)$$

and therefore we have that

$$\begin{aligned}
||\Theta_1-\theta_1||_{\mathcal{L}^\infty} &\leq \sum_{g=1}^{M_t^1}\widetilde{\mathbb{E}}_*\left[\ \left|\left|c_t^{1,g}\right|\right|_\infty\ \right]+\mathbb{E}_*\left[\ \sum_{g=1}^{M_t^1}\left|\left|c_t^{1,g}\right|\right|_\infty\ \right]\\
&\leq \sum_{g=1}^{M_t^1}f^\uparrow(T)+\mathbb{E}_*\left[\ \sum_{g=1}^{M_t^1}f^\uparrow(T)\ \right]\ =\ f^\uparrow(T)(M_t^1+\lambda t)
\end{aligned}$$

And thus we can write

[2] If the 1- and ∞-norm of $\Theta_1-\theta_1$ are finite, which will be proved

$$
\begin{aligned}
&\mathbb{E}\left[\ \left|\Phi_p^*(k,i)-\phi_p^*(k,i)\right|^q\ \right]\leq\\
&\leq \mathrm{K}_{V,\lambda,q}\sum_{j=0}^{q}\binom{q}{j}\ \mathbb{E}\left[\ ||\Theta_1-\theta_1||_{\mathcal{L}^1}^{q-j}\ \ (f^{\uparrow}(T))^j\ (M_t^1+\lambda t)^j\ \right]\\
&\leq \mathrm{K}_{V,\lambda,q}\sum_{j=0}^{q}(q!)\mathbb{E}\left[\ ||\Theta_1-\theta_1||_{\mathcal{L}^1}^{q-j}\ \ (f^{\uparrow}(T))^q\ \sum_{u=1}^{j}\binom{j}{u}\ (M_t^1)^u(\lambda t)^{j-u}\ \right]\\
&\leq \mathrm{K}_{V,\lambda,q}\ (q!)\ (f^{\uparrow}(T))^q\ \sum_{j=0}^{q}\ \mathbb{E}\left[\ ||\Theta_1-\theta_1||_{\mathcal{L}^1}^{q-j}\ \sum_{u=1}^{j}(j!)\ (M_t^1)^j(\lambda t)^j\ \right]\\
&\leq \mathrm{K}_{V,\lambda,q}\ (q!)\ (f^{\uparrow}(T))^q\ \sum_{j=0}^{q}\ j\ (j!)\ (\lambda t)^j\ \mathbb{E}\left[\ ||\Theta_1-\theta_1||_{\mathcal{L}^1}^{q-j}\ \ (M_t^1)^j\ \right]\\
&\leq \underbrace{\mathrm{K}_{V,\lambda,q}\ (q!)\ q\ (q!)\ \lambda^q}\ t^q f^{\uparrow}(q,T)\sum_{j=0}^{q}\ \mathbb{E}\left[\ ||\Theta_1-\theta_1||_{\mathcal{L}^1}^{q-j}\ \ (M_t^1)^j\ \right]\\
&= \mathrm{K}_{V,\lambda,q}\ f^{\uparrow}(q,T)\ \sum_{j=0}^{q}\ \mathbb{E}\left[\ \left|\left|\widetilde{\mathbb{E}}_*\left[\ \sum_{g=1}^{M_t^1}c_t^{1,g}(y-\overline{A}_t^k)\ \right]-\mathbb{E}_*\left[\ \sum_{g=1}^{M_t^1}c_t^{1,g}(y-\overline{A}_t^k)\ \right]\right|\right|_{\mathcal{L}^1}^{q-j}\ (M_t^1)^j\ \right]
\end{aligned}
$$

Now we will use the notation $c_t^{1,g}=c_t^{T^{1,g}}$, to make explicit the fact that such density depends on the birth time $T^{1,g}$.

Observe also that the random objects $M_t^1,T^{1,g}$ of θ_1 are completely averaged out *inside* θ_1, and essentially do not have a direct relationship with the correspondent objects $M_t^1,T^{1,g}$ of Θ_1. To avoid such confusion we exploit the fact that $M_t^1\sim M_t^2,T^{1,g}\sim T^{2,g}$ and then we can write

$$
\theta_1=\mathbb{E}_*\left[\ \sum_{g=1}^{M_t^2}c_t^{T^{2,g}}(y-\overline{A}_t^k)\ \right]=\theta_2=:\theta \tag{5.27}
$$

obtaining thus that

$$
\begin{aligned}
&\mathbb{E}\left[\ \left|\Phi_p^*(k,i)-\phi_p^*(k,i)\right|^q\ \right]\leq\\
&\leq \mathrm{K}_{V,\lambda,q}\ f^{\uparrow}(q,T)\ \sum_{j=0}^{q}\ \mathbb{E}\left[\ \left|\left|\widetilde{\mathbb{E}}_*\left[\ \sum_{g=1}^{M_t^1}c_t^{1,g}(y-\overline{A}_t^k)\ \right]-\mathbb{E}_*\left[\ \sum_{g=1}^{M_t^2}c_t^{2,g}(y-\overline{A}_t^k)\ \right]\right|\right|_{\mathcal{L}^1}^{q-j}\ (M_t^1)^j\ \right]\\
&= \mathrm{K}_{V,\lambda,q}\ f^{\uparrow}(q,T)\ \sum_{j=0}^{q}\ \mathbb{E}\left[\ \left|\left|\widetilde{\mathbb{E}}_*\left[\ \sum_{g=1}^{M_t^1}c_t^{T^{1,g}}(y-\overline{A}_t^k)\ \right]-\mathbb{E}_*\left[\ \sum_{g=1}^{M_t^2}c_t^{T^{2,g}}(y-\overline{A}_t^k)\ \right]\right|\right|_{\mathcal{L}^1}^{q-j}\ (M_t^1)^j\ \mathbb{1}_{M_t^1\neq 0}\ \right]+\\
&\quad+\mathrm{K}_{V,\lambda,q}\ f^{\uparrow}(q,T)\ \sum_{j=0}^{q}\ \mathbb{E}\left[\ \left|\left|0-\mathbb{E}_*\left[\ \sum_{g=1}^{M_t^2}c_t^{T^{2,g}}(y-\overline{A}_t^k)\ \right]\right|\right|_{\mathcal{L}^1}^{q-j}\ (M_t^1)^j\ \mathbb{1}_{M_t^1=0}\ \right]
\end{aligned}
$$

Let us calculate the second term first:

$$
\begin{aligned}
&\mathrm{K}_{V,\curlywedge,q} f^{\uparrow}(q,T) \sum_{j=0}^{q} \mathbb{E}\left[||\theta||_{\mathcal{L}^1}^{q-j} \ (M_t^1)^j \ \mathbb{1}_{M_t^1=0} \right] = \\
&= \mathrm{K}_{V,\curlywedge,q} f^{\uparrow}(q,T) \ \mathbb{E}\left[||\theta||_{\mathcal{L}^1}^{q} \ (M_t^1)^0 \ \mathbb{1}_{M_t^1=0} \right] = \\
&= \mathrm{K}_{V,\curlywedge,q} f^{\uparrow}(q,T) \ \mathbb{E}\left[\left|\left| \mathbb{E}_* \left[\sum_{g=1}^{M_t^2} c_t^{T^{2,g}} (y - \overline{A}_t^k) \right] \right|\right|_{\mathcal{L}^1}^{q} \mathbb{1}_{M_t^1=0} \right] = \\
&= \mathrm{K}_{V,\curlywedge,q} f^{\uparrow}(q,T) \ \mathbb{E}\left[\left|\left| \mathbb{E}_* \left[\sum_{g=1}^{+\infty} \underbrace{c_t^{T^{2,g}} (y - \overline{A}_t^k)}_{\geq 0} \underbrace{\mathbb{1}_{g \leq M_t^2}}_{\mathbb{1}_{T^{2,g} \leq t}} \right] \right|\right|_{\mathcal{L}^1}^{q} \mathbb{1}_{M_t^1=0} \right] \\
&= \mathrm{K}_{V,\curlywedge,q} f^{\uparrow}(q,T) \ \mathbb{E}\left[\left|\left| \sum_{g=1}^{+\infty} \mathbb{E}_* \left[\int_0^t c_t^{\tau} (y - \overline{A}_t^k) \ \mathbb{P}\left(T^{2,g} = d\,\tau \right) \right] \right|\right|_{\mathcal{L}^1}^{q} \mathbb{1}_{M_t^1=0} \right] =
\end{aligned}
$$

Since all the functions involved are positive we can move the infinity sum freely in and out of the integral:

$$
\begin{aligned}
&= \mathrm{K}_{V,\curlywedge,q} f^{\uparrow}(q,T) \ \mathbb{E}\left[\left|\left| \mathbb{E}_* \left[\int_0^t c_t^{\tau} (y - \overline{A}_t^k) \underbrace{\sum_{g=1}^{+\infty} \mathbb{P}\left(T^{2,g} = d\,\tau \right)}_{\lambda\, d\tau} \right] \right|\right|_{\mathcal{L}^1}^{q} \mathbb{1}_{M_t^1=0} \right] \\
&= \mathrm{K}_{V,\curlywedge,q} f^{\uparrow}(q,T) \ \mathbb{E}\left[\left|\left| \mathbb{E}_* \left[\int_0^t c_t^{\tau} (y - \overline{A}_t^k) \ \lambda \ d\tau \right] \right|\right|_{\mathcal{L}^1}^{q} \mathbb{1}_{M_t^1=0} \right] \\
&= \mathrm{K}_{V,\curlywedge,q} f^{\uparrow}(q,T) \ \mathbb{E}\left[\left(\int \left| \mathbb{E}_* \left[\int_0^t c_t^{\tau} (y - \overline{A}_t^k) \ \lambda \ d\tau \right] \right| dy \right)^q \mathbb{1}_{M_t^1=0} \right] =
\end{aligned}
$$

Then, since c_t^{τ} is non-negative, we can leave out the absolute value and use Fubini directly:

$$
\begin{aligned}
&= \mathrm{K}_{V,\curlywedge,q} f^{\uparrow}(q,T) \ \mathbb{E}\left[\left(\mathbb{E}_* \left[\int_0^t \lambda \ \overbrace{\int c_t^{\tau} (y - \overline{A}_t^k) \ dy}^{1} \ d\tau \right] \right)^q \mathbb{1}_{M_t^1=0} \right] \\
&= \mathrm{K}_{V,\curlywedge,q} f^{\uparrow}(q,T) \ \mathbb{E}\left[(t\lambda)^q \, \mathbb{1}_{M_t^1=0} \right] = \mathrm{K}_{V,\curlywedge,q} \ f^{\uparrow}(q,T) \ (\lambda \ t)^q \ \mathbb{P}\left(\ M_t^1 = 0 \ \right) = \\
&= \mathrm{K}_{V,\curlywedge,q,\lambda} f^{\uparrow}(q,T) \ t^q \ \underbrace{e^{-t\lambda}}_{\leq 1} = \mathrm{K}_{V,\curlywedge,q,\lambda} f^{\uparrow}(q,T)
\end{aligned}
$$

The first term will require a bit more effort. Proceeding as before to calculate θ, and using Fubini and the tower property we obtain:

$$\mathrm{K}_{V,\curlywedge,q}\, f^{\uparrow}(q,T)\ \sum_{j=0}^{q}\ \mathbb{E}\left[\ (M_t^1)^j\ \mathbb{1}_{M_t^1\neq 0}\ \left|\left|\widetilde{\mathbb{E}}_*\left[\ \sum_{g=1}^{M_t^1} c_t^{T^{1,g}}(y-\overline{A}_t^k)\ \right]-\mathbb{E}_*\left[\ \sum_{g=1}^{M_t^2} c_t^{T^{2,g}}(y-\overline{A}_t^k)\ \right]\right|\right|_{\mathcal{L}^1}^{q-j}\ \right]=$$

$$=\mathrm{K}_{V,\curlywedge,q}\, f^{\uparrow}(q,T)\ \sum_{j=0}^{q}\ \mathbb{E}\Bigg[\ (M_t^1)^j\ \mathbb{1}_{M_t^1\neq 0}\cdot$$

$$\underbrace{\cdot\,\mathbb{E}\left[\ \left|\left|\sum_{g=1}^{M_t^1}\widetilde{\mathbb{E}}_*\left[\ c_t^{T^{1,g}}(y-\overline{A}_t^k)\ \right]-\lambda\int_0^t\mathbb{E}_*\left[\ c_t^{\tau}(y-\overline{A}_t^k)\ \right]d\tau\right|\right|_{\mathcal{L}^1}^{q-j}\ \middle|\ M_t^1\ \right]\ \Bigg]}_{=:\text{Ж}}=$$

Let us focus only on the term Ж. Since such term is being conditioned on the set $\{M_t^1\neq 0\}$ we are able to do this:

$$\text{Ж}=\mathbb{E}\left[\ \left|\left|\sum_{g=1}^{M_t^1}\widetilde{\mathbb{E}}_*\left[\ c_t^{T^{1,g}}(y-\overline{A}_t^k)\ \right]-\frac{M_t^1}{M_t^1}\lambda\int_0^t\mathbb{E}_*\left[\ c_t^{\tau}(y-\overline{A}_t^k)\ \right]d\tau\right|\right|_{\mathcal{L}^1}^{q-j}\ \middle|\ M_t^1\ \right]$$

$$=\mathbb{E}\left[\ \left|\left|\sum_{g=1}^{M_t^1}\left(\widetilde{\mathbb{E}}_*\left[\ c_t^{T^{1,g}}(y-\overline{A}_t^k)\ \right]-\frac{\lambda}{M_t^1}\int_0^t\mathbb{E}_*\left[\ c_t^{\tau}(y-\overline{A}_t^k)\ \right]d\tau\right)\right|\right|_{\mathcal{L}^1}^{q-j}\ \middle|\ M_t^1\ \right]$$

$$=\mathbb{E}\left[\ \left(\int\left|\sum_{g=1}^{M_t^1}\left(\widetilde{\mathbb{E}}_*\left[\ c_t^{T^{1,g}}(y-\overline{A}_t^k)\ \right]-\frac{\lambda}{M_t^1}\int_0^t\mathbb{E}_*\left[\ c_t^{\tau}(y-\overline{A}_t^k)\ \right]d\tau\right)\right|dy\right)^{q-j}\ \middle|\ M_t^1\ \right]$$

$$\leq\mathbb{E}\left[\ \left(\int\sum_{g=1}^{M_t^1}\left|\widetilde{\mathbb{E}}_*\left[\ c_t^{T^{1,g}}(y-\overline{A}_t^k)\ \right]-\frac{\lambda}{M_t^1}\int_0^t\mathbb{E}_*\left[\ c_t^{\tau}(y-\overline{A}_t^k)\ \right]d\tau\right|dy\right)^{q-j}\ \middle|\ M_t^1\ \right]$$

$$=\mathbb{E}\left[\ \left(\sum_{g=1}^{M_t^1}\underbrace{\int\left|\widetilde{\mathbb{E}}_*\left[\ c_t^{T^{1,g}}(y-\overline{A}_t^k)\ \right]\left(\frac{\int_0^t d\tau}{t}\right)-\frac{\lambda}{M_t^1}\int_0^t\mathbb{E}_*\left[\ c_t^{\tau}(y-\overline{A}_t^k)\ \right]d\tau\right|dy}\right)^{q-j}\ \middle|\ M_t^1\ \right]$$

Let us focus on each single addend:

$$
\begin{aligned}
&\int \left| \widetilde{\mathbb{E}}_* \left[\, c_t^{T^{1,g}}(y - \overline{A}_t^k) \, \right] \left(\frac{\int_0^t d\tau}{t} \right) - \frac{\lambda}{M_t^1} \int_0^t \mathbb{E}_* \left[\, c_t^\tau (y - \overline{A}_t^k) \, \right] d\tau \right| dy = \\
&= \int \left| \frac{1}{t} \int_0^t \widetilde{\mathbb{E}}_* \left[\, c_t^{T^{1,g}}(y - \overline{A}_t^k) \, \right] d\tau - \frac{\lambda}{M_t^1} \int_0^t \mathbb{E}_* \left[\, c_t^\tau (y - \overline{A}_t^k) \, \right] d\tau \right| dy \\
&= \int \left| \int_0^t \frac{1}{t} \widetilde{\mathbb{E}}_* \left[\, c_t^{T^{1,g}}(y - \overline{A}_t^k) \, \right] - \frac{\lambda}{M_t^1} \mathbb{E}_* \left[\, c_t^\tau (y - \overline{A}_t^k) \, \right] d\tau \right| dy \\
&\leq \int \int_0^t \left| \frac{1}{t} \widetilde{\mathbb{E}}_* \left[\, c_t^{T^{1,g}}(y - \overline{A}_t^k) \, \right] - \frac{\lambda}{M_t^1} \mathbb{E}_* \left[\, c_t^\tau (y - \overline{A}_t^k) \, \right] \right| d\tau \; dy \\
&= \int_0^t \int \left| \frac{1}{t} \widetilde{\mathbb{E}}_* \left[\, c_t^{T^{1,g}}(y - \overline{A}_t^k) \, \right] - \frac{\lambda}{M_t^1} \mathbb{E}_* \left[\, c_t^\tau (y - \overline{A}_t^k) \, \right] \right| dy \; d\tau \\
&\leq \int_0^t \int \left| \frac{1}{t} \widetilde{\mathbb{E}}_* \left[\, c_t^{T^{1,g}}(y - \overline{A}_t^k) \, \right] \right| + \left| \frac{\lambda}{M_t^1} \mathbb{E}_* \left[\, c_t^\tau (y - \overline{A}_t^k) \, \right] \right| dy \; d\tau =
\end{aligned}
$$

And since the two quantities are already ≥ 0 we can forget the absolute value and obtain, by the linearity of the integral and Fubini:

$$
\begin{aligned}
&= \int_0^t \int \frac{1}{t} \widetilde{\mathbb{E}}_* \left[\, c_t^{T^{1,g}}(y - \overline{A}_t^k) \, \right] dy \; d\tau + \int_0^t \int \frac{\lambda}{M_t^1} \mathbb{E}_* \left[\, c_t^\tau (y - \overline{A}_t^k) \, \right] dy \; d\tau \\
&= \int_0^t \frac{1}{t} \widetilde{\mathbb{E}}_* \left[\int c_t^{T^{1,g}}(y - \overline{A}_t^k) dy \, \right] d\tau + \int_0^t \frac{\lambda}{M_t^1} \mathbb{E}_* \left[\int c_t^\tau (y - \overline{A}_t^k) \, dy \right] d\tau \\
&= \int_0^t \frac{1}{t} d\tau + \int_0^t \frac{\lambda}{M_t^1} d\tau \; = \; 1 + \frac{t\lambda}{M_t^1} \; \leq 1 + t\lambda
\end{aligned}
$$

At this point we can write

$$
\begin{aligned}
\text{Ж} &\leq \mathbb{E} \left[\left(\sum_{g=1}^{M_t^1} \int \left| \widetilde{\mathbb{E}}_* \left[\, c_t^{T^{1,g}}(y - \overline{A}_t^k) \, \right] - \frac{\lambda}{M_t^1} \int_0^t \mathbb{E}_* \left[\, c_t^\tau (y - \overline{A}_t^k) \, \right] d\tau \right| dy \right)^{q-j} \, \middle| \, M_t^1 \right] \\
&\leq \mathbb{E} \left[\left(\sum_{g=1}^{M_t^1} (1 + t\lambda) \right)^{q-j} \, \middle| \, M_t^1 \right] \; = \; (M_t^1)^{q-j} \; (1 + t\lambda)^{q-j} \\
&\leq \; (M_t^1)^{q-j} \; (q-j) \; (q-j)! \; (t\lambda)^{q-j} = (M_t^1)^{q-j} \; \mathrm{K}_{q,\lambda} \; f^{\uparrow}(q,t)
\end{aligned}
$$

Resuming everything, we have that

$$\mathbb{E}\left[\ \left|\Phi_p^*(k,i)-\phi_p^*(k,i)\right|^q\ \right]\leq$$

$$\leq \mathrm{K}_{V,\lambda,q}\ f^{\uparrow}(q,T)\ \sum_{j=0}^{q}\ \mathbb{E}\left[\ \left|\left|\widetilde{\mathbb{E}}_*\left[\ \sum_{g=1}^{M_t^1} c_t^{T^{1,g}}(y-\overline{A}_t^k)\ \right]-\mathbb{E}_*\left[\ \sum_{g=1}^{M_t^2} c_t^{T^{2,g}}(y-\overline{A}_t^k)\ \right]\right|\right|_{\mathcal{L}^1}^{q-j}\ (M_t^1)^j\ \mathbb{1}_{M_t^1\neq 0}\ \right]+$$

$$+\ \mathrm{K}_{V,\lambda,q}\ f^{\uparrow}(q,T)\ \mathbb{E}\left[\ \left|\left|\mathbb{E}_*\left[\ \sum_{g=1}^{M_t^2} c_t^{T^{2,g}}(y-\overline{A}_t^k)\ \right]\right|\right|_{\mathcal{L}^1}^{q}\ \mathbb{1}_{M_t^1=0}\ \right]$$

$$\leq \mathrm{K}_{V,\lambda,q,\lambda} f^{\uparrow}(q,T)\ \sum_{j=0}^{q}\ \mathbb{E}\left[\ (M_t^1)^{q-j}\ \mathrm{K}_{q,\lambda}\ f^{\uparrow}(q,T)(M_t^1)^j\ \mathbb{1}_{M_t^1\neq 0}\ \right]+\mathrm{K}_{V,\lambda,q,\lambda} f^{\uparrow}(q,T)$$

$$\leq \mathrm{K}_{V,\lambda,q,\lambda}\ \left(\ f^{\uparrow}(q,T)\sum_{j=0}^{q}\ \mathbb{E}\left[\ (M_t^1)^q\ \mathbb{1}_{M_t^1\neq 0}\ \right]+f^{\uparrow}(q,T)\ \right)$$

$$\leq \mathrm{K}_{V,\lambda,q,\lambda}\ f^{\uparrow}(q,T)\left(\mathbb{E}\left[\ (M_t^1)^q\ \right]+1\right)\leq$$

Again we use the result A.2.1, together with the assumption 5.4.1 to ensure that the Poisson parameter is ≥ 1, to obtain that

$$\leq \mathrm{K}_{V,\lambda,q,\lambda}\ f^{\uparrow}(q,T)\left(f^{\uparrow}(q)\ (t\lambda)^q+1\right)\ \leq\ \mathrm{K}_{V,\lambda,q,\lambda}\ f^{\uparrow}(q,T)t^q\ (1+1)\ \leq\ \mathrm{K}_{V,\lambda,q,\lambda}\ f^{\uparrow}(q,T)\leq$$

And the proof is finally finished.

5.4. Composing All the Pieces Together

Proposition 5.4.1. *[Law of large numbers]*

Let $b(t,\cdot)$ be the generalized drift of parameter $0\leq\alpha<\frac{1}{2}$ and constant $K_\lambda>0$ and let $\overline{X}_t^{(N)}$ be the solution to the mean-field system with N amoebas, as defined in 5.1.1. Let $t\leq T$ and let $\xi>0$ be such that $0<\alpha+\xi<\frac{1}{2}$. Choose a value $0<K_\xi\leq 1-2(\alpha+\xi)$.

Then it is valid the law of the large numbers, i.e., there exists $\overline{N}=\overline{N}(\alpha,K_\lambda)>0$ such that for all $n\in\mathbb{N}$ exists an increasing positive continuous function $f_n^{\uparrow}$ (independent from N,α,ξ) so that for all $N>\overline{N}$

$$\mathbb{P}\left(\ \left|b^{(N)}(t,\overline{X}_t^{(N)})-\overline{b}^{(N)}(t,\overline{X}_t^{(N)})\right|_\infty\geq 2N^{-\xi}\ \right)\leq f_n^{\uparrow}(T)\ N^{-K_\xi n+1}\tag{5.28}$$

In particular, if $b(t,\cdot)$ is the usual drift of 2.2.2, then α can be chosen equal to 0.

Proof. Once again we make the assumptions of remark 5.2.1, i.e., that $t,\lambda\geq 1$. Recall that from 5.3 we want to use the strategy "divide et impera". For the "divide" part we use the simple principle A.2.3:

$$
\begin{aligned}
&\mathbb{P}\left(\left| b^{(N)}(t, \overline{X}_t^{(N)}) - \overline{b}^{(N)}(t, \overline{X}_t^{(N)}) \right|_\infty \geq 2N^{-\xi} \right) = \\
&= \mathbb{P}\left(\sup_{1\leq k\leq N} \left| b_k^{(N)}(t, \overline{X}_t^{(N)}) - \overline{b}_k^{(N)}(t, \overline{X}_t^{(N)}) \right|_\infty \geq 2N^{-\xi} \right) = \\
&= \mathbb{P}\Bigg(\sup_{1\leq k\leq N} \left| b_k^{(N)}(t, \overline{X}_t^{(N)}) - \mathbb{E}\left[\; b_k^{(N)}(t, \overline{X}_t^{(N)}) \;\middle|\; \overline{A}_t, \;\; (T^{p,g})_{p,g} \right] \right|_\infty + \\
&\qquad + \sup_{1\leq k\leq N} \left| \mathbb{E}\left[\; b_k^{(N)}(t, \overline{X}_t^{(N)}) \;\middle|\; \overline{A}_t, \;\; (T^{p,g})_{p,g} \right] - \mathbb{E}\left[\; b_k^{(N)}(t, \overline{X}_t^{(N)}) \;\middle|\; \overline{A}_t \right] \right|_\infty \geq N^{-\xi} + N^{-\xi} \Bigg) \\
&\leq \mathbb{P}\left(\sup_{1\leq k\leq N} \left| b_k^{(N)}(t, \overline{X}_t^{(N)}) - \mathbb{E}\left[\; b_k^{(N)}(t, \overline{X}_t^{(N)}) \;\middle|\; \overline{A}_t, \;\; (T^{p,g})_{p,g} \right] \right|_\infty \geq N^{-\xi} \right) + \\
&\qquad + \mathbb{P}\left(\sup_{1\leq k\leq N} \left| \mathbb{E}\left[\; b_k^{(N)}(t, \overline{X}_t^{(N)}) \;\middle|\; \overline{A}_t, \;\; (T^{p,g})_{p,g} \right] - \mathbb{E}\left[\; b_k^{(N)}(t, \overline{X}_t^{(N)}) \;\middle|\; \overline{A}_t \right] \right|_\infty \geq N^{-\xi} \right) \leq
\end{aligned}
\tag{5.29}
$$

For the "impera" part of the strategy we need to use the bounds 5.2.1 and 5.3.1. A bit of care is necessary regarding 5.2.1: we showed a law of large numbers for a general set of particles which had densities uniformly bounded in $\mathcal{L}^\infty(\mathbb{R}^d)$ by a constant K_{dens}

$$
||c^m||_\infty, ||a^p||_\infty < \mathrm{K}_{\text{dens}}
$$

This condition is fullfilled by the densities of the mean-field particles $\overline{X}_t^{(N)}$ on $[0,T]$ with the lemma 3.3.2:

$$
\forall t, \tau \leq T. \;\; ||c_t^\tau||_\infty, ||a_t^p||_\infty < f^\uparrow(T)
$$

Then the 5.11 gives us

$$
\forall N > \overline{N}. \qquad \mathbb{P}\left(\; S_*^{N,\xi} \; \right) \leq \mathrm{K}_n \; f_n^\uparrow(f^\uparrow(T)) \; f_n^\uparrow(t) \; N^{-\mathrm{K}_\xi n+1} \;\leq\; f_n^\uparrow(T) \; N^{-\mathrm{K}_\xi n+1} \tag{5.30}
$$

Now we can proceed with the "impera" part of the strategy applying 5.30 and 5.18 to 5.29:

$$
\leq f_n^\uparrow(T) \; N^{-\mathrm{K}_\xi n+1} \; + \mathrm{K}_n \; f_n^\uparrow(T) \; N^{-\mathrm{K}_\xi n+1} \; \leq f_n^\uparrow(T) \; N^{-\mathrm{K}_\xi n+1}
$$

And the result is proved. □

6. The Proof of the Main Result

Proposition 6.0.1. *Let $X_t^{(N)}$ be the solution to the model 2.6 and let $\overline{X}_t^{(N)}$ be the solution to the mean-field model 3.18, each with N amoebas. Let $T > 0$ and let $0 < \delta < \xi < \frac{1}{2}$. Then there exists an $\overline{N} > 0$ and an increasing continuous function $f^{\uparrow} : \mathbb{R}_+ \longrightarrow \mathbb{R}_+$ independent from $\overline{N}, N$ such that for every $N > \overline{N}$ we have that*

$$\mathbb{P}\left(\sup_{0\leq t\leq T} \left| X_t^{(N)} - \overline{X}_t^{(N)} \right|_\infty \geq N^{-\delta} \right) \leq \; f^{\uparrow}(T)\; N^{-(\xi-\delta)} \tag{6.1}$$

The proof is based on the usage of the Groenwall argument on the auxiliary process

$$J_T^N := 1 \wedge \left(N^{\delta} \sup_{0\leq t\leq T} \left| X_t^{(N)} - \overline{X}_t^{(N)} \right|_\infty \right) \tag{6.2}$$

In fact we have that

$$\begin{aligned}
\mathbb{P}\left(\sup_{0\leq t\leq T} \left| X_t^{(N)} - \overline{X}_t^{(N)} \right|_\infty \geq N^{-\delta} \right) &= \mathbb{P}\left(N^{\delta} \sup_{0\leq t\leq T} \left| X_t^{(N)} - \overline{X}_t^{(N)} \right|_\infty \geq 1 \right) \\
&= \mathbb{P}\left(\left(N^{\delta} \sup_{0\leq t\leq T} \left| X_t^{(N)} - \overline{X}_t^{(N)} \right|_\infty \right) \wedge 1 \geq 1 \right) \\
&= \mathbb{P}\left(\; J_T^N \geq 1 \; \right) \; \leq \; \mathbb{E}\left[\; J_T^N \; \right]
\end{aligned}$$

where in the latter inequality we applied Markov with $f(x) = |x|$. Then we need to find a bound for $\mathbb{E}\left[\; J_T^N \; \right]$.

Remark 6.0.1. From now on we will omit the superscript (N) for $X_t^{(N)}, \overline{X}_t^{(N)}, b^{(N)}, \overline{b}^{(N)}$ whenever possible, since the dependence on N is now clear.

6.1. Stochastic Estimate with Groenwall

Bear in mind that the functions involved are not continuous but càdlàg, thus neither the classic Groenwall argument nor the one generalised for dini derivatives [24, Theorem 1.4.1] can be applied. We will instead of that use the subsequent result from [26, Chapter 1, Theorem 8.1]:

Lemma 6.1.1. *Let $T > 0, c > 0$. Let $u(\cdot)$ be a Borel-measurable bounded non-negative function on $[0, S]$ and let $v(\cdot)$ be a non-negative integrable function in $[0, S]$. If for all $T \in [0, S]$*

$$u(T) \leq \int_0^T u(t)v(t)dt + c$$

then for all $T \in [0, S]$

$$u(T) \leq c \int_0^T \exp(v(t))dt$$

Consider $0 < t \leq T$ and calculate then:

$$\overline{D}_t^+ J_t^N = \left(\mathbb{1}_{\{\overline{D}_t^+ J_t^N > 0\}} + \mathbb{1}_{\{\overline{D}_t^+ J_t^N \leq 0\}}\right) \cdot \overline{D}_t^+ J_t^N \leq \mathbb{1}_{\{\overline{D}_t^+ J_t^N > 0\}} \ \overline{D}_t^+ \left(1 \wedge \left(N^\delta \sup_{0 \leq s \leq t} \left|X_s - \overline{X}_s\right|_\infty\right)\right) \leq$$

$$\leq \mathbb{1}_{\{\overline{D}_t^+ J_t^N > 0\}} \ \left(\overline{D}_t^+ 1 \wedge \overline{D}_t^+ \left(N^\delta \sup_{0 \leq s \leq t} \left|X_s - \overline{X}_s\right|_\infty\right)\right)$$

$$\overset{1}{\leq} \mathbb{1}_{\{\overline{D}_t^+ J_t^N > 0\}} \ \left[0 \vee \left(N^\delta d \left|b(t, X_t) - \bar{b}(t, \overline{X}_t)\right|_\infty\right)\right]$$

$$\leq \mathbb{1}_{\{\overline{D}_t^+ J_t^N > 0\}} \ N^\delta d \left|b(t, X_t) - b(t, \overline{X}_t)\right|_\infty + \mathbb{1}_{\{\overline{D}_t^+ J_t^N > 0\}} \ N^\delta d \left|b(t, \overline{X}_t) - \bar{b}(t, \overline{X}_t)\right|_\infty \leq$$

Recall that b is Lipschitz (2.2.1), therefore:

$$\leq d \ \mathrm{K}_V \ \frac{M_t}{N} \ \underbrace{\mathbb{1}_{\overline{D}_t^+ J_t^N > 0} \ \cdot N^\delta \ \left|X_t - \overline{X}_t\right|_\infty}_{} + N^\delta \ d \left|b(t, \overline{X}_t) - \bar{b}(t, \overline{X}_t)\right|_\infty \leq$$

Now observe that if $N^\delta \left|X_t - \overline{X}_t\right|_\infty \geq 1$ then $J_t^N = 1$, which is the maximum value that J_t^N can reach. This means that in such case we have $\overline{D}_t^+ J_t^N \leq 0$, because the function, in the near future either remains constant or decreases. This means that

$$\left\{\overline{D}_t^+ J_t^N > 0\right\} \subseteq \left\{N^\delta \left|X_t - \overline{X}_t\right|_\infty < 1\right\}$$

and consequently

$$\mathbb{1}_{\overline{D}_t^+ J_t^N > 0} \cdot N^\delta \ \left|X_t - \overline{X}_t\right|_\infty \leq \left(1 \vee \left(N^\delta \ \left|X_t - \overline{X}_t\right|_\infty\right)\right) \leq 1 \vee \left(N^\delta \sup_{s \in [0,t]} \left|X_s - \overline{X}_s\right|_\infty\right) = J_t^N$$

Therefore we can write

$$\overline{D}_t^+ J_t^N \leq d \ \mathrm{K}_V \ \frac{M_t}{N} \ \mathbb{1}_{\overline{D}_t^+ J_t^N > 0} \ \cdot N^\delta \ \left|X_t - \overline{X}_t\right|_\infty + N^\delta \ d \left|b(t, \overline{X}_t) - \bar{b}(t, \overline{X}_t)\right|_\infty$$

$$\leq d \ \mathrm{K}_V \ \frac{M_t}{N} \ J_t^N + N^\delta \ d \left|b(t, \overline{X}_t) - \bar{b}(t, \overline{X}_t)\right|_\infty$$

Integrating over $[0, T]$ we then obtain:

$$J_T^N - J_0^N = J_T^N = \int_0^T \overline{D}_t^+ J_t^N dt$$

$$= \int_0^T d \ \mathrm{K}_V \ \frac{M_t}{N} \ J_t^N + N^\delta \ d \left|b(t, \overline{X}_t) - \bar{b}(t, \overline{X}_t)\right|_\infty dt$$

$$\leq \int_0^T \mathrm{K}_{V,d} \ \frac{M_T}{N} \ J_t^N dt + N^\delta \ \mathrm{K}_d \int_0^T \left|b(t, \overline{X}_t) - \bar{b}(t, \overline{X}_t)\right|_\infty dt \leq$$

Choose now some $S > T$ to obtain

$$\leq \int_0^T \mathrm{K}_{V,d} \frac{M_T}{N} \ J_t^N dt + N^\delta \ \mathrm{K}_d \underbrace{\int_0^S \left|b(t, \overline{X}_t) - \bar{b}(t, \overline{X}_t)\right|_\infty dt}_{=: \mathrm{IO}}$$

[1] by 4.1.2

Notice now that if Ю is positive almost surely, then we can proceed to apply 6.1.1.

Lemma 6.1.2. $\mathbb{P}\left(\int_0^S \left|b(t,\overline{X}_t) - \overline{b}(t,\overline{X}_t)\right|_\infty dt = 0 \right) = 0$

Proof. Again we proceed similarly to 4.3.4: we just need to prove that the result holds with respect to $\mathbb{P}(\quad | \quad (T^{p,g})_{p,g}\ )$. First notice that

$$\begin{aligned}
&\mathbb{P}\left(\int_0^S \left|b(t,\overline{X}_t) - \overline{b}(t,\overline{X}_t)\right|_\infty dt = 0 \;\middle|\; (T^{p,g})_{p,g} \right) = \\
&= \mathbb{P}\left(\forall p,i.\ \int_0^S \left|b_{p;i}(t,\overline{X}_t) - \overline{b}_{p;i}(t,\overline{X}_t)\right| dt = 0 \;\middle|\; (T^{p,g})_{p,g} \right) \\
&\leq \sum_{p,i} \mathbb{P}\left(\int_0^S \left|b_{p;i}(t,\overline{X}_t) - \overline{b}_{p;i}(t,\overline{X}_t)\right| dt \;\middle|\; (T^{p,g})_{p,g} \right) \\
&= \sum_{p,i} \int \mathbb{1}_{\{0\}}\Big(\underbrace{\int_0^S \left|b_{p;i}(t,x) - \overline{b}_{p;i}(t,x)\right| dt}_{Ж_S(x,y)} \Big) \cdot \mathbb{P}\left(\overline{X}_t = dx \;\middle|\; (T^{p,g})_{p,g} \right)
\end{aligned}$$

Let us show that $\{Ж_S = 0\}$ has measure 0. Here we will write "$\forall^{\text{Leb}} t \in [0,S]$" with the meaning of "for almost every $t \in [0,S]$ with respect to the measure Leb".

$$\begin{aligned}
x \in \{Ж_S = 0\} \quad &\iff \forall^{\text{Leb}} t \in [0,S]. \quad \left|b_{p;i}(t,x) - \overline{b}_{p;i}(t,x)\right| = 0 \\
&\iff \forall^{\text{Leb}} t \in [0,S]. \quad \frac{1}{N}\sum_{m=1}^{M_t} V(x^p - x^{N+m}) - \lambda\left(V_i \star \int_0^t c_t^\tau d\tau\right)(x^p) = 0 \\
&\iff \forall^{\text{Leb}} t \in [0,S]. \quad \frac{1}{N}\sum_{m=1}^{M_t} \left(V \star \delta_{x^{N+m}}\right)(x^p) - \left(V_i \star \lambda\int_0^t c_t^\tau d\tau\right)(x^p) = 0 \\
&\iff \forall^{\text{Leb}} t \in [0,S]. \quad \left[V \star \left(\frac{1}{N}\sum_{m=1}^{M_t} \delta_{x^{N+m}} - \lambda\int_0^t c_t^\tau d\tau\right)\right](x^p) = 0 \\
&\iff \forall^{\text{Leb}} t \in [0,S]. \quad \frac{1}{N}\sum_{m=1}^{M_t} \delta_{x^{N+m}} = \lambda\int_0^t c_t^\tau d\tau
\end{aligned}$$

Now observe that if $M_t = 0$, the last equality is trivially not satisfied because its L.H.S. is 0 while its R.H.S. is positive. If instead $M_t \neq 0$ we would have a pure distribution on the L.H.S. and $\mathcal{L}^1$ function on the R.H.S. , which is of course also not attainable. Then $\{Ж_S = 0\}$ is the empty set, and the thesis is proved. □

With this lemma we have that the conditions of 6.1.1 are fullfilled for the inequality

$$J_T^N \leq \int_0^T \mathrm{K}_{V,d}\frac{M_T}{N}\ J_t^N dt + N^\delta\ \mathrm{K}_d \int_0^S \left|b(t,\overline{X}_t) - \overline{b}(t,\overline{X}_t)\right|_\infty dt$$

Therefore we obtain that

$$\begin{aligned}
J_T^N &\leq N^\delta\ \mathrm{K}_d \int_0^S \left|b(t,\overline{X}_t) - \overline{b}(t,\overline{X}_t)\right|_\infty dt \cdot \exp\left(\int_0^T \mathrm{K}_{V,d}\frac{M_T}{N} dt\right) \\
&\leq N^\delta\ \mathrm{K}_d \int_0^S \left|b(t,\overline{X}_t) - \overline{b}(t,\overline{X}_t)\right|_\infty dt \cdot \exp\left(T\mathrm{K}_{V,d}\frac{M_T}{N}\right)
\end{aligned}$$

6.2. Bound for the Average of the Estimate

Define the set

$$\mathcal{B}_t := \left\{ \left| b(t, \overline{X}_t) - \overline{b}(t, \overline{X}_t) \right|_\infty \leq 2N^{-\xi} \right\}$$

for some $0 < \xi < \frac{1}{2}$. Exploiting the definition of such set and the bounds 2.2.1 and 3.3.2 for the drift and the mesoscopic drift we obtain

$$\begin{aligned}
\left| b(t, \overline{X}_t) - \overline{b}(t, \overline{X}_t) \right|_\infty &= \left| b(t, \overline{X}_t) - \overline{b}(t, \overline{X}_t) \right|_\infty \mathbb{1}_{\mathcal{B}_t} + \left| b(t, \overline{X}_t) - \overline{b}(t, \overline{X}_t) \right|_\infty \mathbb{1}_{\mathcal{B}_t^c} \\
&\leq 2N^{-\xi} + \left(||b(t,\cdot)||_\infty + \left|\left| \overline{b}(t,\cdot) \right|\right|_\infty \right) \mathbb{1}_{\mathcal{B}_t^c} \\
&\leq 2N^{-\xi} + \left(\mathrm{K}_V \frac{M_t}{N} + \mathrm{K}_{V,\lambda} t \right) \mathbb{1}_{\mathcal{B}_t^c} \\
&\leq 2N^{-\xi} + \mathrm{K}_{V,\lambda} \left(\frac{M_t}{N} + t \right) \mathbb{1}_{\mathcal{B}_t^c} \quad \leq \quad 2N^{-\xi} + \mathrm{K}_{V,\lambda} \left(\frac{M_S}{N} + S \right) \mathbb{1}_{\mathcal{B}_t^c}
\end{aligned}$$

where the probability of $\mathcal{B}_t^c$ is described by the law of large numbers 5.4.1. Then, averaging on the inequality 6.1 we can calculate

$$\begin{aligned}
\mathbb{E}\left[J_T^N \right] &\leq \mathbb{E}\left[N^\delta \mathrm{K}_d \int_0^S \left| b(t, \overline{X}_t) - \overline{b}(t, \overline{X}_t) \right|_\infty dt \cdot \exp\left(T\mathrm{K}_{V,d} \frac{M_T}{N} \right) \right] \\
&\leq N^\delta \mathrm{K}_d \, \mathbb{E}\left[\int_0^S \left[2N^{-\xi} + \mathrm{K}_{V,\lambda} \left(\frac{M_S}{N} + S \right) \mathbb{1}_{\mathcal{B}_t^c} \right] dt \cdot \exp\left(T\mathrm{K}_{V,d} \frac{M_T}{N} \right) \right] \\
&= \overbrace{N^{-(\xi-\delta)} \, \mathrm{K}_d \, \mathbb{E}\left[\exp\left(T\mathrm{K}_{V,d} \frac{M_T}{N} \right) \right]}^{\text{Term 1}} + \overbrace{N^\delta \, \mathrm{K}_{V,\lambda,d} \mathbb{E}\left[\frac{M_S}{N} \int_0^S \mathbb{1}_{\mathcal{B}_t^c} \, dt \cdot \exp\left(T\mathrm{K}_{V,d} \frac{M_T}{N} \right) \right]}^{\text{Term 2}} \\
&\quad + \underbrace{N^\delta \, \mathrm{K}_{V,\lambda,d} S \, \mathbb{E}\left[\int_0^S \mathbb{1}_{\mathcal{B}_t^c} \, dt \cdot \exp\left(T\mathrm{K}_{V,d} \frac{M_T}{N} \right) \right]}_{\text{Term 3}}
\end{aligned} \tag{6.3}$$

Now observe: the first term has a negative power of N, which means that if the expectation value of the only random element present in it does not depend on N, then the whole term is tamed (i.e., it tends to 0 as N grows). The other two terms instead have a positive power of N, thus will need that the r.v. present in $\mathbb{E}[\ ]$, when evaluated, will yields some estimate with a sufficiently strong negative power of N to contrast N^δ. Luckily inside $\mathbb{E}[\ ]$ appears $\mathbb{1}_{\mathcal{B}_t^c}$ which, with the law of large numbers, serves exactly this purpose. Let us see this strategy in action. Use Fubini on the last two terms to get to:

$$\begin{aligned}
&= N^{\delta-\xi} \, \mathrm{K}_d \, \mathbb{E}\left[\exp\left(T\mathrm{K}_{V,d} \frac{M_T}{N} \right) \right] + N^\delta \, \mathrm{K}_{V,\lambda,d} \int_0^S \mathbb{E}\left[\frac{M_S}{N} \, \mathbb{1}_{\mathcal{B}_t^c} \, \exp\left(T\mathrm{K}_{V,d} \frac{M_T}{N} \right) \right] dt + \\
&\quad + N^\delta \, \mathrm{K}_{V,\lambda,d} S \int_0^S \mathbb{E}\left[\mathbb{1}_{\mathcal{B}_t^c} \, \exp\left(T\mathrm{K}_{V,d} \frac{M_T}{N} \right) \right] dt
\end{aligned}$$

At this point we do need to separate the stochastic variables in pieces of which are able to calculate the average. Choose $p, q, u, v \in (1, +\infty)$ such that $\frac{1}{u} + \frac{1}{v} = \frac{1}{p} + \frac{1}{q} = 1$ and apply Hoelder twice to the second addend's expectation:

$$\mathbb{E}\left[\ \frac{M_S}{N}\ \mathbb{1}_{\mathcal{B}_t^c}\ \exp\left(T\mathrm{K}_{V,d}\frac{M_T}{N}\ \right)\ \right]\ \leq \mathbb{E}\left[\ \left(\frac{M_S}{N}\ \right)^p\ \right]^{\frac{1}{p}}\cdot\mathbb{E}\left[\ (\mathbb{1}_{\mathcal{B}_t^c})^q\ \left(\exp\left(T\mathrm{K}_{V,d}\frac{M_T}{N}\ \right)\right)^q\ \right]^{\frac{1}{q}}\leq$$

$$\leq N^{-1}\mathbb{E}\left[\ (M_S)^p\ \right]^{\frac{1}{p}}\cdot\mathbb{E}\left[\ (\mathbb{1}_{\mathcal{B}_t^c})^{qu}\ \right]^{\frac{1}{qu}}\cdot\mathbb{E}\left[\ \left(\exp\left(T\mathrm{K}_{V,d}\frac{M_T}{N}\ \right)\right)^{qv}\ \right]^{\frac{1}{qv}}$$

$$= N^{-1}\mathbb{E}\left[\ (M_S)^p\ \right]^{\frac{1}{p}}\cdot\mathbb{P}\left(\ \mathcal{B}_t^c\ \right)^{\frac{1}{qu}}\cdot\mathbb{E}\left[\ \exp\left(\frac{qvT\mathrm{K}_{V,d}}{N}\ M_T\right)\ \right]^{\frac{1}{qv}}$$

and analogously, the last expectation value yields:

$$\mathbb{E}\left[\ \mathbb{1}_{\mathcal{B}_t^c}\ \exp\left(T\mathrm{K}_{V,d}\frac{M_T}{N}\ \right)\ \right]\leq\mathbb{E}\left[\ (\mathbb{1}_{\mathcal{B}_t^c})^u\ \right]^{\frac{1}{u}}\cdot\mathbb{E}\left[\ \left(\exp\left(T\mathrm{K}_{V,d}\frac{M_T}{N}\ \right)\right)^v\ \right]^{\frac{1}{v}}$$

$$=\mathbb{P}\left(\ \mathcal{B}_t^c\ \right)^{\frac{1}{u}}\cdot\mathbb{E}\left[\ \exp\left(\frac{vT\mathrm{K}_{V,d}}{N}\ M_T\right)\ \right]^{\frac{1}{v}}$$

Then we can write:

$$\begin{aligned}\mathbb{E}\left[\ J_T^N\ \right]&\leq N^{\delta-\xi}\ \mathrm{K}_d\mathbb{E}\left[\ \exp\left(T\mathrm{K}_{V,d}\frac{M_T}{N}\ \right)\ \right]\\&\quad+N^{\delta}\ \mathrm{K}_{V,\lambda,d}\int_0^S\ \mathbb{E}\left[\ \frac{M_S}{N}\ \mathbb{1}_{\mathcal{B}_t^c}\ \exp\left(T\mathrm{K}_{V,d}\frac{M_T}{N}\ \right)\ \right]\ dt+\\&\quad+N^{\delta}\ \mathrm{K}_{V,\lambda,d}S\ \int_0^S\mathbb{E}\left[\ \mathbb{1}_{\mathcal{B}_t^c}\ \exp\left(T\mathrm{K}_{V,d}\frac{M_T}{N}\ \right)\ \right]dt\\&\leq N^{\delta-\xi}\ \mathrm{K}_d\mathbb{E}\left[\ \exp\left(\frac{T\mathrm{K}_{V,d}}{N}\ M_T\right)\ \right]+\\&\quad+N^{\delta}\ \mathrm{K}_{V,\lambda,d}\int_0^S\ N^{-1}\mathbb{E}\left[\ (M_S)^p\ \right]^{\frac{1}{p}}\cdot\mathbb{P}\left(\ \mathcal{B}_t^c\ \right)\ \mathbb{E}\left[\ \exp\left(\frac{qvT\mathrm{K}_{V,d}}{N}\ M_T\right)\ \right]^{\frac{1}{qv}}dt+\\&\quad+N^{\delta}\ \mathrm{K}_{V,\lambda,d}S\ \int_0^S\mathbb{P}\left(\ \mathcal{B}_t^c\ \right)\ \mathbb{E}\left[\ \exp\left(\frac{vT\mathrm{K}_{V,d}}{N}\ M_T\right)\ \right]^{\frac{1}{v}}dt\end{aligned}\tag{6.4}$$

Now we can evaluate the probability of $\mathcal{B}_t^c$ with the law of large numbers (5.4.1): for all $t\leq S$ and for each $n\in\mathbb{N}$ there is some $\overline{N}$ such that for every $N\geq\overline{N}$ we have that

$$\mathbb{P}\left(\ \mathcal{B}_t^c\ \right)\ \leq f_n^{\uparrow}(S)\ N^{-\mathrm{K}_\xi n+1}$$

for some constant K_ξ. Regarding the higher moments of M_T, they can be evaluated with A.2.1 obtaining (assume N big enough, so that the Poisson parameter is ≥ 1 and the notation is smoother)

$$\mathbb{E}\left[\ (M_S)^p\ \right]\leq f^{\uparrow}(p)(\lambda NS)^p$$

There is only one r.v. of which we still need to evaluate the expectation value:

Lemma 6.2.1. *Consider, for $a>0$, the quantity*

$$\mathbb{E}\left[\ \exp\left(\frac{a}{N}\ M_T\right)\ \right]$$

where $M_T\sim$ Poisson(λNt). Then $\forall\epsilon>0$ there exists $\overline{N}=\overline{N}_{a,\epsilon}>0$ such that, if $N>\overline{N}$

$$\mathbb{E}\left[\ \exp\left(\frac{a}{N}\ M_T\right)\ \right] \leq e^{a\lambda T} + \epsilon\ =\ f^{\uparrow}_{a,\lambda,\epsilon}(T)$$

Proof. The expectation

$$\mathbb{E}\left[\ \exp\left(\frac{a}{N}\ M_T\right)\ \right]$$

is nothing else than an evaluation in $\frac{a}{N}$ of the moment generating function of the Poisson random variable M_t. By the result of A.2.2 such quantities are:

$$\mathbb{E}\left[\ \exp\left(\frac{a}{N}\ M_T\right)\ \right] = \exp\left[\lambda NT(e^{\frac{a}{N}} - 1)\right]$$

Then we can use De l'Hopital to see that:

$$\lim_N\ \exp\left[\lambda NT(e^{\frac{a}{N}} - 1)\right] = \exp\left[\lim_N\ \frac{e^{\frac{a}{N}} - 1}{\left(\frac{1}{\lambda NT}\right)}\right]\ = \exp\left[\lim_N\ \frac{e^{\frac{a}{N}}(-1)\frac{a}{N^2}}{\left(\frac{-1}{\lambda T N^2}\right)}\right] = \exp\left[a\lambda T e^0\right] = e^{a\lambda T}$$

This means that if we choose $\epsilon > 0$ than there exists $\overline{N} = \overline{N}_{a,\epsilon} > 0$ such that, if $N > \overline{N}$, then

$$\mathbb{E}\left[\ \exp\left(\frac{a}{N}\ M_T\right)\ \right] \leq e^{a\lambda T} + \epsilon\ =\ f^{\uparrow}_{a,\lambda,\epsilon}(T)$$

□

Fix $\epsilon > 0$. Then, continuing from 6.4, for N big enough we have that

$$\begin{aligned}
\mathbb{E}\left[\ J^N_T\ \right] &\leq N^{\delta-\xi}\ \mathrm{K}_d\mathbb{E}\left[\ \exp\left(\frac{T\mathrm{K}_{V,d}}{N}\ M_T\right)\ \right] + \\
&\quad + N^{\delta}\ \mathrm{K}_{V,\lambda,d}\int_0^S\ N^{-1}\mathbb{E}\left[\ (M_S)^p\ \right]^{\frac{1}{p}} \cdot \mathbb{P}\left(\ \mathcal{B}^c_t\ \right)\ \mathbb{E}\left[\ \exp\left(\frac{qvT\mathrm{K}_{V,d}}{N}\ M_T\right)\ \right]^{\frac{1}{qv}} dt + \\
&\quad + N^{\delta}\ \mathrm{K}_{V,\lambda,d}S\ \int_0^S \mathbb{P}\left(\ \mathcal{B}^c_t\ \right)\ \mathbb{E}\left[\ \exp\left(\frac{vT\mathrm{K}_{V,d}}{N}\ M_T\right)\ \right]^{\frac{1}{v}} dt \\
&\leq N^{\delta-\xi}\ \mathrm{K}_d(e^{\mathrm{K}_{V,d}\lambda T^2} + \epsilon) + \\
&\quad + N^{\delta}\ \mathrm{K}_{V,\lambda,d}\ \int_0^S\ N^{-1}\left[f^{\uparrow}(p)(\lambda NS)^p\right]^{\frac{1}{p}}\ f^{\uparrow}_n(S)\ N^{-\mathrm{K}_\xi n+1}(e^{qv\mathrm{K}_{V,d}\lambda T^2} + \epsilon)^{\frac{1}{qv}} dt + \\
&\quad + N^{\delta}\ \mathrm{K}_{V,\lambda,d}S\ \int_0^S f^{\uparrow}_n(S)\ N^{-\mathrm{K}_\xi n+1}(e^{v\mathrm{K}_{V,d}\lambda T^2} + \epsilon)^{\frac{1}{v}} dt =
\end{aligned}$$

Now there are no factors dependent on S anymore, thus the integral on $[0, S]$ is just a multiplication for a factor S. Besides, taking generic increasing continuous functions in t we can simplify everything to

$$
\begin{aligned}
&= N^{\delta-\xi}\ \mathrm{K}_d f^{\uparrow}_{V,d,\epsilon,\lambda}(T) + \\
&\quad + N^{\delta}\ \mathrm{K}_{V,\lambda} N^{-1} N\ \mathrm{K}_{p,\lambda}\ S\ f^{\uparrow}_{n}(S)\ N^{-\mathrm{K}_\xi n+1} f^{\uparrow}_{q,v,V,d,\epsilon,\lambda}(T)\ S + \\
&\quad + N^{\delta}\ \mathrm{K}_{V,n,\lambda}\ S\ f^{\uparrow}_{n}(T)\ N^{-\mathrm{K}_\xi n+1}\ f^{\uparrow}_{v,V,d,\epsilon,\lambda}(T)\ S \\
&\leq\ N^{\delta-\xi}\ f^{\uparrow}_{V,d,\epsilon,\lambda}(S) \\
&\quad +\ N^{\delta}\ N^{-\mathrm{K}_\xi n+1} f^{\uparrow}_{n,p,q,v,V,d,\epsilon,\lambda,\epsilon}(S) + \\
&\quad +\ N^{\delta}\ N^{-\mathrm{K}_\xi n+1}\ f^{\uparrow}_{n,v,V,d,\epsilon,\lambda}(S)\ \leq
\end{aligned}
$$

Choose now n so that $N^{\delta-\mathrm{K}_\xi n+1} < N^{\delta-\xi}$. Then

$$
\leq N^{\delta-\xi}\ \left[f^{\uparrow}_{V,d,\epsilon,\lambda}(S)\ + f^{\uparrow}_{n,p,q,v,V,d,\epsilon,\lambda}(S)\ + f^{\uparrow}_{n,v,V,d,\epsilon,\lambda}(S)\ \right] = N^{\delta-\xi}\ f^{\uparrow}_{n,p,q,v,V,d,\epsilon,\lambda}(S)
$$

Resuming we get

$$
\mathbb{E}\left[\ J^N_T\ \right] \leq N^{\delta-\xi}\ f^{\uparrow}_{n,p,q,v,V,d,\epsilon,\lambda}(S)
$$

Since the R.H.S. is $\mathcal{C}^0$ and is valid for all $S > T$ we also have that

$$
\mathbb{E}\left[\ J^N_T\ \right] \leq N^{\delta-\xi}\ f^{\uparrow}_{n,p,q,v,V,d,\epsilon,\lambda}(T)
$$

And thus the main result has been proven.

7. Generalize: the Pulsing Waves Model

Till now we treated the simplest model, where at the moment of spawning the cAMP was a lonely molecule simply roaming around of B.m. . This is though very far from reality, where the chemoattractant is released in waves expanding radially around the amoeba. In fact, marking the molecules radioactively, one is able to see clearly these cAMP circles expanding around the amoebas. In this chapter we build a generalized version of the previous simple system with the purpose of describing more accurately this phenomenon, and we sketch how one should be able to derive effective equations which are only the scaled version of the ones in 3.19.

7.1. Micromodel and Meso- and Macrodescriptions

Consider the same Poisson process $(M_t^p)_p$ and their correspondent M_t, with the respective birth times. Let $L \in \mathbb{N}$ be the (deterministic) number of cAMP molecules being spawned at each birth time $T^{p,g} = T_m$ and let $\frac{v_C}{N} \in \mathbb{R}_+$ be the modulus of their drift velocity at each of these instants. Notice that such speed is chosen to be deterministic and constant, and that the factor $\frac{1}{N}$ is the necessary coupling constant. Then, at each $T^{p,g} = T_m$ we would release the particles

$$\{C^{p,g,1}, \dots, C^{p,g,l}, \dots, C^{p,g,L}\} = \{C^{m,1}, \dots, C^{m,l}, \dots, C^{m,L}\}$$

While the dynamic equation for the N amoebas remains the same, at this point the dynamic for the cAMP changes. First, each of them will have its quasi-B.m. . Besides that, at each fixed birth time T_m the molecules $(C^{p,g,l})_{l=1}^L = (C^{m,l})_{l=1}^L$ should expand radially with respect to the position $A^p_{T_m}$. A solution to this would be that each $C^{p,g,l} = C^{m,l}$ expands in a constant random direction $\widehat{\Theta}_{p,g,l} = \widehat{\Theta}^{m,l} \sim \text{Unif}(\mathbb{S}^d)$, i.e., uniformly-distributed on the d-dimensional sphere, as in the figure 7.1.

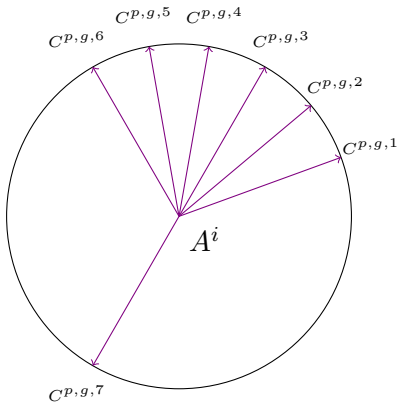


Figure 7.1.: The molecules expanding, each at a random direction, with $d = 2, L = 7$ (7 children). Notice the not uniform wave front.

This might seem to allow also configurations which are too chaotic, with a not uniform wave front (meaning that we could have $L - 1$ particles all running in approximately the

same direction and just 1 in the opposite one), contrary to what happens in nature. The solution is to fix the relative position of the drifting velocities and require that the they are, at least approximately, uniformly arranged on the unit sphere (see Fig. 7.2 for the case $d = 2$).

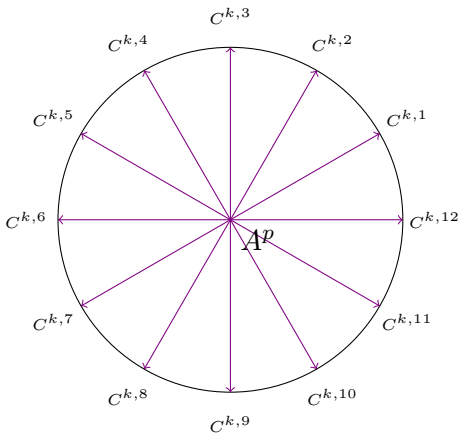


Figure 7.2.: The molecules expanding with equidistant directions on the unit sphere in dimension $d = 2, L = 10$.

The stochasticity is then given, at each birth time, by rotating randomly the whole directions all together in the same way. With $d = 2$ the solution of arranging L points evenly on the circle is trivial: just leave a point on the circle every $\frac{2\pi}{L}$ radiants. With $d = 3$ there is considerable literature about distributing L points evenly on a sphere, depending also on the definition of "uniformly arranged" (see for example [14]), but as we will see the particular arrangement does not influence the final solution, so fix a set of points on the sphere which are at least approximately evenly spaced, for the sake of reproducing nature (e.g., take inspiration from the spherical Fibonnaci points set in [28]). Let these points be

$$(e_1, \theta_2, \dots, \theta_L) \in \mathbb{S}^d$$

where e_1 is the unit vector $(1, 0, 0)$ or $(1, 0)$ depending on the dimension. For each l fix a rotation R_l such that $R_l(e_1) = \theta_l$ (then $R_1 = \mathrm{id}$) and notice then that:

$$(e_1, \theta_2, \dots, \theta_L) = (R_1, R_2, \dots, R_L)\,(e_1, e_1, \dots, e_1)$$

i.e., we can identify the arrangement with the rotation of the unit vector e_1 that are needed to find the other θ_l. Then let $\widehat{\Theta}_{p,g,1} = \widehat{\Theta}_{m,1} \sim \mathrm{Unif}(\mathbb{S}^d)$ and for each $l \neq 1$ let $\widehat{\Theta}_{p,g,l} := R_l(\widehat{\Theta}_{p,g,1})$, i.e., the whole arrangement of direction at the time $T^{p,g} = T_m$ is given by

$$(\widehat{\Theta}_{p,g,1}, \widehat{\Theta}_{p,g,2}, \dots, \widehat{\Theta}_{p,g,L}) = (R_1, R_2, \dots, R_L)\,(\widehat{\Theta}_{p,g,1}, \widehat{\Theta}_{p,g,1}, \dots, \widehat{\Theta}_{p,g,1})$$

Then the model is described as

$$\begin{cases} (M_t^p)_{p=1}^N \text{ i.i.d. and } \sim \text{Poisson}_{\lambda t} \qquad \perp\!\!\!\perp \qquad (\widehat{\Theta}_{p,g,1})_{p,g}^{N,+\infty} \quad \text{i.i.d. and } \sim \text{Unif}(\mathbb{S}^d) \\ \quad T^{p,g} := \inf\{t : M_t^p \geq g\} \qquad\qquad (\widehat{\Theta}_{p,g,l})_{p,g,l}^{N,+\infty,L} := \big(R_l(\widehat{\Theta}_{p,g,1})\big)_{p,g,l}^{N,+\infty,L} \\ \quad A_t^p = A_0^p - \displaystyle\int_0^t \frac{1}{N} \sum_{l=1}^{L} \sum_{q=1}^{N} \sum_{g=1}^{M_s^q} V(A_s^p - C_s^{q,g,l}) ds + \sigma^A B_t^p \\ C_t^{p,g,l} \quad \text{does not exist} \qquad\qquad \text{if} \quad t < T^{p,g} \\ C_t^{p,g,l} = A_{T^{p,g}}^p + \displaystyle\int_0^t \mathbb{1}_{\geq T^{p,g}}(s) \frac{\upsilon_C}{N} \widehat{\Theta}_{p,g,l} ds + \sigma_C Q_t^{p,g,l} \qquad \text{if} \quad t \geq T^{p,g} \\ \qquad = A_{T^{p,g}}^p + (t - T^{p,g}) \frac{\upsilon_C}{N} R_l(\widehat{\Theta}_{p,g,1}) ds + \sigma_C Q_t^{p,g,l} \end{cases} \tag{7.1}$$

As in the case of the basic model, we can eventually relabel the molecules according to

$$M_t := \sum_{p=1}^{N} M_t^p, \qquad T_m := \text{arrival times of } M_t, \qquad (p,g) \rightsquigarrow m$$

Similarly to the case of the model 2.6, this is a piecewise SDE under the random probability $\mathbb{P}\Big(\cdot \mid (\widehat{\Theta}_{m,l})_{m,l},\ (T_m)_m \Big)$, and thus to obtain the existence and uniqueness of solutions for fixed initial conditions (recall, the conditions of 2.2.1), we only need to verify the local lipschitzianity and the sublinear growth (both in space for a fixed time t) of the drift.

Remark 7.1.1. Now the dimension of the solution is given by:

$$D_t := N + L \cdot \sum_{p=1}^{N} M_t^p = \overbrace{N}^{\#\text{amoebas}} + \overbrace{LM_t}^{\#\text{cAMP}}$$

and the drift by:

$$b_p(s,x) = -\frac{1}{N} \sum_{l=1}^{L} \sum_{q=1}^{N} \sum_{g=1}^{M_s^q} V(x^p - x^{q,g,l})$$
$$b_{p,g,l}(s,x) = \mathbb{1}_{\geq T^{p,g}}(s) \frac{\upsilon_C}{N} \widehat{\Theta}_{p,g,l} = \mathbb{1}_{\geq T^{p,g}}(s) \frac{\upsilon_C}{N} R_l(\widehat{\Theta}_{p,g,1})$$

Lemma 7.1.1. *The function $b = b^{(N)}$ is Lipschitz in space with fixed time*

$$|b(s,x) - b(s,y)|_\infty \leq \frac{K_{V,L}}{N} M_s \ |x - y|_\infty$$

where $K_{V,L} > 0$ is a constant dependent on V and L. Furthermore $b = b^{(N)}$ is bounded

$$||b(s,\cdot)||_\infty \leq \frac{1}{N} K_{L,V,\upsilon_C} \ (M_s + 1)$$

and it grows therefore sublinearly in space with fixed time.

Proof. Let us see the Lipschitzianity first. Use the relabelling notation for the cAMP part with $(p,g) = m$. Let l be the Lipschitz constant of V. Then:

$$
\begin{aligned}
|b(s,x)-b(s,y)|_\infty &= \max\left\{ \max_{p\le N} |b_p(s,x)-b_p(s,y)|_\infty ,\right. \\
&\qquad \left. \max_{1\le m\le M_s}\max_{1\le l\le L} \overbrace{\left|\mathbb{1}_{\ge T_m}(s)\frac{v_C}{N}\widehat{\Theta}_{m,l}-\mathbb{1}_{\ge T_m}(s)\frac{v_C}{N}\widehat{\Theta}_{m,l}\right|_\infty}^{=0} \right\} \\
&= \max_{p\le N}\left|\frac{1}{N}\sum_{l=1}^{L}\sum_{q=1}^{N}\sum_{g=1}^{M_s^q} V(x^p-x^{q,g,l})-\frac{1}{N}\sum_{l=1}^{L}\sum_{q=1}^{N}\sum_{g=1}^{M_s^q} V(y^p-y^{q,g,l})\right|_\infty \\
&\le \frac{1}{N}\sum_{q=1}^{N}\sum_{l=1}^{L}\sum_{g=1}^{M_s^q} \max_{p\le N}\left|V(x^p-x^{q,g,l})-V(y^p-y^{q,g,l})\right|_\infty \\
&\le \frac{1}{N}\sum_{q=1}^{N}\sum_{l=1}^{L}\sum_{g=1}^{M_s^q} \max_{p\le N} l\left|(x^p-y^p)-(x^{q,g,l}-y^{q,g,l})\right|_\infty \\
&\le \frac{l}{N}\sum_{q=1}^{N}\sum_{l=1}^{L}\sum_{g=1}^{M_s^q} \left(\max_{p\le N}|(x^p-y^p)|_\infty+\left|(x^{q,g,l}-y^{q,g,l})\right|_\infty\right) \\
&\le \frac{l}{N}\sum_{q=1}^{N}\sum_{l=1}^{L}\sum_{g=1}^{M_s^q} (|x-y|_\infty+|x-y|_\infty) \le \frac{lL2}{N}\sum_{q=1}^{N} M_s^q\ |x-y|_\infty \\
&= \frac{\mathrm{K}_{V,L}}{N} M_s\ |x-y|_\infty
\end{aligned}
$$

Regarding the bound we have that

$$
\begin{aligned}
||b(s,\cdot)||_\infty &= \sup_x\left\{\max_{p\le N}|b_p(s,x)|_\infty, \max_{1\le m\le M_s}\max_{1\le l\le L}\left|\mathbb{1}_{\ge T_m}(s)\frac{v_C}{N}\widehat{\Theta}_{m,l}\right|_\infty\right\} \\
&\le \max\left\{\sup_x\max_{p\le N}\left|\frac{1}{N}\sum_{l=1}^{L}\sum_{q=1}^{N}\sum_{g=1}^{M_s^q} V(x^p-x^{q,g,l})\right|_\infty, \frac{v_C}{N}\right\} \\
&\le \max\left\{\frac{1}{N}\sum_{l=1}^{L}\sum_{q=1}^{N}\sum_{g=1}^{M_s^q}\sup_x\left|V(x^p-x^{q,g,l})\right|, \frac{v_C}{N}\right\} \\
&\le \frac{1}{N}\left(LM_s\,||V||_\infty+v_C\right) \le \frac{1}{N}\mathrm{K}_{L,V}\ (M_s+v_C) \le \frac{1}{N}\mathrm{K}_{L,V,v_C}\ (M_s+1)
\end{aligned}
$$

□

Remark 7.1.2. Notice that now the system 2.6 is just a specific case of the system 7.1 with $L=1$ and cAMP drift $v_C=0$.

7.1.1. Mean-field particles candidate

The mean-field particles candidates are derived similarly to the case of the basic model. Let us start with the amoebas' part:

$$
\begin{aligned}
\overline{b}_p(t,x) :&= \mathbb{E}\left[\ b_p(t,\overline{X}_t)\ \middle|\ \overline{X}_t^p\ \right](x) \ = \mathbb{E}\left[\ \ \mathbb{E}\left[\ b_p(t,\overline{X}_t)\ \middle|\ \overline{A}_t^p = x,\ (T^{p,g})_{p,g}\ \right]\ \middle|\ \overline{A}_t^p = x\ \right] \\
&= -\mathbb{E}\left[\ \ \mathbb{E}\left[\ \frac{1}{N}\sum_{l=1}^{L}\sum_{q=1}^{N}\sum_{g=1}^{M_t^q} V(\overline{A}_t^p - \overline{C}_t^{q,g,l})\ \middle|\ \overline{A}_t^p = x,\ (T^{p,g})_{p,g}\right]\ \middle|\ \overline{A}_t^p = x\right] \\
&= -\frac{1}{N}\sum_{q=1}^{N}\mathbb{E}\left[\ \sum_{l=1}^{L}\sum_{g=1}^{M_t^p}\mathbb{E}\left[\ V(x-\overline{C}_t^{q,g,l})\ \middle|\ (T^{p,g})_{p,g}\ \right]\ \right] \\
&= -\frac{1}{N}\sum_{q=1}^{N}\mathbb{E}\left[\ \sum_{l=1}^{L}\sum_{g=1}^{M_t^q}\left(V\star c_t^{T^{q,g},l}\right)(x)\ \right] =
\end{aligned}
$$

where $c_t^{T^{q,g},l}$ is the density of the mean-field cAMP particle $\overline{C}_t^{q,g,l}$ released at the birth time $T^{q,g}$. Now recall that $M^q \sim M^1$ and thus also $T^{q,g} \sim T^{1,g}$, therefore

$$
\begin{aligned}
&= -\frac{1}{N}\sum_{q=1}^{N}\mathbb{E}\left[\ \sum_{l=1}^{L}\sum_{g=1}^{M_t^1}\left(V\star c_t^{T^{1,g},l}\right)(x)\ \right] = -\sum_{l=1}^{L}\mathbb{E}\left[\ \sum_{g=1}^{M_t^1}\left(V\star c_t^{T^{1,g},l}\right)(x)\ \right] \\
&= -\sum_{l=1}^{L}\mathbb{E}\left[\ \sum_{g=1}^{\infty}\underbrace{\overbrace{\mathbb{1}_{T^{1,g}\le t}}^{\mathbb{1}_{g\le M_t^1}}\left(V\star c_t^{T^{1,g},l}\right)(x)}_{\text{(coordinatewise)}\ge 0}\ \right] = -\sum_{l=1}^{L}\sum_{g=1}^{\infty}\mathbb{E}\left[\ \mathbb{1}_{T^{1,g}\le t}\left(V\star c_t^{T^{1,g},l}\right)(x)\ \right] \\
&= -\sum_{l=1}^{L}\sum_{g=1}^{\infty}\int \mathbb{1}_{\tau\le t}\left(V\star c_t^{\tau,l}\right)(x)d\underbrace{\mathbb{P}(T^{1,g}=\tau)}_{\text{Gamma}_{\lambda,g}} \\
&= -\sum_{l=1}^{L}\sum_{g=1}^{\infty}\int_0^t \left(V\star c_t^{\tau,l}\right)(x)\frac{\lambda^g}{(g-1)!}x^{g-1}\ e^{-\lambda x}d\tau \\
&= -\sum_{l=1}^{L}\int_0^t \left(V\star c_t^{\tau}\right)(x)\lambda\underbrace{\sum_{g=1}^{\infty}\frac{(\lambda x)^{g-1}}{(g-1)!}}_{e^{\lambda x}}e^{-\lambda x}d\tau \\
&= -\sum_{l=1}^{L}\lambda\int_0^t\left(V\star c_t^{\tau,l}\right)(x)d\tau = -\left(V\star\int_0^t\sum_{l=1}^{L}\lambda c_t^{\tau,l}d\tau\right)(x) \quad = \overline{b}_p(t,x) \equiv \overline{b}_p(t,x)
\end{aligned}
$$

For the cAMP's part we can obtain quickly a result with these two observations:

- Since $\widehat{\Theta}_{p,g,1}$ is uniformly distributed on the sphere $\mathbb{S}^d$, which has a central symmetry around its center $0 \ni \mathbb{R}^d$ (i.e., the sphere is invariant under the map $x \longmapsto -x$), we have that $\mathbb{E}\left[\ \widehat{\Theta}_{p,g,1}\ \right] = (\text{center of mass of } \mathbb{S}^d) = 0$.

- Since $\widehat{\Theta}_{p,g,1}$ is uniformly distributed it is easy to prove that since every rotation R_l is deterministic we have that

$$
\mathbb{E}\left[\ \widehat{\Theta}_{p,g,l}\ \right] = \mathbb{E}\left[\ R_l(\widehat{\Theta}_{p,g,1})\ \right] = R_l\left(\mathbb{E}\left[\ \widehat{\Theta}_{p,g,1}\ \right]\right) = R_l(0) = 0
$$

Recall that the random direction/angle $\widehat{\Theta}_{p,g,1}$ and the birth time $T^{p,g}$ are independent by hypothesis. Then we have that

$$\overline{b}_{p,g,l}(t,x) = \mathbb{E}\left[\, b_{p,g,l}(t,x)\,\right] = \mathbb{E}\left[\, \mathbb{1}_{\geq T^{p,g}}(t)\frac{v_C}{N}\widehat{\Theta}_{p,g,l}\,\right]$$

$$= \mathbb{E}\left[\, \mathbb{1}_{\geq T^{p,g}}(t)\frac{v_C}{N}R_l(\widehat{\Theta}_{p,g,1})\,\right] = \frac{v_C}{N}\mathbb{E}\left[\, \mathbb{1}_{\geq T^{p,g}}(t)\,\right] \overbrace{\mathbb{E}\left[\, R_l(\widehat{\Theta}_{p,g,1})\,\right]}^{=0} = 0 \equiv: \overline{b}_{p,g}(t,x)$$

Then the particles $\left(C_t^{p,g,l}\right)_{l=1}^{L}$ are i.i.d. , and therefore we can observe that $c^{T^{p,g},l} \equiv: c^{T^{p,g}}$ are independent from l. Then we can further simplify the expression of the mean-field for the amoeba's part:

$$\overline{b}_p(t,x) = -L\ \lambda \int_0^t \left(V \star c_t^\tau\right)(x)d\tau = -\left(V \star \int_0^t L\lambda c_t^\tau d\tau\right)(x)$$

Then we have that the mean-field of the pulsing wave model is exactly the mean-drift of the basic model (see 3.18) but scaled by a factor L. This is true not only for the amoebas' part, but also for the cAMP part, because $L \cdot 0 = 0$ (with the difference in dimension between the two models, because with the pulsing wave one we have L times more cAMP molecules). Then we can write exactly the same models that we set for the basic model at 3.19, 3.18 and 3.20, but scaled of a factor L and with one more index l for the cAMP part:

$$\begin{cases} \overline{A}_t^0 \sim a_0 \\ \overline{A}_t^p = -\displaystyle\int_0^t \left(V \star \int_0^s L\lambda c_s^\tau d\tau\right)(\overline{A}_s^p)ds + \sigma_A B_t^p \\ \overline{C}_t^{p,g,l} = \overline{A}_{T^{p,g}}^p + \sigma_C Q_t^{p,g} \qquad \text{from time } T^{p,g} \end{cases} \tag{7.2}$$

and the respective effective equations are given by

$$\begin{cases} a(0,\cdot) = a_0 \\ \quad \partial_t a = \nabla\left[\left(V \star \displaystyle\int_0^t L\lambda c_t^\tau d\tau\right)a\right] + \frac{d\sigma_A^2}{2} \triangle a \\ c^\tau(t,\cdot) \equiv 0 \qquad \text{if } t < \tau \\ c^\tau(\tau,\cdot) = a(\tau,\cdot) \\ \quad \partial_t c^\tau = \dfrac{d\sigma_C^2}{2} \triangle c^\tau \qquad \text{if } t \geq \tau \end{cases} \tag{7.3}$$

or, expressing the density of cAMP molecules analytically (with the heat kernel, exactly as in 3.15) we have

$$\begin{cases} a(0,\cdot) = a_0 \\ \quad \partial_t a = L\lambda\nabla\left[\left(\displaystyle\int_0^t V \star \Phi_{t-\tau}^{\frac{d\sigma_C^2}{2}} \star a(\tau,\cdot)d\tau\right)a\right] + \dfrac{d\sigma_A^2}{2} \triangle a \end{cases} \tag{7.4}$$

We make exactly the same assumption as 3.3.2 on the densities a_t, c_t. It is easy to show that then the existence and uniqueness of solutions for the mean-field particles can be proven in the same way as for the basic model, and that we have a similar bound for the mean-field drift:

Lemma 7.1.2. *The mean-field drift $\overline{b}(t,x)$ defined in 3.18 coordinate-wise as*

$$\overline{b}_p(t,x) = -\left(V \star \int_0^t L\lambda c_t^\tau d\tau\right)(x)$$

is Lipschitz and bounded in space with bound

$$\left|\left|\overline{b}(t,x)\right|\right|_\infty \leq K_{V,\lambda,L}t$$

for a constant $K_{V,\lambda,L} > 0$.

Remark 7.1.3. Notice how the macroscopic description is completely independent from the rotations R_l, from the angles $(\widehat{\Theta}^{m,l})$ and from the constant v_C. This means that the only determinant parameter for the macroscopic behaviour is given by L, the number of molecules released at each birth time.

7.1.2. Convergence Condition

Notice that since also in the case of the pulsing waves model the amoebas $A_t^{(N)} = (A_t^1, \ldots, A_t^N)$ are exchangeable, the condition

$$\mu_t^{(A),N} \xrightarrow{N\longrightarrow+\infty} a(t,\cdot)$$

can be proven with the proposition of Sznitman 3.2.1, as for the basic model. Analogously the lemma 3.2.1 works smoothly as before, and so 3.9. Then, to obtain the convergence condition above, it is enough to prove that

$$\mathbb{P}\left(\sup_{0\leq t\leq T}\left|X_t - \overline{X}_t\right|_\infty \geq N^{-\delta}\right) \leq \; f^{\uparrow}(T)\; N^{-(\xi-\delta)} \tag{7.5}$$

7.2. Preliminaries for the Final Proof

7.2.1. The Dini Derivation

The paths of the distance process

$$\left|X_t - \overline{X}_t\right|_\infty$$

for the pulsing wave model solution X_t and its mean-field counterpart $\overline{X}_t$ are $\mathcal{C}^0$ and thus also càdlàg, therefore the lemma 4.2.1 remains valid:

$$\overline{D}_T^+ \sup_{0\leq t\leq T}\left|X_t - \overline{X}_t\right|_\infty \leq \max_{s\in\{0,T\}}\left\{\overline{D}_s^+\left|X_s - \overline{X}_s\right|_\infty\right\} \tag{7.6}$$

We have though to be slightly more careful regarding the lemma 4.3.1. By signing with $b^{(A)}, \overline{b}^{(A)}, B^{(A)}$ the amoeba's part of the drift, the mean-field drift and the B.m. respectively we obtain:

$$
\begin{aligned}
\left|X_t - \overline{X}_t\right|_\infty &= \max\left\{\left|A_t^{(N)} - \overline{A}_t^{(N)}\right|_\infty, \left|C_t^{(N)} - \overline{C}_t^{(N)}\right|_\infty\right\} \\
&= \max\left\{ \left| A_0^{(N)} + \int_0^t b^{(A)}(s, X_s)ds + \sigma B_t^{(A)} - \left(\overline{A}_0^{(N)} + \int_0^t \overline{b}^{(A)}(s, \overline{X}_s^N)ds + \sigma B_t^{(A)}\right)\right|_\infty, \right. \\
&\qquad \left. \max_{1\le m\le M_t}\max_{1\le l\le L}\left|A_{T_m}^{P(m)} + (t-T^m)\frac{\upsilon_C}{N}R_l(\widehat{\Theta}_{m,1}) + Q_t^m - (\overline{A}_{T_m}^{P(m)} + Q_t^m)\right|_\infty \right\} \\
&= \max\left\{ \left| \int_0^t b^{(A)}(s, X_s) - \overline{b}^{(A)}(s, \overline{X}_s)ds\right|_\infty, \right. \\
&\qquad \left. \max_{1\le m\le M_t}\max_{1\le l\le L}\left|A_{T_m}^{P(m)} - \overline{A}_{T_m}^{P(m)} + (t-T^m)\frac{\upsilon_C}{N}R_l(\widehat{\Theta}_{m,1})\right|_\infty \right\}
\end{aligned}
$$

Let us define

$$
\begin{aligned}
[A] &:= \left\{\left|\int_0^t b^{(A)}(s, X_s) - \overline{b}^{(A)}(s, \overline{X}_s)ds\right|_\infty = \left|X_t - \overline{X}_t\right|_\infty\right\} \\
[C] &:= \left\{\max_{1\le l\le L}\left|A_{T_m}^{P(m)} - \overline{A}_{T_m}^{P(m)} + (t-T^m)\frac{\upsilon_C}{N}R_l(\widehat{\Theta}_{m,1})\right|_\infty = \left|X_t - \overline{X}_t\right|_\infty\right\}
\end{aligned}
$$

Then by recalling the form of the derivative of the infinity norm 4.3.2 we can obtain that

$$
\begin{aligned}
\overline{D}_t^+ \left|X_t - \overline{X}_t\right|_\infty = \mathbb{1}_{[A]}\ \overline{D}_t^+ &\left|\int_0^t b^{(A)}(s, X_s) - \overline{b}^{(A)}(s, \overline{X}_s)ds\right|_\infty + \\
&+ \mathbb{1}_{[C]}\ \overline{D}_t^+ \max_{1\le m\le M_t}\max_{1\le l\le L}\left|A_{T_m}^{P(m)} - \overline{A}_{T_m}^{P(m)} + (t-T^m)\frac{\upsilon_C}{N}R_l(\widehat{\Theta}_{m,1})\right|_\infty
\end{aligned}
$$

It is quite easy to prove that the amoebas' part yields the same result as in the basic model case:

$$
\overline{D}_t^+\left|\int_0^t b^{(A)}(s, X_s) - \overline{b}^{(A)}(s, \overline{X}_s)ds\right|_\infty \le d\left|b^{(A)}(t, X_t) - \overline{b}^{(A)}(t, \overline{X}_t)\right|_\infty
$$

This because the only difference between this model and the previous one for the amoebas is the finite sum $\sum_{l=1}^L$ and thus an analogous form of lemma 4.3.4 for this distance is valid as well. We need though to take care also of the cAMP part this time. Let

$$
\begin{aligned}
I_t &:= \left[A_{T_m}^{P(m);i} - \overline{A}_{T_m}^{P(m);i} + (t-T^m)\frac{\upsilon_C}{N}\pi_i\left(R_l(\widehat{\Theta}_{m,1})\right)\right]_{m,l} && \in (\mathbb{R}^d)^{M_t+L} \\
I_t^{m,l;i} &:= A_{T_m}^{P(m);i} - \overline{A}_{T_m}^{P(m);i} + (t-T^m)\frac{\upsilon_C}{N}\pi_i\left(R_l(\widehat{\Theta}_{m,1})\right) && \in \mathbb{R}
\end{aligned}
$$

Then by the derivative of the infinity norm 4.3.2 we have that

$$\overline{D}_t^+ \max_{1\le m\le M_t}\max_{1\le l\le L}\left|A_{T_m}^{P(m)} - \overline{A}_{T_m}^{P(m)} + (t-T^m)\frac{\upsilon_C}{N}R_l(\widehat{\Theta}_{m,1})\right|_\infty =$$
$$= \sum_{1\le i\le d}\sum_{1\le m\le M_t}\sum_{1\le l\le L} \mathbb{1}_{\left|I_t^{m,l;i}\right|=|I_t|_\infty}\ \overline{D}_t^+ I_{m,l;i}$$
$$= \sum_{1\le i\le d}\sum_{1\le m\le M_t}\sum_{1\le l\le L} \mathbb{1}_{\left|I_t^{m,l;i}\right|=|I_t|_\infty}\ \overline{D}_t^+ \left[A_{T_m}^{P(m);i} - \overline{A}_{T_m}^{P(m);i} + (t-T^m)\frac{\upsilon_C}{N}\pi_i\left(R_l(\widehat{\Theta}_{m,1})\right)\right] =$$

Notice, as in the basic model case, that if $t \neq T_m$ the term $A_{T_m}^{P(m);i} - \overline{A}_{T_m}^{P(m);i}$ is independent from t and it is thus erased with the derivation. As for the basic model case, since $t = T_m$ happens with probability 0, we simply remove from the space Ω the correspondent paths. Then we can write:

$$= -\sum_{1\le i\le d}\sum_{1\le m\le M_t}\sum_{1\le l\le L} \mathbb{1}_{\left|I_t^{m,l;i}\right|=|I_t|_\infty}\ \frac{\upsilon_C}{N}\pi_i\left(R_l(\widehat{\Theta}_{m,1})\right)$$

Again, as for the amoebas' part in the lemma 4.3.4, we want to get rid of $\sum_{1\le m\le M_t}$, namely to show that only one addend for such sum survives i.e.,

Lemma 7.2.1. *For all $t > 0$ and for all $m \neq n$ we have that*

$$\mathbb{P}\left(\ \exists i,j,l,k.\ I_t^{m,l;i} = I_t^{n,k;j}\ \right) = 0$$

We sketch only the proof of such result, because we expect it to work.

Sketch of a possible proof. All the variables involved are continuous, thus their laws are non-atomic. We expect that the condition $I_t^{m,l;i} = I_t^{n,k;j}$ requires that some quantities involved are the same on manifolds of dimension inferior to the whole configuration space, yielding the result. We can reduce the complexity of the problem conditioning the probability with respect to the angles $\widehat{\Theta}_{m,1}, \widehat{\Theta}_{n,1}$. □

Then we can resume the whole proof as

$$\begin{aligned}
\overline{D}_t^+ \left|X_t - \overline{X}_t\right|_\infty &= \mathbb{1}_{[A]}\ \overline{D}_t^+\ \left|\int_0^t b^{(A)}(s,X_s) - \overline{b}^{(A)}(s,\overline{X}_s)ds\right|_\infty + \\
&\quad + \mathbb{1}_{[C]}\ \overline{D}_t^+\ \max_{1\le m\le M_t}\max_{1\le l\le L}\left|A_{T_m}^{P(m)} - \overline{A}_{T_m}^{P(m)} + (t-T^m)\frac{\upsilon_C}{N}R_l(\widehat{\Theta}_{m,1})\right|_\infty \\
&\le d\left|b^{(A)}(t,X_t) - \overline{b}^{(A)}(t,\overline{X}_t)\right|_\infty - \sum_{1\le i\le d}\sum_{1\le l\le L}\ \frac{\upsilon_C}{N}\pi_i\left(R_l(\widehat{\Theta}_{m,1})\right) \\
&\le d\left|b^{(A)}(t,X_t) - \overline{b}^{(A)}(t,\overline{X}_t)\right|_\infty + \sum_{1\le i\le d}\sum_{1\le l\le L}\ \max_{m,l;i}\left|\frac{\upsilon_C}{N}\pi_i\left(R_l(\widehat{\Theta}_{m,1})\right)\right| \\
&= d\left|b^{(A)}(t,X_t) - \overline{b}^{(A)}(t,\overline{X}_t)\right|_\infty + dL\ \left|b^{(C)}(t,X_t) - \overline{b}^{(C)}(t,\overline{X}_t)\right|_\infty \\
&\le \mathrm{K}_{d,L}\left|b(t,X_t) - \overline{b}(t,\overline{X}_t)\right|_\infty
\end{aligned}$$

where in the last passages $b^{(C)}, \overline{b}^{(C)}$ are the correspondent drifts for the cAMP molecules.

7.2.2. Sketch of Proof for the Law of Large Numbers

Again V could be easily here generalized as in 5.4.1, but for the sake of clarity we stick to the more classic attraction.

Proposition 7.2.2 (Law of large numbers for the pulsing wave model)**.** *Consider the microscopic drift $b = b^{(N)}$ and the particles $X_t = X_t^{(N)}$ and the correspondent mean-field drift and particles $\overline{b} = \overline{b}^{(N)}, \overline{X}_t = \overline{X}_t^{(N)}$. Let $t \leq T$. Then there exists $\overline{N}, K_\xi > 0$ such that for all $n \in \mathbb{N}$ there exists an increasing positive continuous function $f_{n,L}^{\uparrow}(t)$ in t (independent from N, ξ) so that for all $N > \overline{N}$ we have that*

$$\mathbb{P}\left(\left| b^{(N)}(t, X_t^{(N)}) - \overline{b}^{(N)}(t, \overline{X}_t^{(N)}) \right|_\infty \geq (1+2L)N^{-\xi} \right) \leq f_{n,L}^{\uparrow}(T) N^{-K_\xi n + 1}$$

Sketch of proof. Notice that

$$\begin{aligned}
(1+L)N^{-\xi} &\leq \left| b^{(N)}(t, X_t^{(N)}) - \overline{b}^{(N)}(t, \overline{X}_t^{(N)}) \right|_\infty \\
&= \max\left\{ \left| \left(\frac{1}{N} \sum_{l=1}^{L} \sum_{q=1}^{N} \sum_{g=1}^{M_t^q} V(\overline{A}_t^p - \overline{C}_t^{q,g,l}) - \left(V \star \int_0^t L\lambda c_t^\tau \right)(\overline{A}_t^p) \right)_p \right|_\infty, \right. \\
&\qquad \left. \left| \left(\mathbb{1}_{\geq T^{p,g}}(t) \frac{v_C}{N} \widehat{\Theta}_{p,g,l} \right)_{p,g,l} \right|_\infty \right\} \\
&\leq \sum_{l=1}^{L} \left| \left(\frac{1}{N} \sum_{q=1}^{N} \sum_{g=1}^{M_t^q} V(\overline{A}_t^p - \overline{C}_t^{q,g,l}) - \left(V \star \int_0^t \lambda c_t^\tau \right)(\overline{A}_t^p) \right)_p \right|_\infty + \frac{v_C}{N}
\end{aligned}$$

With these consideration in mind and using the trivial principle A.2.3 we can then write

$$\begin{aligned}
&\mathbb{P}\left(\left| b^{(N)}(t, X_t^{(N)}) - \overline{b}^{(N)}(t, \overline{X}_t^{(N)}) \right|_\infty \geq (1+2L)N^{-\xi} \right) \leq \\
&\leq \mathbb{P}\left(\sum_{l=1}^{L} \left| \left(\frac{1}{N} \sum_{q=1}^{N} \sum_{g=1}^{M_t^q} V(\overline{A}_t^p - \overline{C}_t^{q,g,l}) - \left(V \star \int_0^t \lambda c_t^\tau \right)(\overline{A}_t^p) \right)_p \right|_\infty + \frac{v_C}{N} \geq (1+2L)N^{-\xi} \right) \\
&\leq \mathbb{P}\left(\sum_{l=1}^{L} \left| \left(\frac{1}{N} \sum_{q=1}^{N} \sum_{g=1}^{M_t^q} V(\overline{A}_t^p - \overline{C}_t^{q,g,l}) - \left(V \star \int_0^t \lambda c_t^\tau \right)(\overline{A}_t^p) \right)_p \right|_\infty \geq 2LN^{-\xi} \right) + \mathbb{P}\left(\frac{v_C}{N} \geq N^{-\xi} \right) \leq
\end{aligned}$$

Now focus on the cAMP part and see that, since $\xi < \frac{1}{2}$, for N big enough we have that

$$\mathbb{P}\left(\frac{v_C}{N} \geq N^{-\xi} \right) = \mathbb{P}\left(v_C \geq N^{1-\xi} \right) = 0$$

and therefore, again by A.2.3, we can write that

$$\begin{aligned}
&\leq \mathbb{P}\left(\sum_{l=1}^{L} \left| \left(\frac{1}{N} \sum_{q=1}^{N} \sum_{g=1}^{M_t^q} V(\overline{A}_t^p - \overline{C}_t^{q,g,l}) - \left(V \star \int_0^t \lambda c_t^\tau \right)(\overline{A}_t^p) \right)_p \right|_\infty \geq 2LN^{-\xi} \right) \\
&\leq \sum_{l=1}^{L} \mathbb{P}\left(\left| \left(\frac{1}{N} \sum_{q=1}^{N} \sum_{g=1}^{M_t^q} V(\overline{A}_t^p - \overline{C}_t^{q,g,l}) - \left(V \star \int_0^t \lambda c_t^\tau \right)(\overline{A}_t^p) \right)_p \right|_\infty \geq 2N^{-\xi} \right) \leq
\end{aligned}$$

The addends have now the same shape as in 5.4.1. Not only, the random variables involved have densities respecting the exact same conditions by hypothesis and their drifts

for the amoebas are simply scaled by a factor L with respect to the basic model. Thus, even though the statement 5.4.1 cannot be directly used to obtain a bound for such addends (in 5.4.1 by hypothesis we require that $\overline{X}_t$ are the solution specifically to the mean-field basic model), the proof can be followed exactly as in the basic model case but scaled of L, obtaining

$$\leq \sum_{l=1}^{L} \mathrm{K}_n f_n^{\uparrow}(T) N^{-\mathrm{K}\xi n+1} = f_{n,L}^{\uparrow}(T) N^{-\mathrm{K}\xi n+1}$$

□

7.3. Final Proof

We expect the condition

$$\mathbb{P}\left(\sup_{0\leq t\leq T} \left| X_t^{(N)} - \overline{X}_t^{(N)} \right|_\infty \geq N^{-\delta} \right) \leq \ f^{\uparrow}(T)\ N^{-(\xi-\delta)} \tag{7.7}$$

to be provable exactly as in chapter 6, even with the same symbol, with the only difference being that the function $f^{\uparrow}(T)$ is dependent also on L. This because the lemmas for the Dini derivation still have the same form, and the law of large numbers has only a scaling for some constant, and we still have bounds for both the drift and the mean-field drift, thus all the proof concept holds on the exact same principles.

Stochastic Estimate with Groenwall

Define

$$J_T^N := 1 \wedge \left(N^{\delta} \sup_{0\leq t\leq T} \left| X_t^{(N)} - \overline{X}_t^{(N)} \right|_\infty \right) \tag{7.8}$$

Exactly as before, we have that

$$\mathbb{P}\left(\sup_{0\leq t\leq T} \left| X_t^{(N)} - \overline{X}_t^{(N)} \right|_\infty \geq N^{-\delta} \right) \leq \ \ldots \ \leq \ \mathbb{E}\left[\ J_T^N\ \right]$$

Again, we will omit the superscript (N). Consider $0 < t \leq T$ and have then, as previously:

$$\overline{D}_t^+ J_t^N \leq \mathbb{1}_{\{\overline{D}_t^+ J_t^N > 0\}}\ N^{\delta} d \left| b(t, X_t) - b(t, \overline{X}_t) \right|_\infty + \mathbb{1}_{\{\overline{D}_t^+ J_t^N > 0\}}\ N^{\delta} d \left| b(t, \overline{X}_t) - \bar{b}(t, \overline{X}_t) \right|_\infty \leq$$

Again we exploit the lipschitzianity of b (7.1.1) but simply with a different constant with respect to the basic model case:

$$\leq d\ \frac{\mathrm{K}_{V,L}}{N} M_t \mathbb{1}_{\overline{D}_t^+ J_t^N > 0}\ \cdot N^{\delta}\ \left| X_t - \overline{X}_t \right|_\infty + N^{\delta}\ d \left| b(t, \overline{X}_t) - \bar{b}(t, \overline{X}_t) \right|_\infty \leq$$

As previously, we can infer that $\mathbb{1}_{\overline{D}_t^+ J_t^N > 0}\ \cdot N^{\delta}\ \left| X_t - \overline{X}_t \right|_\infty \leq J_t^N$ and therefore

$$\overline{D}_t^+ J_t^N \leq \mathrm{K}_{V,L,d} \frac{M_t}{N} \; J_t^N + N^\delta \; \mathrm{K}_d \left| b(t, \overline{X}_t) - \overline{b}(t, \overline{X}_t) \right|_\infty$$

Integrating over $[0, T]$ and choosing some $S > T$ we then obtain:

$$J_T^N = J_T^N - J_0^N \leq \int_0^T \mathrm{K}_{V,L,d} \; \frac{M_T}{N} \; J_t^N dt + N^\delta \; \mathrm{K}_d \underbrace{\int_0^S \left| b(t, \overline{X}_t) - \overline{b}(t, \overline{X}_t) \right|_\infty dt}_{=: \text{IO}}$$

We should again prove that the term IO is different from 0 with probability 1, but this should hold since $\overline{b}$ is simply L scaled with respect to the basic model case, and b is the sum of L processes distributed very similarly to the basic model case. We assume this holds, and by applying the Groenwall lemma in his form 6.1.1 we obtain:

$$J_T^N \leq N^\delta \; \mathrm{K}_d \int_0^S \left| b(t, \overline{X}_t) - \overline{b}(t, \overline{X}_t) \right|_\infty dt \cdot \exp\left(T \mathrm{K}_{V,L,d} \; \frac{M_T}{N} \right) \tag{7.9}$$

Bound for the Average of the Estimate

Again we can now use the law of large numbers and the bounds on the drift b and mean-field drift $\overline{b}$, which are essentially the same bounds as before, but with different constants. Define the set

$$\mathcal{B}_t := \left\{ \left| b(t, \overline{X}_t) - \overline{b}(t, \overline{X}_t) \right|_\infty \leq 2N^{-\xi} \right\}$$

for some $0 < \xi < \frac{1}{2}$. Then, for N big enough, by the law of large numbers 7.2.2 and exploiting the bounds from the lemmas 7.1.1 and 7.1.2 we have that

$$\begin{aligned}
\left| b(t, \overline{X}_t) - \overline{b}(t, \overline{X}_t) \right|_\infty &= \left| b(t, \overline{X}_t) - \overline{b}(t, \overline{X}_t) \right|_\infty \; \mathbb{1}_{\mathcal{B}_t} + \left| b(t, \overline{X}_t) - \overline{b}(t, \overline{X}_t) \right|_\infty \; \mathbb{1}_{\mathcal{B}_t^c} \\
&\leq 2N^{-\xi} + \left(||b(t, \cdot)||_{\mathcal{L}^\infty(\mathbb{R}^d)} + \left|\left| \overline{b}(t, \cdot) \right|\right|_{\mathcal{L}^\infty(\mathbb{R}^d)} \right) \mathbb{1}_{\mathcal{B}_t^c} \\
&\leq 2N^{-\xi} + \left(\frac{1}{N} \mathrm{K}_{L,V} \; (M_t + v_C) + \mathrm{K}_{V,\lambda,L} t \right) \mathbb{1}_{\mathcal{B}_t^c} \\
&\leq 2N^{-\xi} + \mathrm{K}_{V,\lambda,L,v_C} \left(\frac{M_t}{N} + 1 + t \right) \mathbb{1}_{\mathcal{B}_t^c} \\
&\leq \; 2N^{-\xi} + \mathrm{K}_{V,\lambda,L,v_C} \left(\frac{M_S}{N} + 1 + S \right) \mathbb{1}_{\mathcal{B}_t^c}
\end{aligned}$$

Then, averaging on the inequality 7.9 we can calculate

$$
\begin{aligned}
\mathbb{E}\left[\ J_T^N\ \right] &\leq \mathbb{E}\left[\ N^{\delta}\ \mathrm{K}_d \int_0^S \left|b(t,\overline{X}_t) - \overline{b}(t,\overline{X}_t)\right|_{\infty} dt \cdot \exp\left(T\mathrm{K}_{V,L,d}\ \frac{M_T}{N}\right)\ \right] \\
&\leq N^{\delta}\ \mathrm{K}_d \mathbb{E}\left[\ \int_0^S \left[2N^{-\xi} + \mathrm{K}_{V,\lambda,L,v_C,d}\left(\frac{M_S}{N} + 1 + S\right)\mathbb{1}_{\mathcal{B}_t^c}\right] dt \cdot \exp\left(T\mathrm{K}_{V,L,d}\ \frac{M_T}{N}\right)\ \right] \\
&= \overbrace{N^{-(\xi-\delta)}\ \mathrm{K}_d \mathbb{E}\left[\ \exp\left(T\ \mathrm{K}_{V,L,d}\ \frac{M_T}{N}\ \right)\ \right]}^{\text{Term 1}} \\
&\quad + \underbrace{N^{\delta}\ \mathrm{K}_{V,\lambda,L,v_C,d}\ \mathbb{E}\left[\ \frac{M_S}{N}\int_0^S \mathbb{1}_{\mathcal{B}_t^c}\ dt \cdot \exp\left(T\mathrm{K}_{V,L,d}\ \frac{M_T}{N}\ \right)\ \right]}_{\text{Term 2}} + \\
&\quad + \underbrace{N^{\delta}\ \mathrm{K}_{V,\lambda,L,v_C,d}\ (1+S)\ \mathbb{E}\left[\ \int_0^S \mathbb{1}_{\mathcal{B}_t^c}\ dt \cdot \exp\left(T\mathrm{K}_{V,L,d}\ \frac{M_T}{N}\ \right)\ \right]}_{\text{Term 3}}
\end{aligned}
\tag{7.10}
$$

And at this point the we are exactly at the same stage as in the equation 6.3:

- Sure, we have different constants and different continuous increasing positive functions $f^{\uparrow}(S)$ in S, but they are still constants and continuous increasing positive functions in S.
- The processes
$$\exp\left(T\mathrm{K}\ \frac{M_T}{N}\ \right) \quad \text{and} \quad \frac{M_S}{N}$$
under the integral $\mathbb{E}[\ ]$ are the same as in 6.3, differing just for the constant K.
- The strongest difference is given by $\mathbb{1}_{\mathcal{B}_t^c}$, which is the indicator function of the set tamed by 7.2.2, but this is anyway tamed with a law of the large numbers which differ from 5.4.1 only by the choice of the constants, and not by the powers of N.

Then the proof goes on, from here on, exactly as in the section 6.2, with different constants and a different function $f^{\uparrow}(T)$. And thus the proof is concluded.

8. Conclusion

8.1. The Revelations of the Pulsing Wave Model

Starting from the pulsing wave system 7.1, it is clear that its macroscopic description, i.e.,

$$\begin{cases} a(0,\cdot) = a_0 \\ \quad \partial_t a = \nabla\left[\left(V \star \int_0^t L\lambda c_t^\tau d\tau\right) a\right] + \frac{d\sigma_A^2}{2} \triangle a \\ c^\tau(t,\cdot) \equiv 0 & \text{if } t < \tau \\ c^\tau(\tau,\cdot) = a(\tau,\cdot) \\ \quad \partial_t c^\tau = \frac{d\sigma_C^2}{2} \triangle c^\tau & \text{if } t \geq \tau, \end{cases}$$

is independent from certain mathematical objects of the microscopic model:

- The effective description is not influenced by the constant deterministic velocity v_C of the cAMP molecules. This might seem strange at first glance, but the actual drift for these particles is $\frac{v_C}{N}$, which goes to 0 as N grows to infinity: the result is intuitively justified.
- The system is independent from the choice of the chosen rotations $(R_l)_{l=1}^L$, meaning that even if every amoeba A_t^p at every time T_m shoots all its L children in the same random direction $\widehat{\Theta}_{m,1}$ (i.e., we set $\forall l. R_l = \mathrm{id}$), the macroscopic description remains the same one. This is somehow expected because what counts is that the r.v. $\widehat{\Theta}_{m,1}$ is uniformly distributed on the d-dimensional sphere, and thus each direction $R_l(\widehat{\Theta}_{m,1})$ is in average equal to 0.

Then, the only parameters which do have a say also macroscopically are the number of children L at each birth time and the rate λ with which such birth times occur. With one simple idea for both parameters, we can say that the speed $L \cdot \lambda$ at which the chemoattractant is released is telling us how much $\partial_t a$ is proportional to the member

$$\left[\left(V \star \int_0^t c_t^\tau d\tau\right) a\right].$$

The other thing which we can immediately observe, is that the term above represents a dependence of $\partial_t a$ (the present evolution) from the past, namely from all the state of $a(\cdot,\cdot\cdot)$ in $[0,t]$. This happens because by integrating over $[0,t]$ in τ the expression $c_t^\tau = c^\tau(t,x)$ we are asking what happens at time t to all the cAMP particles created along the times $\tau \in [0,t]$. This is even more evident if we substitute c_t^τ with its analytical solution given by 3.15 and obtain the P.I.D.E. 7.4:

$$\partial_t a = L\lambda\nabla\left[\left(\int_0^t V \star \Phi_{t-\tau}^{\frac{d\sigma_C^2}{2}} \star a(\tau,\cdot) d\tau\right) a\right] + \frac{d\sigma_A^2}{2} \triangle a.$$

Here a appears also under the integral, and thus the memory effect (the direct dependence of the present t from the events happening in $[0,t]$) is explicit.

8.1.1. Comparison with the Equation of Keller-Segel

The original paper from Cañizares García [11] modeled the movement of amoebas as individuals attracting each others with a Coulomb-like potential (the cAMP particles were absent). From this she derived the notorious Keller-Segel equation [23] for the specific case in which the chemoattractant diffuses much faster than the organisms releasing it [19]:

$$\begin{cases} \partial_t a = -\chi \nabla(a \nabla c) + \triangle a \\ -\triangle c = a \end{cases}, \tag{8.1}$$

or, solving c_t analytically:

$$\partial_t a = \chi \nabla((V_\lambda \star a)a) + \triangle a, \tag{8.2}$$

where $V_\lambda := \frac{x}{2\pi|x|^2}$ is the attraction coming from the previously mentioned Coulomb-like potential and χ is the *chemosensitivity*, namely a positive constant indicating the sensibility of the organisms to the chemoattractant. We modeled a slightly different scenario: the function V taken into consideration is Gaussian-like, and the diffusion coefficients σ_A, σ_C of the amoebas and of the cAMP molecules are both finite (i.e., of comparable order of magnitude). Let us ignore the difference regarding V for now. Then, to compare the two models, we want to ask what happens to 7.4 if we modulate the diffusions coefficient to be analogous to the ones treated in [11], i.e., when $\lambda >> 1$ and $\sigma_C >> \sigma_A$. To simplify the situation as Cañizares García does, let us set $\sigma_A := \sqrt{\frac{2}{d}}$ so that $\frac{d\sigma_A^2}{2} \triangle a = \triangle a$.

Slow Release of a Chemoattractant with a High Diffusion

Let us for now leave λ as it is. Consider the modified heat-kernel from 3.15:

$$\Phi_{t-\tau}^{\frac{d\sigma_C^2}{2}}(x) = \frac{1}{\left(2\pi d\sigma_C^2(t-\tau)\right)^{\frac{d}{2}}} \exp\left(-\frac{x^2}{2d\sigma_C^2(t-\tau)}\right) \mathbb{1}_{t-\tau\geq 0} = \frac{d\,\mathcal{N}\left(0, d\sigma_C^2(t-\tau)\right)}{d\,\mathrm{Leb}}(x)\mathbb{1}_{t\geq\tau}.$$

Then in the limit case we do obtain

$$\Phi_{t-\tau}^{\frac{d\sigma_C^2}{2}}(x) \xrightarrow{\sigma_C \to +\infty} \frac{d\,\mathcal{N}(0, "+\infty")}{d\,\mathrm{Leb}}(x)\mathbb{1}_{t\geq\tau} \equiv 0$$

and therefore the effective equation 7.3 becomes

$$\partial_t a = L\lambda\nabla\left[\left(\int_0^t V \star 0 \star a(\tau,\cdot)d\tau\right)a\right] + \frac{d\frac{2}{d}}{2} \triangle a = \triangle a,$$

which is the heat equation. This can be interpreted as the consequence of the fact that the chemoattractant is diffusing with a very high speed but is released with a slow rate λ, and therefore escapes at infinity too fast to have an influence on the amoebas.

Fast Release of a Chemoattractant with a High Diffusion

We expect our model to reach a description similar to the one of [11] in the moment in which both λ and σ_C tend to infinity. The idea is that as the release frequency λ of the cAMP and as its diffusion σ_C grow, the movement of the amoebas in the time interval $[0, t]$ becomes in scale less and less relevant, which means that we can approximate $a(\tau, \cdot)$ with $a(t, \cdot)$:

$$L\lambda\nabla\left[\left(V\star\int_0^t\Phi_{t-\tau}^{\frac{d\sigma_C^2}{2}}\star a(\tau,\cdot)d\tau\right)a\right]\;\approx\;L\lambda\nabla\left[\left(V\star\int_0^t\Phi_{t-\tau}^{\frac{d\sigma_C^2}{2}}d\tau\star a(t,\cdot)\right)a\right]=$$
$$=L\nabla\left[\left(V\star\int_0^t\frac{\lambda}{\left(2\pi d\sigma_C^2(t-\tau)\right)^{\frac{d}{2}}}\exp\left(-\frac{x^2}{2d\sigma_C^2(t-\tau)}\right)d\tau\star a\right)a\right].$$

At this point one might take a look at the function

$$\widetilde{V}(t,x)\;:=\left(V\star\int_0^t\frac{\lambda}{\left(2\pi d\sigma_C^2(t-\tau)\right)^{\frac{d}{2}}}\exp\left(-\frac{(\cdot)^2}{2d\sigma_C^2(t-\tau)}\right)d\tau\right)(x)$$
$$=\left(V\star\int_0^t\frac{1}{\left[2\pi d\left(\frac{\sigma_C}{\lambda^{\frac{1}{d}}}\right)^2(t-\tau)\right]^{\frac{d}{2}}}\exp\left(-\frac{(\cdot)^2}{2d\sigma_C^2(t-\tau)}\right)d\tau\right)(x)$$

and notice that if λ,σ_C are very big, than the time integration is relevant mainly when $\tau\in[0,\epsilon]$ for some very small ϵ. In general we expect that if $\lambda,\sigma_C\to+\infty$ in a way such that the ratio $\frac{\sigma_C}{\lambda^{\frac{1}{d}}}$ remains constant, then the member $\widetilde{V}(t,x)$ will approach the time independent gradient of some potential $\frac{\mathrm{K}}{|x|}$ (for some constants K). In such case, by signing this time-independent attraction as $\overline{V}(x)$, the effective equation would become

$$\partial_t a=\chi\nabla((\overline{V}\star a)a)+\triangle a,\tag{8.3}$$

which is the equation 8.2 from [11]. The interpretation of this limit case is that the cAMP is released so fast, that it compensates for the very high diffusion.

Slow Release of a Chemoattractant with no Diffusion

It is interesting to observe what happens in the other limit case, namely when $\sigma_C\to0^+$ (i.e., the cAMP does not diffuse):

$$\Phi_s^{\frac{d\sigma_C^2}{2}}(x)\xrightarrow{\sigma_C\to0^+}\frac{d\,\mathcal{N}(0,0)}{d\,\mathrm{Leb}}(x)\mathbb{1}_{s\geq0}=\delta(x)\mathbb{1}_{s\geq0}.$$

In such case the macroscopic description of the amoebas' density becomes

$$\begin{aligned}\partial_t a&=\overbrace{L\lambda}^{=:\chi}\nabla\left[\left(\nabla\int_0^t v\star\delta(x)\mathbb{1}_{t-\tau\geq0}\star a(\tau,\cdot)d\tau\right)a\right]+\triangle a\\&=\chi\nabla\left[\left(V\star\int_0^t a(\tau,\cdot)d\tau\right)a\right]+\triangle a.\end{aligned}\tag{8.4}$$

Now notice that the expression obtained is similar to the equation 8.1: they differ in the convolution term, which is $(V_\lambda\star a)$ in 8.1 and is $(V\star\int_0^t a(\tau,\cdot)d\tau)$ in 8.4. Though we can see that even if we would have the same attraction V, the partial integro-differential equation (PIDE) 8.4 still would have a memory of the past (the integration on $[0,t]$) that the Keller-Segel equation does not possess.

8.2. Further Research Possibilities

8.2.1. The Most Enticing Question

The portrait depicted by the pulsing wave models is that of amoebas which are moving towards the chemotactic agent, where the latter one is released in *independent* bursts by the population. We mean that in our image 7.1, the time at which a burst happen (T_m) is not dependent on the contact with the chemotactic agent (C_t), or in other words the amoebas are not communicating with each others, but are simply shouting at random times and are following the noise they hear. This is just half of what happens in nature, where the organisms involved are not only aiming at the direction of the chemical cry, but are also actively answering to it, i.e., they *react* to a pulse of cAMP with another pulse, implementing a mechanism of *chemical response*. The really interesting challenge would therefore be to build a mathematical model of this phenomenon and search for its effective description.

Some Ideas on an Hypothetical "Answering Pulses" Model A mathematical depiction of such an idea would be particularly difficult because the processes $(M_t^p)_p$ describing the release of cAMP molecules would now be dependent on $X_t^{(N)} = (A_t^{(N)}, C_t^{(N)})$. This fact would hinder the usage of many mathematical tactics which were adopted along this work to calculate the effective equation candidate and to prove that such smart guess was correct. One would probably like to let such processes be Poisson distributed in case of no contact between the organisms and the chemotactic agent, but also to forcibly jump up at the moment in which a particle of substance C_t^m travels near enough to one of the amoebas. One could then define a sequence of stopping times $(S_j)_j$ describing (in chronological order) all the impacts between a cAMP molecule and an amoeba. We expect that this would let us say that the system of answering amoebas so described, between two successive impacts, behaves exactly as the pulsing wave model 7.1 aforethought. We also expect the impacts' frequency to grow with a high probability, the more the time flows, namely we expect $f(j) := S_j - S_{j-1}$ to be with a high probability a strictly decreasing function. Of course such ideas definitely need a more rigorous framework to be discussed properly.

8.2.2. "Nice" Technical Properties to Have for the Pulsing Wave Model

Assume a Coulomb-like Potential As in the original PhD thesis [11], it would be interesting to use a Coulomb-like potential (the analogous to the generalized attraction V_G used in the law of large numbers 5.4.1, but without cutoff), and prove that the equation 7.3 still holds with V substituted with V_G. As seen in other works previously [11], [5], the problem that arises is that V_G is not globally Lipschitz anymore, spoiling this property also for the drift $b(t, x)$. The solution is to show that $b(t, \cdot)$ is locally Lipschitz with a high probability (using essentially the same argument as the law of large numbers 5.4.1), and use such condition to bound the term $b(t, X_t) - b(t, \overline{X}_t)$ which spawns in the proof of the final result 7.7. We expect that using the techniques of [5], this proof should be not too difficult to construct.

Prove a Higher Degree of Convergence Again, as in [11], it would be interesting to obtain a higher degree of convergence for the distance between the microscopic and the mesoscopic description, namely going from the condition 7.7 (for the basic model more precisely stated in 6.0.1) to a result of this kind:

Proposition 8.2.1. *Let $0 \leq t \leq T$. Let $X_t^{(N)} = (A_t^{(N)}, C_t^{(N)})$ and $\overline{X}_t^{(N)} = (\overline{A}_t^{(N)}, \overline{C}_t^{(N)})$ be the solutions to the systems 7.1 and 7.2 respectively. Then, for each $\gamma > 0$ and there is a positive, increasing and continuous function $f_\gamma^\uparrow$ independent of N such that*

$$\mathbb{P}\left(\sup_{0\leq t\leq T} \left| X_t^{(N)} - \overline{X}_t^{(N)} \right|_\infty \geq N^{-\delta} \right) \leq \ f_\gamma^\uparrow(T)\ N^{-\gamma}. \tag{8.5}$$

Contrary to the previous point, we expect this to be a tougher goal to pursue.

8.2.3. Numerics and Simulations on the Pulsing Wave Model

It could be an exciting task to search numerical solutions both for the PIDE and for the stochastic system 7.1, or to create simulations for the latter one. Of course, we expect this not be easy because the PIDEs are generally considered very wild beasts to deal with, and because the system 7.3, as underlined in chapter 2, is too stochastic to be precisely an SDE, therefore one could not directly apply the Euler-Maruyama method to it, but would likely first need to sample the spawning Poisson processes $(M_t^p)_p$. Still, it is not excluded that the author will one day set off on one of such adventures (keep an eye on **this Github profile**).

8.2.4. More Technical and (Probably) less Interesting Questions

Higher (Finite) Dimensions If somebody would like to know what happens with the Dictyostelium discoideum amoebas moving in a 42-dimensional space (or whatever higher dimension), we expect the macroscopic description proved in this master thesis to still be valid, since along this work the dimension was a finite dimensional parameter d. We also expect this to be a very futile question, unless somebody will discover some 42-dimensional amoeba or some natural phenomenon which can be modeled exactly in the same way as 7.1 but with $d = 42$.

The Solution(s) to the PIDE and its Properties As suggested in chapter 3.3.2, one might want to prove that the partial integro-differential equation 7.4 has a solution (with the required conditions) exploiting the Schauder fixed point theorem 3.3.1. We expect this solution to exist, thus we regard this more as a technical verification rather than an epiphany bringing brilliant new insights. One might also inquire about the properties of such $a(t, x)$, which would probably be a more interesting adventure, but since the object considered is a PIDE, we expect this not to be an easy task.

The Macroscopic Description of the cAMP Molecules As suggested in 3.5, one might inquire about proving formally that

$$\text{"}\mathbb{1}_{M_t\neq 0}\frac{1}{M_t}\sum_{m=1}^{M_t}\delta_{C_t^m}\text{"} =: \mu_t^{C,(N)} \xrightarrow[\text{in law}]{N\longrightarrow+\infty} c(t,\cdot) := \frac{1}{t}\int_0^t c_t^\tau d\tau,$$

where the measure considered are treated as r.v. in the space of Borel probability measures with finite moments $\mathcal{BP}_1$.

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A. Appendix

A.1. Notation

A.1.1. Common Sets and Measures

In this work it will adopted the convention that the set of natural numbers $\mathbb{N}$ starts with the number 1. If we want to include the 0 we will write $\mathbb{N}_0$, therefore

$$\mathbb{N} = \{1, \ldots\}, \quad \mathbb{N}_0 = \{0, 1, \ldots\}$$

Whenever we have a measure μ on a space, to indicate a property $P(x)$ valid almost everywhere with respect to μ we write

$$\forall^{\mu} x.\ P(x)$$

The Lebesgue measure is signed as Leb.

A.1.2. Norms

For the great majority of cases we must take into consideration some function

$$\begin{aligned} f : \mathbb{R}_+ \times \mathbb{R}^m &\longrightarrow \mathbb{R}^n \\ (t, x) &\longmapsto f(t, x) \end{aligned}$$

and consider some kind of norm $||f(t,x)||$. We adopt the subsequent convention, where f_i are the components of f:

$$\begin{cases} |f(t,x)|_{\infty} := \max\limits_{1 \le i \le n} |f_i(t,x)| & \text{where } x, t \text{ are fixed} \\ ||f(t,x)||_{\infty} := \max\limits_{1 \le i \le n} ||f_i(t,\cdot)||_{\mathcal{L}^{\infty}(\mathbb{R}^m)} & \text{where } t \text{ is fixed} \end{cases}$$

In the cases where it is needed to consider the supremum also on some time interval $[s_1, s_2]$ this will be explicitly written down:

$$\sup_{s_1 \le t \le s_2} ||f(t,x)||_{\infty} = \sup_{s_1 \le t \le s_2} \max_{1 \le i \le n} ||f_i(t,\cdot)||_{\mathcal{L}^{\infty}(\mathbb{R}^m)}$$

In some cases where the $\mathcal{L}^1$-norm appears together with the $\mathcal{L}^{\infty}$-norm, for symmetry, the latter one will be also signed as $||\cdot||_{\mathcal{L}^{\infty}}$.

A.1.3. Generic Positive Constants and Increasing Continuous Functions

We will indicate generic constants with the symbol K, and a subscript will indicate on which mathematical objects such constants depend:

$$\mathrm{K}_{x,y,z} := \text{ generic const} > 0 \text{ dependent on } x, y, z$$

Sometimes we will operate on a fixed time t, considering certain quantities as constants $\mathrm{K}_{t,y}$, but then it will be necessary to consider an object of the type

$$\sup_{t\in[0,T]} \mathrm{K}_{t,y}$$

and show that this is bounded. Of course here the problem arises because $[0,T]$ is an infinite set, and thus the sup might be $+\infty$. In such situation it is usually necessary to show that the function $f_y(t) := \mathrm{K}_{t,y}$ is increasing and continuous on $\mathbb{R}$ so that we can affirm

$$\sup_{t\in[0,T]} \mathrm{K}_{t,y} = \sup \mathrm{K}_{T,y} < +\infty$$

Similarly is the case when we want to bound quantities on finite sets in $\mathbb{N}$ with the value assumed in the upper extremum:

$$¿ \quad \sup_{0\leq q\leq Q} \mathrm{K}_q \leq \mathrm{K}_Q \quad ?$$

or do both things at the same time, where the constant might depend on some further objects a, b, c:

$$¿ \quad \sup_{t\in[0,T]} \sup_{0\leq q\leq Q} \mathrm{K}_{q,t,a,b,c} \leq \mathrm{K}_{Q,T,a,b,c} < +\infty \quad ?$$

In all these cases it is good solution to show that for every choice of a, b, c the function $f_{a,b,c}(q,t) := \mathrm{K}_{q,t,a,b,c}$ is increasing in both t and q and it is continuous in t. For this reasons we indicate with

$$f^{\uparrow}_{a,b,c}(t,q)$$

a generic positive function dependent from a, b, c, *increasing* in each of its arguments and continuous (it is trivially $\mathcal{C}^0$ on q if we endow $\{1,\ldots,Q\}$ with the discrete topology).

A.1.4. System, Drift and Diffusion

Components

The drift b and diffusion σ of the systems taken into account are defined per-particle, and each particle is d-dimensional, e.g., the expression

$$b_p(t, X_t)$$

indicates the drift in $\mathbb{R}^d$ for the particle A^p_t. If we want to pin down then the second component of the drift defined for such A^p_t, the temptation is to then write

$$b_{p,2}(t, X_t)$$

This create ambiguity though, because the cAMP molecule (and the drift that they feel) are identified with a double index p, g, and thus the notation above might indicate the drift to which the cAMP molecule $C^{p,2}$ is subjected. For this reason the components of the drift and the diffusion will be written as

$$< d\text{-dimensional particle} >; < 1 - \text{dimensional component of the } d\text{-dimensional vector} >$$

i.e., if we indicate with $\pi_i : \mathbb{R}^d \longrightarrow \mathbb{R}$ the projection on the i-th component, the expressions

$$b_{p;i}(t, X_t) := \pi_i(b_p(t, X_t))$$
$$b_{p,g;i}(t, X_t) := \pi_i(b_{p,g}(t, X_t))$$

indicate the i-th 1-dimensional component of the drift for the amoeba A^p and of the drift for the cAMP molecule $C^{p,g}$. If necessary, the same notation will be used on stochastic processes $A_t^p = (A_t^{p;i})_{i=1}^d$.

Superscript

Starting from 3 it will be important in many occasions to keep in mind the amount of amoebas considered in the system for many mathematical objects. In such case the number of amoebas involved is signed a superscript with parenthesis, e.g., $b_p^{(N)}(t, X_t)$ is the drift for A^p where the model has N amoebas.

This shall not create ambiguity, in the case in which we want to sign the components of a multivariate stochastic process $X_t^{(N)}$: they shall be $X_t^{(N),p}$ and $X_t^{(N),p,g}$ for the d-dimensional position of the amoebas and the cAMP respectively. The 1 dimensional coordinates will then be $X_t^{(N),p;i}$ and $X_t^{(N),p,g;i}$ with $1 \leq i \leq d$.

A few non-ambiguous exception are present, the most notable one being the auxiliary process of the final proof: J_t^N.

A.1.5. Conditional Expectations and Probabilities

Mainly in the proofs of the laws of large numbers we will need to do long calculations with conditional expectations and probabilities. For this reason the subsequent less heavy notation has been adopted: a tilde above $\mathbb{E}[\,\cdot\,], \mathbb{P}$ will indicate conditioning with respect to $(T^{p,g})_{p,g}$, and a subscript k will indicate a conditioning with respect to the k-th amoeba (usually $\overline{A}^k$, but depending on the context it could be also A^k):

$$\widetilde{\mathbb{E}}[\,\cdot\,] := \mathbb{E}[\;\cdot\;|\;(T^{p,g})_{p,g}\,] \qquad \widetilde{\mathbb{P}}(\cdot) := \mathbb{P}(\;\cdot\;|\;(T^{p,g})_{p,g}\,)$$
$$\widetilde{\mathbb{E}}_k[\,\cdot\,] := \mathbb{E}[\;\cdot\;|\;k\text{-th amoeba},\,(T^{p,g})_{p,g}\,] \qquad \widetilde{\mathbb{P}}_k(\cdot) := \mathbb{P}(\;\cdot\;|\;k\text{-th amoeba}\,,\,(T^{p,g})_{p,g}\,)$$
$$\mathbb{E}_k[\,\cdot\,] := \mathbb{E}[\;\cdot\;|\;k\text{-th amoeba}\,] \qquad \mathbb{P}_k(\,\cdot\,) := \mathbb{P}(\;\cdot\;|\;k\text{-th amoeba}\,)$$

Furthermore, a subscript $*$ will mean ambiguity in the presence of the subscript:

$$\mathbb{E}_*[\;] := \text{ either of } \mathbb{E}[\,\cdot\,], \mathbb{E}_k[\,\cdot\,] \qquad \mathbb{P}_*(\,\cdot\,) := \text{ either of } \mathbb{P}(\,\cdot\,), \mathbb{P}_k(\,\cdot\,)$$
$$\widetilde{\mathbb{E}}_*[\;] := \text{ either of } \widetilde{\mathbb{E}}[\,\cdot\,], \widetilde{\mathbb{E}}_k[\,\cdot\,] \qquad \widetilde{\mathbb{P}}_*(\cdot) := \text{ either of } \widetilde{\mathbb{P}}(\cdot), \widetilde{\mathbb{P}}_k(\cdot)$$

A.1.6. Cyrillic Variables

The cyrillic letters Ч, Ж, Б and Ю are used in this master thesis for mathematical objects that need to be temporarily addressed (e.g., in proofs), for which thus we do not want to use greek or latin letters which would be useful in other places. Do not feel intimidated by them: Ч is pronounced as the sound "tsch" in German, Ж as the "j" in the french word "bonjour" and Ю as the english word "you". And Б is just a "b".

A.2. Miscellaneous from Stochastics

Lemma A.2.1 (Estimates of the moments of a Poisson r.v.)**.** *Let P be a Poisson r.v. of parameter μ. Then there is an increasing function $f^{\uparrow} : \mathbb{N} \longrightarrow \mathbb{N}$ independent from μ such that*

$$\mathbb{E}\,[\,|P|^n\,] \leq f^{\uparrow}(n)\,(\mu^n \vee 1)$$

Proof. In [13, Equation (1.3-14)] we have that for each $P \sim \text{Poisson}(\mu)$

$$\mathbb{E}\,[\,|P|^n\,] = \sum_{k=1}^{n} \mu^k \left\{ {n \atop k} \right\}$$

where

$$\left\{ {n \atop k} \right\} := \left(\frac{1}{k!} \sum_{i=0}^{k} (-1)^{k-i} \binom{k}{i} i^n \right)$$

are the Stirling number of the second kind. Then it is sufficient to observe that

$$\mathbb{E}\,[\,|P|^n\,] = \sum_{k=1}^{n} \mu^k \left\{ {n \atop k} \right\} \leq \sum_{k=1}^{n} (\mu \vee 1)^k \left\{ {n \atop k} \right\} \leq (\mu \vee 1)^n \sum_{k=1}^{n} \left\{ {n \atop k} \right\} = (\mu^n \vee 1) \underbrace{\sum_{k=1}^{n} \left\{ {n \atop k} \right\}}_{=: f^{\uparrow}(n)}$$

to obtain the thesis. □

We have also this useful result from [13, Equation (1.6-9)].

Lemma A.2.2 (The moment generating function of a Poisson r.v.)**.** *Let P be a Poisson r.v. of parameter μ. Then its moment generating function is given by:*

$$\mathbb{E}\,[\,\exp(sP)\,] = \exp[\mu(e^s - 1)]$$

The next one is a one line proof that looks absolutely trivial here. Yet in some very cumbersome calculations, where the r.v. involved are particularly nasty, we might miss that some inequalities happen because of this very obvious fact.

Lemma A.2.3. *Let A, B be non-negative r.v. and let $a, b > 0$. Then we have that*

$$\mathbb{P}\,(\,A + B \geq a + b\,) \leq \mathbb{P}\,(\,A \geq a\,) + \mathbb{P}\,(\,B \geq b\,)$$

Proof.

$$\begin{aligned} \mathbb{P}\,(\,A + B \geq a + b\,) &= \mathbb{P}\,(\,A \geq a, A + B \geq a + b\,) + \mathbb{P}\,(\,A < a, A + B \geq a + b\,) \\ &= \mathbb{P}\,(\,A \geq a, A + B \geq a + b\,) + \mathbb{P}\,(\,B \geq b, A + B \geq a + b\,) \\ &\leq \mathbb{P}\,(\,A \geq a\,) + \mathbb{P}\,(\,B \geq b\,) \end{aligned}$$

□

A.3. The Solutions for the Basic Model, More Formally

In this section we want to address a bit more formally two closely related questions:

1. What is the exact space of solutions for the solutions to the basic model? In the definition 2.2.2 did not specify the codomain of the r.v. involved.

2. How can we be sure that the conditioning $\mathbb{P}$ we are obtaining *regular* conditional probability measures ([7, Chapter 10.2, first definition]), i.e., that we obtain mathematical objects that behave as probability measures? This is an important question because for the definition 2.2.2 we need certain objects to be B.m. under $\mathbb{P}(\ \cdot \mid\ (T^{p,g})_{p,g}\ )$, thus we assume implicitly that $\mathbb{P}(\ \cdot \mid\ (T^{p,g})_{p,g}\ )$ is a probability measure.

We will not give a complete answer to the second question, but we will give a direction to search for it, and we expect that with further work one might prove the necessary technicalities required to obtain the desired result.

A.3.1. The Space of Solutions

We first need to check where the spawning Poisson processes $(M_t^p)_p$ are taking values for each fixed time t. This is of course $(\mathbb{N}_0)^N$. Now we define the space of configurations for the cAMP molecules $(C^{p,g})_g$ spawned by a single amoeba A^p. Here the symbol "♠" means that a particle has not been spawned yet.

Definition A.3.1. Define the space of the positions cAMP particles spawned by a single amoeba as:

$$\mathbb{R}_{\text{camp}}(d) := (\{\spadesuit\} \sqcup \mathbb{R}^d)^{\mathbb{N}}$$

Define the space of configurations for all particles (the N amoebas A^p and the cAMP molecules $C^{p,g}$) as

$$\text{Conf} := (\mathbb{R}^d)^N \times \left(\mathbb{R}_{\text{camp}}(d)\right)^N$$

For each configuration of particles $x \in \text{Conf}$ sign its d-dimensional coordinates as

$$\begin{aligned} x &= (x^1, \dots, x^N, \quad (x^{1,g})_{g\in\mathbb{N}}, \dots, (x^{p,g})_{g\in\mathbb{N}}, \dots, \quad (x^{N,g})_{g\in\mathbb{N}}, \dots) \\ &= (a^1, \dots, a^N, \quad c^{1,1}, c^{1,2}, \dots, c^{p,1}, \dots, \qquad\qquad c^{N,1}, \dots) \end{aligned}$$

where the absence of a coordinate in the list means that such coordinate is set to ♠.

Remark A.3.1. One might want to define the configuration space of the cAMP as the more precise set

$$\begin{aligned} &\{x = (x_g)_{g\in\mathbb{N}}\ :\ \exists G \in \mathbb{N}_0.\ \forall g < G.\ x_g \in \mathbb{R}^d \ \text{ and } \ \forall g \geq G.\ x_g = \spadesuit\} \subseteq \\ &\subseteq \bigoplus_{g=1}^{\mathbb{N}} (\{\spadesuit\} \sqcup \mathbb{R}^d) \ := \{x = (x_g)_{g\in\mathbb{N}}\ :\ \exists G \in \mathbb{N}_0.\ \forall g \geq G.\ x_g = \spadesuit\} \end{aligned}$$

But such space is very likely not Polish (the topic of the next section), thus we rely on the previous choice.

And what about the Brownian motion associated to each particle? That is just a countable amount of independent d-dimensional B.m. . With these definitions the solution of 2.2.2 can be seen as a *random càdlàg* function

$$((M^p)_p, X = (A, C), B)(\omega) :\ \mathbb{R}_+ \longrightarrow \underbrace{(\mathbb{N}_0)^N}_{\text{Poisson processes}} \times \underbrace{(\mathbb{R}^d \times \mathbb{R}_{\text{camp}}(d))^N}_{\text{positions particles}} \times \underbrace{(\mathbb{R}^d)^N \times (\mathbb{R}_{\text{camp}}(d))^N}_{\text{B.m.}}$$

or, in other words, the r.v. $((M^p)_p, X, B)$ takes values in the set

$$\left\{ f : \mathbb{R}_+ \longrightarrow (\mathbb{N}_0 \times \mathbb{R}^d \times \mathbb{R}_{\text{camp}}(d) \times \mathbb{R}^d \times \mathbb{R}_{\text{camp}}(d))^N \ :\ f \text{ is càdlàg} \right\} \tag{A.1}$$

It is not necessary to redefine the drift b and the diffusion σ, because they become relevant for the SDE framework only under the conditioning $\mathbb{P}(\ \ |\ (T^{p,g})_{p,g}\ )$. In any case, if we really wanted to see them in this framework, we would have for absent particle they have no mapping, so one could set them to $\spadesuit$ for every cAMP molecule not yet spawned and obtain consistent mappings in the appropriate spaces.

A.3.2. Can we Obtain Regular Probability Measures by Conditioning $\mathbb{P}$?

For this chapter it is assumed the knowledge of [7, Section 10.2]. The main instrument to verify that a measure is regular is [7, Theorem 10.2.2], namely:

Theorem A.3.1. *Let T be any Polish space, $\mathcal{B}$ its σ-algebra of Borel sets. Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space, $Y : \Omega \longrightarrow T$ a $(\mathcal{A}, \mathcal{B})$-measurable function and $\mathcal{C}$ any sub-σ-algebra of $\mathcal{A}$. Then there exists a conditional distribution $\mathbb{P}_{Y|\mathcal{C}}(\cdot, \cdot\cdot) : \mathcal{B} \times \Omega \longrightarrow [0, 1]$ (see [7, Chapter 10.2, first definition]) and it is unique, meaning that if another $\mathbb{P}'(\cdot, \cdot\cdot)$ satisfies the definition of conditional distribution, then*

$$\forall^{\mathbb{P}}\omega \in \Omega. \quad \mathbb{P}_{Y|\mathcal{C}}(\cdot, \omega) = \mathbb{P}'(\cdot, \omega)$$

In other words, given a sub-σ-algebra $\mathcal{C}$ of $\mathcal{A}$ and a r.v. $Y : (\Omega, \mathcal{A}, \mathbb{P}) \longrightarrow (T, \mathcal{B})$, a sufficient condition for $\mathbb{P}(\ Y \in \cdot \mid \mathcal{C}\ )$ to behave like a measure for $\mathbb{P}_{|\mathcal{C}}$-almost every ω is that the codomain T is a Polish space, i.e.,

Definition A.3.2. Let (T, τ) be a topological space. T is *Polish* iff τ is metrizable with respect to a metric d such that (T, d) is a complete separable metric space.

Then the regularity of our measures is depending not on the σ-algebra on which we are conditioning, but on the codomain of the process which is being conditioned, which must be Polish. Most of the processes we take into account are taking values into some subset of A.1 of the form

$$\left\{ f : \mathbb{R}_+ \longrightarrow (\mathbb{R}^d)^n \times (\mathbb{R}_{\text{camp}}(d))^k :\ f \text{ is càdlàg} \right\}$$

for some $n, k \in \mathbb{N}_0$. This is an idea of how somebody might try to prove that this space is Polish:

1. Prove that $\{\spadesuit\} \sqcup \mathbb{R}^d$ is Polish. This can be done by viewing $\{\spadesuit\} \sqcup \mathbb{R}^d$ as a metrized one-point-compactification of $\mathbb{R}^d$ (since $\mathbb{R}^d$ is Polish one can use [22, Theorem 5.3] to obtain the metric), which then is separable (compact and metrizable). Since then $\{\spadesuit\} \sqcup \mathbb{R}^d$ is metrizable and compact, by [29, Theorem 45.1] we obtain that it is complete, and thus a Polish space.

2. Prove that $\mathbb{R}_{\text{camp}}(d)$ is Polish. One might want to generalize the result [29, Theorem 43.4] for $\{\spadesuit\} \sqcup \mathbb{R}^d$ to ensure that the countable product $(\{\spadesuit\} \sqcup \mathbb{R}^d)^{\mathbb{N}}$ is complete. In such case, since the metric ρ of the latter theorem induces the product topology, we get that by the Tychonoff theorem ([29, Theorem 37.3]) the space $(\{\spadesuit\} \sqcup \mathbb{R}^d)^{\mathbb{N}}$ is also compact. Being compact and metrizable we obtain that is also separable, and therefore Polish.

3. Since both the completeness and the separability remain stable under a finite product we obtain that $(\mathbb{R}^d)^n \times (\mathbb{R}_{\text{camp}}(d))^k$ is Polish.

4. Defining $E := (\mathbb{R}^d)^n \times (\mathbb{R}_{\text{camp}}(d))^k$ we have that the space

$$\left\{ f : \mathbb{R}_+ \longrightarrow (\mathbb{R}^d)^n \times (\mathbb{R}_{\text{camp}}(d))^k :\ f \text{ is càdlàg} \right\}$$

can be seen as the *Skorokhod space* $D_E[0,\infty)$ (see [8, Chapter 3, Section 5]), which, if E is Polish, is also Polish ([8, Chapter 3, Theorem 5.6]).

We expect then that the all conditional probability measures are regular, since we expect that the processes considered have Polish spaces as codomains.